Climate Policies under Dynamic Factor Adjustment *

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Abstract

I develop a dynamic general equilibrium model of the global economy with gradual and imperfect capital and labor reallocation, and show that imperfect factor mobility can cause clean energy subsidies to increase rather than decrease global emissions. In the calibrated model, unilateral permanent clean energy subsidies in China and the United States increase global emissions in the long run. This result arises because, under imperfect factor mobility, the output expansion through production linkages and capital accumulation outweighs the decline in emissions per unit of output. Directing model-implied clean energy subsidy expenditure toward capital maximizes domestic emission reductions but incurs the largest welfare costs, while directing it toward labor yields modest reductions and performs best on welfare.

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1 Introduction

Failure to meet the Paris Agreement's target of limiting global warming to 1.5°C above pre-industrial levels is projected to significantly increase the frequency and severity of extreme weather events, such as heat waves, heavy rainfall, and droughts.[1] A key obstacle to the transition toward a low-carbon economy is the persistence of entrenched infrastructure and institutions, which sustains reliance on existing production structures and slows the reallocation toward renewable energy (Lee et al., 2023). Existing quantitative frameworks, however, typically assume frictionless factor reallocation across sectors, limiting their ability to assess how climate-policy effectiveness changes when capital and labor respond gradually and imperfectly.[2]

This paper develops a quantitative framework to examine how imperfect and gradual adjustment of capital and labor across sectors shapes both the effectiveness of climate policies and the ranking of alternative policy instruments. I show that imperfect factor mobility can cause clean energy subsidies to increase rather than decrease global emissions, and that the ranking of alternative instruments depends on economy-specific production structure and factor adjustment.

The quantitative framework is a dynamic general equilibrium model of the global economy with sectoral and international production linkages. The model features multiple sectors, divided into three groups: (i) a brown energy sector that produces energy using raw fossil fuels, (ii) a clean energy sector that produces an alternative to fossil fuel–based energy using intermediate inputs such as solar panels and wind turbines, and (iii) non-energy sectors, such as manufacturing, that require energy produced by the brown and clean energy sectors as intermediate inputs in production. Raw fossil fuel use generates carbon emissions, which reduce economy-specific environmental quality and, in turn, aggregate productivity (Barrage and Nordhaus, 2024).

The overall impact of climate policies on global emissions depends on two forces: (i) a substitution effect and (ii) a scale effect. The substitution effect arises from relative price

[1] If emissions remain at 2023 levels, global temperatures could reach 1.5°C above pre-industrial levels within a decade and exceed 2°C in 27 years (Lee et al., 2023). The remaining carbon budget is estimated at 500 $GtCO_2$ for a 50% chance of limiting warming to 1.5°C and 1150 $GtCO_2$ for a 67% chance of staying below 2°C (Lee et al., 2023). Global carbon emissions in 2024 was approximately 43.18 $GtCO_2$.

[2] Imperfect factor mobility is an empirical regularity extensively documented in the international trade literature. For examples, see Artuç et al. (2010), Dix-Carneiro (2014), Dix-Carneiro and Kovak (2017).

changes that reallocate demand and factors of production away from the brown energy sector; for example, a clean energy subsidy lowers the relative price of clean energy and shifts capital and labor from brown energy production to the clean energy sector. The scale effect operates through aggregate expansion: by lowering energy input costs, the subsidy stimulates energy use in production, reduces intermediate input prices, and raises demand for goods produced in the subsidizing economy, including brown energy, thereby increasing aggregate output and potentially global emissions.[3] Under imperfect factor mobility, capital and labor respond sluggishly to policy changes, weakening the substitution effect and making it more likely that the scale effect dominates and global emissions rise.

I estimate the key parameters governing capital and labor adjustment using country-sector-level data from the World Input–Output Database (WIOD) 2016 Release for capital supply and individual-level longitudinal labor surveys from eight countries for labor supply.[4] The remaining model parameters are calibrated to a global economy consisting of the United States, China, India, the European Union, and the Rest of the World, using multiple datasets including the WIOD 2016 Release.

The estimates reveal (i) significant idiosyncratic heterogeneity in capital efficiency within and across sectors and (ii) strong non-pecuniary motives affecting workers' mobility choices, both of which dampen capital and labor adjustments to sector-specific policy changes. Specifically, I find that a 1 percent decline in the rental rate of capital in the brown energy sector leads to only a 0.31 percent increase in capital supplied to other sectors within the same period. For labor, a 1 percent decline in wages in the brown energy sector increases annual worker outflows to other sectors by only 0.69 percent.

Comparing the baseline model featuring imperfect factor mobility to counterfactual models featuring perfect factor mobility, I find that imperfect factor mobility substantially weakens the substitution effect of climate policies. In each policy experiment, I compare outcomes to a business-as-usual (BAU) scenario where the policy is not implemented. A permanent 10 percentage point increase in the clean energy consumption subsidy in China increases global emissions in both the baseline model and the perfect factor

[3] Refer to Stern (2004) for a detailed discussion of substitution and scale effects.

[4] The countries are the United States, China, India, France (as a proxy for the European Union), and the United Kingdom, Australia, South Korea, and Argentina (as proxies for the Rest of the World).

mobility model, but the emission increase under imperfect mobility is 337 percent larger in the initial year of implementation and 164 percent larger 35 years after implementation, relative to BAU.

For the United States, the clean energy subsidy initially reduces global emissions relative to BAU, but imperfect factor mobility attenuates this effect to 26 percent of the perfect factor mobility benchmark. Moreover, global emissions reverse course and rise above BAU within 10 years. These findings highlight that when capital and labor responses to changes in climate policies are limited, scale effects from lower energy prices dominate substitution effects, causing policies to increase rather than decrease global emissions.

I then examine how instrument choice matters by comparing consumption, capital, and labor subsidies for clean energy while holding fixed the total subsidy expenditure under the model-implied energy-related policy instruments over 2015–2022. These model-implied instruments increase cumulative global emissions relative to BAU. Reallocating the same resources toward clean energy subsidies reduces global emissions for the United States, China, and India under all three alternatives, but raises them for the European Union and the Rest of the World. Among instruments, capital subsidies can deliver the largest emission reductions, but they also impose the largest welfare costs by distorting capital allocation and reducing aggregate capital.

Labor subsidies deliver modest emission reductions, but they perform best on the welfare margin in most cases and even raise welfare in China by directly relaxing labor mobility frictions. No single instrument dominates across all economies. In the European Union, for example, the model-implied instruments outperform all three alternative subsidy designs on both emissions and welfare: the clean energy subsidy lowers domestic energy prices and expands non-energy production, and the resulting increase in output and energy demand abroad through international trade linkages more than offsets the domestic emission reductions. Overall, the results show that effective climate policy design depends on economy-specific production structure, international spillovers, and factor adjustment frictions.

Related Literature This paper contributes to the literature on the economics of climate change (Nordhaus, 1992; Heal, 2005; Stern, 2008; Golosov et al., 2014; Hassler et al.,

2016; Hémous, 2016; Shapiro, 2021; Kortum and Weisbach, 2022; Cruz, 2023; Känzig, 2023; Arkolakis and Walsh, 2023; Xiang, 2023; Bilal and Känzig, 2024; Farrokhi and Lashkaripour, 2024; Casey, 2024; Hong, 2025; Casey et al., 2026). It closely relates to the work of Cruz and Rossi-Hansberg (2023), who examines how inter-regional labor mobility frictions and idiosyncratic preferences influence the welfare impact of climate change. This paper departs from the existing literature's focus on regional labor mobility by incorporating imperfect and gradual adjustment of both capital and labor across energy and non-energy sectors. By incorporating these sectoral adjustments into a global quantitative framework, this paper shows that standard results on climate-policy effectiveness, derived under frictionless reallocation, can reverse in sign once capital and labor respond gradually and imperfectly.

This paper also relates to a large body of literature on imperfect capital mobility, which has documented that capital adjustments are costly and occur gradually over time (Dixit, 1989; Dixit and Pindyck, 1994; Ramey and Shapiro, 2001; Cooper and Haltiwanger, 2006; Dix-Carneiro and Kovak, 2017; Lanteri, 2018; Artuc et al., 2022; Lanteri et al., 2023). The key contribution of the paper is the tractable incorporation of imperfect and dynamic capital adjustment into the analysis of climate policies. This is achieved by extending the capital allocation framework of Kleinman et al. (2023) to a multi-sector setting and incorporating dynamic capital adjustments driven by a dynamic agglomeration externality, in the spirit of the directed technical change literature (Acemoglu et al., 2012, 2016; Aghion et al., 2016; Acemoglu et al., 2023). In contrast to that literature, where subsidies redirect innovation toward clean sectors, the externality here operates through capital allocation directly: past investment in a sector raises its capital efficiency, generating gradual and persistent reallocation dynamics without endogenous innovation. This approach yields tractable estimation of the parameters governing capital adjustment and enables quantification of how imperfect capital mobility alters climate-policy effectiveness in a multi-country setting.

Moreover, this paper relates to the quantitative international trade literature that analyzes the global impact of policies (Armington, 1969; Anderson, 1979; Eaton and Kortum, 2002; Arkolakis et al., 2012; Head and Mayer, 2014; Caliendo and Parro, 2015). Specifically, it contributes to a strand of literature examining the impact of policies when workers adjust gradually and imperfectly to policy changes (Artuç et al., 2010;

Dix-Carneiro, 2014; Artuç and McLaren, 2015; Caliendo et al., 2019; Traiberman, 2019; Caliendo et al., 2021; Dix-Carneiro et al., 2023). Understanding the extent and speed of labor adjustment in major economies responsible for global emissions is central to this paper. Hence, compared to previous work that estimated labor supply parameters for a single country, I merge longitudinal individual-level labor surveys from eight major economies to estimate country-level bilateral labor mobility costs and the strength of non-pecuniary motives for worker mobility. Estimating these parameters across eight economies enables cross-country comparisons of labor adjustment that are central to understanding how factor mobility shapes the global spillovers of climate policies.

This paper also relates to the literature on carbon leakage, which documents that unilateral climate policy can increase emissions in non-abating economies, both by shifting energy-intensive production to less regulated countries and by lowering world fossil fuel prices and stimulating energy demand elsewhere (Copeland and Taylor, 2004; Böhringer et al., 2012; Aichele and Felbermayr, 2015). This paper identifies a distinct scale-effect channel of leakage: a clean energy subsidy lowers the domestic energy composite price, expands production through input-output linkages and capital accumulation, and raises global energy demand through trade, increasing global emissions even when the implementing economy reduces its own.

The remainder of the paper proceeds as follows. Section 2 presents the dynamic general equilibrium model of the global economy. Section 3 describes the model's quantification. Section 4 reports the quantitative results. Section 5 concludes.

2 Model

In this section, I develop a dynamic general equilibrium model of the global economy that incorporates imperfect and gradual adjustments of capital and labor across sectors. I first present the demand and production structures and show where output-side substitution and scale effects of climate policies arise. I then describe capital and labor supply decisions and input-side substitution and scale effects of climate policies. Theoretical derivations for all equations are provided in Appendix A.10.

2.1 Setting

Time is discrete and is indexed by $t \in \mathcal{T}$. The world economy consists of N economies, indexed by $n, m \in \mathcal{N}$. Each economy contains J sectors, indexed by $j, k \in \mathcal{J}$. I categorize these sectors into three categories: non-employment (NE) sector, which does not produce output; energy sectors, denoted by $\mathcal{J}_E$, which produce energy goods; and non-energy sectors, denoted by $\mathcal{J}_N$, which use energy inputs for production. The set $\mathcal{J}_E$ includes both the brown energy (B) sector, which extracts and processes raw fossil fuels such as crude oil, natural gas, and coal, and the clean energy (CL) sector, which produces energy without directly extracting and processing raw fossil fuels. Motivated by the climate policies commonly implemented in practice, I limit the government's policy instruments to the following: (i) sector-specific taxes/subsidies on consumption, labor, and capital ($\{\tau^{Y,j}_{n,t}, \tau^{K,j}_{n,t}, \tau^{L,j}_{n,t}\}_j$), and (ii) taxes on consumption of raw fossil fuels ($\{\tau^{f,h}_{n,t}\}_h$).

Within each sector, a continuum of perfectly competitive producers (i.e., firms) produce economy-sector-specific varieties.[5] The numeraire of the model is one unit of final good in economy n_0, i.e., $P_{n_0,t} = 1$ for all $t \in \mathcal{T}$. Within each economy, a domestic market for riskless assets exists where only domestic agents can participate. Finally, there is no aggregate uncertainty. Agents possess perfect foresight over aggregate variables up to a finite horizon $T_{\max}$; I assume that these variables remain constant beyond that point.[6]

2.2 Demand

Sectoral Goods Sectoral goods aggregate sector-specific products (or varieties) differentiated by the economy of origin (Armington, 1969; Dixit and Stiglitz, 1977). Formally, the sectoral good $Y^j_{n,t}$ in sector j of economy n at time t is a Constant Elasticity of Substitution (CES) aggregate of sector-specific varieties $\{y^j_{nm,t}\}$ produced in economy m and consumed

[5] I assume perfect competition to avoid distortions due to market power. In a model with market power, government intervention through production subsidies would be necessary to boost production and restore efficiency. Since this paper focuses on environmental externalities, I abstract from market power considerations.

[6] This limited-horizon assumption allows agents to make forward-looking decisions over a relevant transition period without requiring implausible expectations about far-future environmental collapse under business-as-usual.

in economy n:

$$Y^j_{n,t} = \left(\sum_{m \in \mathcal{N}} \left(y^j_{nm,t} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where $\sigma > 1$ dictates the elasticity of substitution across varieties.

International Trade The share $\pi^j_{nm,t}$ of sector j expenditures of economy n spent on varieties produced in sector j of economy m is given by:

$$\pi^j_{nm,t} \equiv \frac{p^j_{nm,t} y^j_{nm,t}}{P^j_{n,t} Y^j_{n,t}} = \left(\frac{d^j_{nm,t} p^j_{mm,t}}{P^j_{n,t}} \right)^{1-\sigma}, \tag{2}$$

where $p^j_{nm,t}$ denotes the price of variety produced in sector j of economy m and consumed in economy n at time t; $d^j_{nm,t} \geq 1$ denotes the sector-level bilateral iceberg trade cost of shipping goods from economy m to economy n (Krugman, 1980).[7] Due to this cost, the delivered price satisfies $p^j_{nm,t} = d^j_{nm,t} p^j_{mm,t}$. $P^j_{n,t}$ is the sector-level price index in economy n at time t given by $P^j_{n,t} \equiv \left(\sum_{m \in \mathcal{N}} (p^j_{nm,t})^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$.[8]

Energy Composite Goods Energy composite goods are bundles that represent consumers' demand for energy goods produced by brown energy (B) and clean energy (CL) sectors. Formally, the energy composite good $Y^E_{n,t}$ in economy n at time t is a Constant Elasticity of Substitution (CES) aggregate of the sectoral goods $Y^{CL}_{n,t}$ and $Y^B_{n,t}$:

$$Y^E_{n,t} = \left((\alpha^{CL}_n)^{\frac{1}{\eta_1}} (Y^{CL}_{n,t})^{\frac{\eta_1-1}{\eta_1}} + (\alpha^B_n)^{\frac{1}{\eta_1}} (Y^B_{n,t})^{\frac{\eta_1-1}{\eta_1}} \right)^{\frac{\eta_1}{\eta_1-1}}, \tag{3}$$

where the weights α^{CL}_n and α^B_n represent the reliance of economy n on clean energy and brown energy sectors, respectively, with $\alpha^{CL}_n + \alpha^B_n = 1$; the parameter η_1 governs the elasticity of substitution across these energy goods.

[7] Also, the triangle inequality holds, i.e., $d^j_{nm,t} \leq d^j_{nh,t} \cdot d^j_{hm,t}$ for all $m, n, h \in \mathcal{N}$.

[8] The elasticity of trade flows with respect to changes in trade costs (i.e., trade elasticity) is $1 - \sigma$. Equation (2) is commonly referred to as the gravity equation in the international trade literature (Armington, 1969; Anderson, 1979).

The energy price index $P^E_{n,t}$ is given by:

$$P^E_{n,t} = \left(\alpha^{CL}_n \Big((1+\tau^{Y,CL}_{n,t}) P^{CL}_{n,t} \Big)^{1-\eta_1} + \alpha^B_n \Big((1+\tau^{Y,B}_{n,t}) P^B_{n,t} \Big)^{1-\eta_1} \right)^{\frac{1}{1-\eta_1}}, \tag{4}$$

where $\tau^{Y,CL}_{n,t}$ and $\tau^{Y,B}_{n,t}$ are consumption taxes imposed on the clean energy sector and the brown energy sector, respectively. Equation (4) shows that the government can shift demand away from the brown energy sector, which is referred to as the output-side *substitution effect*, by either increasing the consumption tax on brown energy goods, $\tau^{Y,B}_{n,t}$ (hereafter referred to as a carbon tax), or reducing the consumption tax on clean energy goods, $\tau^{Y,CL}_{n,t}$ (hereafter referred to as a clean energy subsidy).

Raising the carbon tax not only shifts demand away from the brown energy sector but also raises the energy price index $P^E_{n,t}$, thereby reducing overall demand for the energy composite good. I refer to this channel as a negative output-side *scale effect*, which further reduces demand for brown energy. Conversely, a clean energy subsidy lowers the energy price index, increases demand for the energy composite good, and expands aggregate energy use, generating a positive output-side *scale effect* that can raise global emissions.

Final Goods The final good $Y_{n,t}$ in economy n at time t is a Cobb–Douglas aggregate of sectoral goods $\{C^j_{n,t}\}$ produced in non-energy sectors and the energy composite good $C^E_{n,t}$:

$$Y_{n,t} = \mathcal{D}_n(\mathcal{C}_t) \Big(\prod_{j \in \mathcal{J}_N} (C^j_{n,t})^{\alpha^j_n} \Big)^{\alpha_n} (C^E_{n,t})^{1-\alpha_n}, \tag{5}$$

where α^j_n governs the share of final goods expenditure allocated to non-energy sector j, with $\sum_{j \in \mathcal{J}_N} \alpha^j_n = 1$, and $\alpha_n \in [0,1]$ denotes the share of total final goods expenditure allocated to non-energy goods.

Following Barrage and Nordhaus (2024), environmental quality $\mathcal{D}_n(\mathcal{C}_t)$, which depends on the global stock of carbon dioxide $\mathcal{C}_t$, influences the total factor productivity (TFP) of an economy. Consequently, it affects consumption and investment decisions. This term will be explained in detail in Section 2.4. Since consumers and producers do not account for the impact of their activities on the global carbon stock, a *global environmental*

externality arises, opening the door for government intervention to reduce global emissions and prevent the deterioration of environmental quality.

Materials Materials are bundles of sectoral goods reflecting non-energy intermediate inputs used for production.[9] Formally, the material $M^j_{n,t}$ in sector j of economy n is a Cobb-Douglas aggregate of sectoral goods $\{M^{j,k}_{n,t}\}_{k\in\mathcal{J}}$: $M^j_{n,t} = \prod_{k\in\mathcal{J}} \left(M^{j,k}_{n,t}\right)^{\gamma^{j,k}_n}$, where intermediate input shares satisfy $\sum_{k\in\mathcal{J}} \gamma^{j,k}_n = 1$.

2.3 Production

Variety Production The output $y^B_{n,t}$ of brown energy varieties produced in economy n at time t is produced using a Cobb-Douglas function that takes fossil fuel $F_{n,t}$ as an input for production in addition to labor $L^B_{n,t}$, capital $K^B_{n,t}$, and materials $M^B_{n,t}$:

$$y^B_{n,t} = z^B_{n,t}\left((F^B_{n,t})^{\psi^{B,F}_n}(L^B_{n,t})^{\psi^{B,L}_n}(K^B_{n,t})^{\psi^{B,K}_n}\right)^{\gamma^{B,V}_n}\left(M^B_{n,t}\right)^{1-\gamma^{B,V}_n}, \tag{6}$$

where $z^B_{n,t}$ is the total factor productivity of the variety producer in sector B of economy n at time t, and input shares satisfy $(1-\gamma^{B,V}_n) + \sum_{h\in\{L,K,F\}} \gamma^{B,V}_n \psi^{B,h}_n = 1$. Moreover, taxes $\tau^{L,B}_{n,t}$ and $\tau^{K,B}_{n,t}$ are applied to labor and capital inputs, respectively.

In contrast to producers in the brown energy sector, the output $y^j_{n,t}$ produced in sector $j \in \mathcal{J} \setminus \{B\}$ of economy n at time t requires energy inputs, as shown in Equation (3), rather than raw fossil fuel inputs.[10] In particular, the production function is expressed as a Cobb-Douglas function that takes energy composite $Q^{j,E}_{n,t}$, labor $L^j_{n,t}$, capital $K^j_{n,t}$, and

[9] Materials are included in the model for two main reasons. First, intermediate input expenditures account for 20 to 70 percent of gross output across sectors and across economies, making them a quantitatively significant feature needed to match the data. This is particularly important because global emissions depend on gross output rather than value added. Second, materials capture the interconnections between sectors through production linkages. They illustrate how changes in the output price of one sector affect the economy by altering the input prices of other sectors that use these outputs for production. This is crucial for analyzing the scale effects of climate policies that target specific sectors.

[10] The model features roundabout production where the clean energy producers can substitute between intermediate energy inputs produced by the brown energy sector and the clean energy sector. As the price of clean energy goods decreases relative to the price of brown energy goods, the clean energy producers rely less on the inputs produced by the brown energy sector through substitution.

materials $M^j_{n,t}$ as inputs:

$$y^j_{n,t} = z^j_{n,t}\left((Q^{j,E}_{n,t})^{\psi^{j,E}_n}(L^j_{n,t})^{\psi^{j,L}_n}(K^j_{n,t})^{\psi^{j,K}_n}\right)^{\gamma^{j,V}_n}\left(M^j_{n,t}\right)^{1-\gamma^{j,V}_n}, \tag{7}$$

where $z^j_{n,t}$ is the total factor productivity of the variety producer in sector j of economy n at time t, and input shares satisfy $(1-\gamma^{j,V}_n) + \sum_{h\in\{L,K,E\}}\gamma^{j,V}_n\psi^{j,h}_n = 1$. Moreover, taxes $\tau^{L,j}_{n,t}$ and $\tau^{K,j}_{n,t}$ are applied to labor and capital inputs, respectively.

Fossil Fuel Extraction Variety producers in the brown energy sector of economy n use fossil fuel inputs $F_{n,t}$ at time t, which are a Constant Elasticity of Substitution (CES) composite of crude oil $f^o_{n,t}$, natural gas $f^g_{n,t}$, and coal $f^c_{n,t}$.[11] Formally, $F_{n,t} = \left(\sum_{h\in\{o,g,c\}}(\omega^h_n)^{\frac{1}{\eta_2}}\left(f^h_{n,t}\right)^{\frac{\eta_2-1}{\eta_2}}\right)^{\frac{\eta_2}{\eta_2-1}}$, where the weights $\{\omega^h_n\}_{h\in\{o,g,c\}}$ represent the reliance of economy n on different types of fossil fuels for production with $\sum_{h\in\{o,g,c\}}\omega^h_n = 1$.[12] The parameter η_2 governs the elasticity of substitution across these fossil fuels.[13]

To make the model tractable, I assume that either state-owned enterprises or perfectly competitive and atomistic firms extract each type of fossil fuel.[14] These firms solve static extraction problems. Consequently, the before-tax price $s^h_{n,t}$ of fossil fuel type $h \in \{o, g, c\}$ in economy n at time t equals the marginal cost of extraction φ^h_n, measured in units of the final good: $s^h_{n,t} = \varphi^h_n P_{n,t}$ where $P_{n,t}$ denotes the price of the final good in economy n at

[11] Including different types of fossil fuels is necessary to capture differing carbon emission trends across countries due to the changing composition of emissions. In 2022, more than 65% of the emissions in China and India were from coal, while less than 25% of the emissions in the United States and the European Union were from coal, driven by a shift to oil and natural gas.

[12] I assume that crude oil, natural gas, and coal are non-tradable in their raw form and become tradable only after being refined by producers in the brown energy sector. This assumption aligns the model with bilateral trade flows observed in the World Input-Output Database 2016 Release.

[13] When fossil fuels are substitutes (i.e., $\eta_2 > 1$), fuel-specific taxes $\{\tau^{f,h}_{n,t}\}_{h\in\{o,g,c\}}$ can discourage the use of carbon-intensive fossil fuels (e.g., coal) and reduce carbon emissions.

[14] A more realistic model would involve fossil fuel extracting firms with market power solving dynamic extraction problems, considering the future path of fossil fuel reserves. However, this is beyond the scope of this paper, as such an inclusion would introduce additional externalities and inefficiencies (e.g., resource depletion externalities, price distortion). For quantification, I account for any observed differences between retail prices and supply costs of fossil fuels using the fuel-specific consumption tax, which may also reflect markups.

time t. Hence, the price of a fossil fuel input $s_{n,t}$ in economy n at time t is given by:

$$s_{n,t} = \left(\sum_{h \in \{o,g,c\}} \omega_n^h \left((1 + \tau_{n,t}^{f,h}) s_{n,t}^h \right)^{1-\eta_2} \right)^{\frac{1}{1-\eta_2}} \tag{8}$$

where $\tau_{n,t}^{f,h}$ represents the consumption tax specific to fossil fuel type h.[15]

Finally, the law of motion for the reserve $D_{n,t}^h$ of fossil fuel type $h \in \{o, g, c\}$ is given by: $D_{n,t+1}^h = D_{n,t}^h - f_{n,t}^h$ with the constraint $D_{n,t+1}^h \geq 0$.[16]

2.4 Carbon Emissions and Environmental Quality

Carbon dioxide (CO_2) emissions arise from the consumption of fossil fuels $\{f_{n,t}^h\}$ used in the brown energy sector. Specifically, consuming one unit of fossil fuel of type h emits ϕ^h units of CO_2. Hence, the total carbon emissions $E_{n,t}$ due to fossil fuel production in economy n at time t is given by: $E_{n,t} = \sum_{h \in \{o,g,c\}} \phi^h \cdot f_{n,t}^h$. The measure $E_{n,t}$ represents territorial carbon emissions based on fossil fuel production within the economy.[17] The global CO_2 emissions at time t, denoted by E_t, are given by: $E_t = \sum_{n \in \mathcal{N}} E_{n,t}$.

Following Cruz and Rossi-Hansberg (2023), I assume the global stock of carbon dioxide $\mathcal{C}_t$ in the atmosphere evolves as follows: $\mathcal{C}_t = \mathcal{S} + \sum_{\tau=T_{min}}^{t} (1 - \delta_{t-\tau}^{\mathcal{C}}) \left(E_\tau + E^L \right)$, where $\mathcal{S}$ denotes the pre-industrial stock of carbon dioxide; T_{min} denotes the start period for measuring carbon emissions; $(1 - \delta_{t-\tau}^{\mathcal{C}})$ represents the fraction of carbon emissions at time τ that remains in the atmosphere at time t, and E^L denotes exogenous carbon emissions due to land-use change (e.g., deforestation, urbanization, desertification).

Furthermore, following Golosov et al. (2014), the global average temperature $\mathbb{T}(\mathcal{C}_t)$ at time t relative to the global average temperature at time T_{min} is approximated as follows:

[15] Equation (8) shows that the government can reduce carbon emissions through the use of fuel-specific consumption taxes that discourage the use of fuels emitting more carbon dioxide per unit (e.g., coal).

[16] The existence of exhaustible fossil fuel resources implies that the steady state can only be reached when (i) fossil fuel reserves in all economies are depleted, or (ii) the clean energy sector becomes productive enough to asymptotically dominate the brown energy sector. In both scenarios, global emissions will approach zero, highlighting the importance of analyzing climate policies during the transition period rather than at the long-run steady state.

[17] In Appendix A.12.6, I present an alternative measure of carbon emissions based on the economy's total consumption of brown energy, which includes both domestically produced and imported products. However, since countries typically report territorial carbon emissions, I use the production-based measure for the analysis.

$\mathbb{T}(\mathcal{C}_t) = \lambda_T \log_2\left(\frac{\mathcal{C}_t}{\mathcal{S}}\right)$, where λ_T governs the increase in the global average temperature due to the increase in the global stock of carbon.

Following Barrage and Nordhaus (2024), I assume the economy-specific environmental quality function $\mathcal{D}_n(\cdot)$ decreases non-linearly with the global average temperature $\mathbb{T}(\mathcal{C}_t)$: $\mathcal{D}_n(\mathcal{C}_t) = 1 - \xi_n\Big(\mathbb{T}(\mathcal{C}_t)\Big)^2$ where $\xi_n > 0$ captures the environmental damage specific to economy n resulting from changes in global average temperature. The convexity of this function reflects the non-linear impact of rising global carbon stock on the climate system and the global economy, encompassing effects such as natural disasters and tipping points.

2.5 Capitalists

Within each economy, there are infinitely-lived capitalists of measure one who are assumed to be homogeneous in their initial aggregate capital $W_{n,0}$. Given this homogeneity, I simplify the analysis by considering an infinitely-lived representative capitalist.

Capital Allocation At each period t, the representative capitalist in economy n holds $W_{n,t}$ units of aggregate capital, which evolves according to the following law of motion:

$$W_{n,t+1} = (1 - \delta_n^k)W_{n,t} + I_{n,t}^k, \tag{9}$$

where $I_{n,t}^k$ represents aggregate investment in units of the final good at time t, and $\delta_n^k \in [0, 1]$ denotes the economy-specific capital depreciation rate.

To simplify the capital allocation problem and capture the observed allocation and evolution of capital across sectors, I assume that aggregate capital in economy n at time t, $W_{n,t}$, consists of infinitesimal capital assets with heterogeneous returns across and within sectors, and over time. This assumption reflects the idea that observed capital allocation results from the decisions of numerous investors who encounter idiosyncratic and random opportunities with varying returns across and within sectors and over time, due to barriers such as information and search frictions (Gonçalves et al., 2020; Smith et al., 2023). Since returns are opportunity-specific and random, each unit of capital earns

a heterogeneous return.[18]

To tractably capture this notion, I build on Kleinman et al. (2023) and assume that the return on each unit of capital in sector j depends on (i) the sector-specific rental rate of capital $r^j_{n,t}$, and (ii) the sector-specific capital efficiency $\zeta^j_{n,t}$.[19] Formally, each infinitesimal unit of aggregate capital $W_{n,t}$ is characterized by its profile $\vec{\zeta}_{n,t} = \{\zeta^j_{n,t}\}_{j\in\mathcal{J}}$, which contains the idiosyncratic capital efficiency for each sector, independently drawn from a Fréchet distribution $F^j_{n,t}(\zeta^j_{n,t})$ at each period:

$$F^j_{n,t}(\zeta^j_{n,t}) = e^{-(\zeta^j_{n,t}/a^j_{n,t})^{-\rho_K}}, \tag{10}$$

where $a^j_{n,t}$ represents the average capital efficiency in sector j at time t, while the dispersion parameter ρ_K captures the variation in capital efficiencies within sectors due to idiosyncratic factors.[20]

If a capitalist invests an infinitesimal unit of aggregate capital in sector j, the return before depreciation is $\zeta^j_{n,t}P_{n,t}r^j_{n,t}$. The unit of aggregate capital translates to $\zeta^j_{n,t}$ units of capital available for production in sector j. The capitalist allocates a given unit of aggregate capital to sector j if its return is the highest among all sectors; that is, $\zeta^j_{n,t}P_{n,t}r^j_{n,t} = \max_{h\in\mathcal{J}}\{\zeta^h_{n,t}P_{n,t}r^h_{n,t}\}$. As shown in Appendix A.12.5, the probability that one unit of aggregate capital is allocated to sector j is given by:

$$\Phi^j_{n,t} = \exp\left(\rho_K\left(\ln r^j_{n,t} + \ln a^j_{n,t}\right) - \ln\left(\sum_{k\in\mathcal{J}}(r^k_{n,t}a^k_{n,t})^{\rho_K}\right)\right). \tag{11}$$

[18]These idiosyncratic returns may vary along three dimensions. Across sectors, returns may differ due to sector-specific capital efficiencies; for example, fossil fuel production benefits from existing oil rigs and refineries, whereas solar and wind manufacturing are limited by underdeveloped wafer and turbine plants. Within sectors, investors may be randomly matched to producers of varying efficiency. Over time, opportunities may evolve through changes in match quality and search frictions (Hortaçsu and Syverson, 2004).

[19]Kleinman et al. (2023) utilizes this modeling approach in a one-sector model of international trade and capital holdings to match key facts about international capital holdings. I adapt this approach in a multi-sector model to match the allocation of capital stock across sectors, the heterogeneous returns on capital, and the evolution observed in the data.

[20]When ρ_K is low, indicating high dispersion, investment opportunities associated with capital assets vary greatly across sectors, even when sector-specific capital efficiency terms $\{a^j_{n,t}\}$ are equal. On the other hand, when ρ_K is high, indicating low dispersion, investment opportunities are more similar within sectors, and, hence, sector-specific capital efficiencies $\{a^j_{n,t}\}$ primarily determine differences in returns.

By the law of large numbers, the share of aggregate capital allocated to sector j coincides with $\Phi^j_{n,t}$ shown in Equation (11). Furthermore, let $v_{n,t}$ denote the average return per unit of capital investment after accounting for idiosyncratic capital efficiencies. As shown in Appendix A.12.5, $\frac{v_{n,t}}{P_{n,t}} = \Gamma\left(\frac{\rho_K - 1}{\rho_K}\right)\left(\sum_{j\in\mathcal{J}}(a^j_{n,t} r^j_{n,t})^{\rho_K}\right)^{\frac{1}{\rho_K}}$, which by the law of large numbers also equals the average return on investing one unit of aggregate capital.

To allow for gradual capital adjustment across sectors, I introduce a dynamic agglomeration externality in which past capital allocations to a sector influence its current sector-specific capital efficiency.[21] This externality reflects external learning-by-doing and within-sector knowledge spillovers, generating gradual and persistent capital reallocation in response to policy changes (Aghion et al., 2016; Bradt, 2024). It is considered an externality because individual capitalists do not internalize the broader impact of their past capital allocations on current sector-specific capital efficiency.[22]

Formally, the law of motion for sector-specific efficiency, $a^j_{n,t}$, which captures this idea, is given by: $\ln(a^j_{n,t}) = \ln(a^j_n) + \rho_\Phi \ln\left(\frac{K^j_{n,t-1}}{W_{n,t-1}}\right)$. In this equation, a^j_n represents the time-invariant economy-sector-specific efficiency, while the parameter ρ_Φ governs the strength of the dynamic agglomeration externality.[23] I adopt this log-linear form because it yields a tractable empirical counterpart for estimating the capital supply parameters.

As derived in Appendix A.12.5, the optimal sector-level capital allocation condition is

[21] As shown in Equation (11), without this assumption, capitalists would reallocate capital across sectors immediately in response to changes in sector-specific capital demand induced by climate policies. When sector-specific efficiencies $a^j_{n,t}$ are time-invariant, and agents possess perfect foresight, the absence of physical adjustment costs implies that these demand changes translate directly into changes in the rental rate $r^j_{n,t}$ and induce immediate reallocation with limited dynamic effects.

[22] Capitalists do not internalize the agglomeration externality, so policies that promote capital reallocation toward clean energy generate an additional benefit beyond correcting the carbon externality alone. This second market failure could strengthen the case for capital subsidies relative to consumption subsidies, though quantifying this channel is left for future work.

[23] Specifically, ρ_Φ indicates the degree to which capital efficiency improves as a result of allocating aggregate capital to sector j in the previous period. A high value of ρ_Φ suggests that past capital allocation decisions significantly influence current efficiency. Conversely, the externality disappears when $\rho_\Phi = 0$. The inclusion of past aggregate capital, $W_{n,t-1}$, captures the diminishing gains in capital efficiency across all sectors from aggregate investment, similar to the approach of Hémous (2016).

given by

$$\underbrace{\ln(K^j_{n,t})}_{\text{Capital Allocation}} = \underbrace{(\rho_K - 1)}_{\text{Responsiveness}} \cdot \underbrace{\ln\left(\frac{P_{n,t} r^j_{n,t}}{v_{n,t}}\right)}_{\text{Rental Rate of Capital}} + \underbrace{\ln(W_{n,t})}_{\text{Aggregate Capital}} - \underbrace{\rho_K \cdot \rho_\Phi \cdot \ln(W_{n,t-1})}_{\text{Lagged Aggregate Capital}}$$

$$+ \underbrace{\rho_K \cdot \rho_\Phi \cdot \ln(K^j_{n,t-1})}_{\text{Lagged Capital Allocation}} + \underbrace{\rho_K \cdot \ln(a^j_n)}_{\text{Sector-specific Capital Efficiency}} + \rho_K \ln\left(\Gamma\left(\frac{\rho_K - 1}{\rho_K}\right)\right). \tag{12}$$

Equation (12) serves as the key moment condition used to estimate the capital supply parameters ρ_K and ρ_Φ. This condition implies that the extent of capital adjustment and the short-run, input-side *substitution* effect of climate policies depend on the dispersion parameter ρ_K.[24] Additionally, the speed of capital adjustment and the long-run, input-side *substitution* effect of climate policies depend on the magnitude of the externality parameter ρ_Φ.[25] The qualitative finding that imperfect capital mobility weakens the substitution effect of climate policies does not depend on this specific functional form: any mechanism limiting the extent or speed of capital reallocation would produce the same result.

Aggregate Capital Accumulation The representative capitalist at time 0 with initial aggregate capital $W_{n,0}$ chooses a sequence of future aggregate capital $\{W_{n,t+1}\}$, in units of final good, to maximize her life-time utility $V^K_{n,0}$:

$$V^K_{n,0} = \max_{W_{n,t+1}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}_0\left[\ln(C^K_{n,t})\right] \tag{13}$$

[24] If ρ_K is low, capital allocation responds inelastically to changes in the rental rate of capital induced by climate policies, as allocation decisions are primarily driven by idiosyncratic investment opportunities, which vary significantly even within sectors.

[25] A high ρ_Φ implies that past capital allocations across sectors would strongly persist over time in the absence of policy intervention, while also leading to persistent dynamic responses to temporary policy changes.

s.t.

$$C^K_{n,t} = \Big(\frac{v_{n,t}}{P_{n,t}} + (1-\delta^k_n)\Big)W_{n,t} - W_{n,t+1} + \frac{\Omega_{n,t}}{P_{n,t}}, \tag{14}$$

$$0 = \lim_{t\to\infty}\Big(\prod_{s=0}^{t}\Big(\frac{v_{n,s}}{P_{n,s}} + 1 - \delta^k_n\Big)^{-1}\Big)W_{n,t+1}, \tag{15}$$

where $C^K_{n,t}$ denotes consumption of final good by the capitalist; β denotes the discount factor; $\Omega_{n,t}$ denotes the lump-sum transfer from the government in economy n at time t; Equation (15) represents the transversality condition of the representative capitalist.[26] Aggregate capital accumulation generates an input-side *scale* effect of climate policies by altering aggregate productive capacity and, in turn, emissions.

2.6 Workers

Within each economy, there are infinitely-lived hand-to-mouth workers of measure $L_{n,t}$ who are assumed to be homogeneous in their ability but heterogeneous in their sector of employment.[27] At each period, workers make dynamic discrete choice decisions to move from sector j (origin) to sector k (destination) which depends on (i) the destination-specific wage ($w^k_{n,t}$), (ii) the origin-destination-specific mobility costs ($\kappa^{j,k}_n$), (iii) the discounted expected life-time utility from working in sector k ($\beta\mathbb{E}_t[\mathcal{U}^k_{n,t+1}]$), (iv) the sector-specific non-pecuniary benefit (μ^j_n), and (v) the destination-specific idiosyncratic preference ($\varepsilon^k_{n,t}$).[28]

[26]Although not explicitly shown in Equation (14), there exists a domestic financial asset market where capitalists can trade risk-free bonds, $A_{n,t+1}$, at a gross interest rate $R_{n,t+1}$. The no-arbitrage condition implies the gross interest rate on bonds equals the net return on aggregate capital: $R_{n,t+1} = \frac{v_{n,t+1}}{P_{n,t+1}} + (1-\delta^k_n)$.

[27]The assumption that workers are hand-to-mouth is a standard assumption made in the labor market dynamics literature for tractability (Artuç et al., 2010). While allowing workers to make saving decisions is both desirable and realistic, it complicates the analysis because it requires keeping track of the asset distribution of workers, where each worker's asset would depend on the full path of her past income and employment history. The non-employment sector generates unemployment-like transition dynamics, in which workers temporarily exit production while reallocating between sectors, a feature empirically relevant for energy-sector displacement.

[28]The inclusion of the sector-specific non-pecuniary benefit, μ^j_n, primarily serves a quantitative purpose. Specifically, it captures factors like better work-life balance and job security that rationalize the inter-sectoral wage differentials that cannot be explained by estimated mobility frictions.

Formally, the lifetime utility $\mathcal{U}^j_{n,t}$ of a worker in sector j at time t is given by:

$$\mathcal{U}^j_{n,t} = \ln(c^j_{n,t}) + \ln(\mu^j_n) + \max_{k\in\mathcal{J}}\{\beta\mathbb{E}_t[\mathcal{U}^k_{n,t+1}] + \ln(1-\kappa^{j,k}_n) + \rho_L\varepsilon^k_{n,t}\}, \tag{16}$$

where $c^j_{n,t}$ denotes the consumption of final goods; the parameter $\kappa^{j,k}_n \in [0,1)$ captures inter-sectoral mobility costs including the psychological costs, imperfect inter-sectoral transferability of experience, and training costs (Kambourov, 2009; Artuç et al., 2010; Dix-Carneiro, 2014).[29] Since future idiosyncratic preference shocks $\{\varepsilon^k_{n,t+h}|h>0\}$ are not realized at time t, workers make moving decisions based on the expected life-time utility from working at sector k at time $t+1$, $\mathbb{E}_t[\mathcal{U}^k_{n,t+1}]$. At each period, a worker draws an i.i.d. idiosyncratic preference shock $\varepsilon^k_{n,t}$ for each sector from a standard Gumbel distribution with mean zero, i.e., $\varepsilon^k_{n,t} \overset{\text{i.i.d.}}{\sim} \text{Gumbel}(\gamma,1)$, where γ is the Euler-Mascheroni constant.[30] The parameter ρ_L governs the dispersion of idiosyncratic non-pecuniary preferences.[31]

Since workers are hand-to-mouth, the consumption of a worker in sector j in economy n at time t equals real wage, i.e., $c^j_{n,t} = \frac{w^j_{n,t}}{P_{n,t}}$. Moreover, I assume workers in the non-employment (NE) sector receive income $w^{NE}_{n,t}$ from the government. In particular, $w^{NE}_{n,t}$ is assumed to depend on the economy-specific component w^{NE}_n, and the average wage of the economy: $w^{NE}_{n,t} = w^{NE}_n \times \left(\frac{\sum_{j\in\mathcal{J}\setminus\{NE\}} w^j_{n,t}L^j_{n,t}}{\sum_{j\in\mathcal{J}\setminus\{NE\}} L^j_{n,t}}\right)$.[32]

Denote $V^j_{n,t}$ as the expected lifetime utility of a worker in sector j of economy n at time t, i.e., $V^j_{n,t} \equiv \mathbb{E}_t[\mathcal{U}^j_{n,t}]$. As derived in Appendix A.10, the probability that a worker in

[29]The cost of staying in the origin sector is normalized to 0, i.e., $\kappa^{j,j}_n = 0$ for all $j \in \mathcal{J}$. As $\kappa^{j,k}_n$ approaches 1, the utility of moving from sector j to sector k approaches negative infinity, reflecting extreme difficulty in moving to sector k. Utility from consumption and the utility loss from moving can be combined as follows: $\ln\left((1-\kappa^{j,k}_n)c^j_{n,t}\right) = \ln(1-\kappa^{j,k}_n) + \ln(c^j_{n,t})$. Therefore, for a worker in sector j, moving from sector j to k is equivalent to sacrificing $\kappa^{j,k}_n$ fraction of her current consumption.

[30]The functional form is assumed for the tractability of the model. For examples of papers that use similar methodology, see Artuç et al. (2010), Dix-Carneiro (2014), Artuç and McLaren (2015), Caliendo et al. (2019), Dix-Carneiro et al. (2023). The idiosyncratic preference reflects various reasons associated with changing sectors, including but not limited to pursuing passion, relocation due to family reasons, and tedium.

[31]A larger ρ_L indicates a greater influence of non-pecuniary reasons on labor mobility.

[32]The parameter w^{NE}_n will be estimated in Section 3.

sector j at time t moves to sector k at time $t+1$ is given by:

$$m_{n,t}^{j,k} = \frac{\exp\left(\left(\frac{\beta}{\rho_L}\right)V_{n,t+1}^{k} + \left(\frac{1}{\rho_L}\right)\ln(1-\kappa_n^{j,k})\right)}{\sum_{h\in\mathcal{J}}\exp\left(\left(\frac{\beta}{\rho_L}\right)V_{n,t+1}^{h} + \left(\frac{1}{\rho_L}\right)\ln(1-\kappa_n^{j,h})\right)}. \tag{17}$$

By the law of large numbers, $m_{n,t}^{j,k}$ is also the share of workers in sector j at time t moving to sector k at time $t+1$. Therefore, the law of motion for labor supply in sector j is given by $L_{n,t+1}^{j} = \sum_{k\in\mathcal{J}} L_{n,t}^{k,j}$ where $L_{n,t+1}^{j}$ denotes the employment in sector j at time $t+1$, and $L_{n,t}^{k,j}$ denotes the number of workers moving from sector k at time t to sector j at time $t+1$, i.e., $L_{n,t}^{k,j} \equiv m_{n,t}^{k,j} L_{n,t}^{k}$.

Combining equations (16) and (17) leads to the following inter-sectoral labor flow condition:

$$\underbrace{\ln\left(\frac{L_{n,t}^{j,k}}{L_{n,t}^{j,j}}\right)}_{\text{Labor Flows}} = \underbrace{\left(\frac{\beta}{\rho_L}\right)}_{\text{Responsiveness}} \cdot \underbrace{\ln\left(\frac{w_{n,t+1}^{k}}{w_{n,t+1}^{j}}\right)}_{\text{Future Wage Differential}} + \underbrace{\left(\frac{1-\beta}{\rho_L}\right)\ln(1-\kappa_n^{j,k})}_{\text{Bilateral Mobility Cost}} \tag{18}$$
$$+ \underbrace{\beta\ln\left(\frac{\mu_n^{k}}{\mu_n^{j}}\right)}_{\text{Non-pecuniary Preferences}} + \underbrace{\beta\ln\left(\frac{L_{n,t+1}^{j,k}L_{n,t+1}^{k}}{L_{n,t+1}^{j}L_{n,t+1}^{k,k}}\right)}_{\text{Option Value}}.$$

Equation (18) serves as the key moment condition describing the relationship between labor flows and key labor supply parameters. The extent of labor adjustment and the short-run, input-side *substitution* effect of climate policies are governed by the dispersion of non-pecuniary preferences ρ_L.[33] Furthermore, the speed of adjustment and the long-run, input-side *substitution* effect of climate policies depend on the bilateral mobility costs.[34]

[33]When ρ_L is high, workers will reallocate less to the destination sector because economic incentives matter less for labor mobility decisions.

[34]In particular, higher bilateral mobility costs, $\kappa_n^{j,k}$, reduce the size of labor flows from the origin to the destination, causing labor adjustments in response to sector-level policy changes to occur more slowly over time.

2.7 Governments

Since this paper does not aim to solve for optimal climate policies, I assume that governments do not make endogenous decisions to maximize domestic social welfare. Instead, I model two types of pre-implemented sector-specific policy instruments that reflect climate policies used in practice: (i) sector-specific taxes or subsidies on consumption, labor, and capital $\{\tau_{n,t}^{H,j}\}_{H\in\{Y,L,K\},j\in\mathcal{J}}$, and (ii) fuel-specific taxes $\{\tau_{n,t}^{f,h}\}_{h\in\{o,g,c\}}$. I assume the government maintains a balanced budget each period through a lump-sum transfer $\Omega_{n,t}$ to the representative capitalist conditional on the set of policy instruments for all $t \in \mathcal{T}$:

$$\Omega_{n,t} = \sum_{j\in\mathcal{J}} \tau_{n,t}^{Y,j} P_{n,t}^{j} Y_{n,t}^{j} + \sum_{j\in\mathcal{J}} \tau_{n,t}^{L,j} w_{n,t}^{j} L_{n,t}^{j} + \sum_{j\in\mathcal{J}} \tau_{n,t}^{K,j} P_{n,t} r_{n,t}^{j} K_{n,t}^{j} + \sum_{h\in\{o,g,c\}} \tau_{n,t}^{f,h} s_{n,t}^{h} f_{n,t}^{h} - w_{n,t}^{NE} L_{n,t}^{\mathrm{NE}}. \tag{19}$$

As shown in Equation (19), the lump-sum government transfer increases with tax revenues and decreases with expenditure towards workers in the non-employment sector.

2.8 General Equilibrium

In equilibrium, all markets in every economy must clear at every period. For readability, detailed descriptions of the sets of parameters Θ, prices $\mathbb{P}$, and endogenous variables $\mathbb{Y}$ are relegated to Appendix A.1.

Definition 1. ***Sequential Equilibrium.***
Given an initial distribution of aggregate capital and labor allocations $\{W_{n,0}, L_{n,0}^{j}\}_{n\in\mathcal{N},j\in\mathcal{J}}$, fossil fuel reserves $\{D_{n,0}^{h}\}_{h\in\{o,g,c\},n\in\mathcal{N}}$, and the set of parameters Θ, the sequential equilibrium of the model is characterized by a sequence of factor prices, interest rates, goods prices, and taxes $\mathbb{P}$, and the sequence of consumption, investment, saving, output, supply and demand for labor, capital, and fossil fuel, trade shares, and transfers $\mathbb{Y}$, that (i) maximize the profits of producers (6), (7), *(ii) maximize the life-time utilities of capitalists* (13) *and workers* (16) *(iii) satisfy governments' balanced budget conditions* (19), *and (iv) satisfy the conditions for bilateral trade shares* (2), *output market clearing, factor market clearing as well as the evolution of capital and labor allocation across sectors, and the law of motion for the stock of global carbon dioxide.*

3 Quantification

The model is calibrated at an annual frequency to a global economy consisting of three countries and two country aggregates: the United States (US), China, India, the European Union (EU), and the Rest of the World (ROW).[35] I employ the industrial classification from the World Input-Output Database (WIOD) 2016 Release (Timmer et al., 2015) to group 56 industries into five sectors, excluding the non-employment sector: Agriculture (AG), Manufacturing (MN), Services (SR), Clean Energy (CL), and Brown Energy (B).[36] Following O'Mahony and Timmer (2009), the clean energy sector includes the industry supplying electricity, and the brown energy sector includes the mining and quarrying industry as well as the industry manufacturing coke and refined petroleum products. See Appendix A for a detailed description of how economies and sectors are categorized.

I categorize the model parameters into three categories: (i) parameters that are imposed based on values previously used in the literature (summarized in Table 1), (ii) parameters calibrated outside of the model (summarized in Table 2), and (iii) parameters calibrated using the method of simulated moments (MSM) (summarized in Table 5). I assume agents have perfect foresight over aggregate variables for the next 200 years by setting $T_{\max} = 200$. All parameters are calibrated using data from 2014, the final year available in the WIOD, or the closest available year for other sources.

3.1 Imposed Parameters

The imposed parameters are derived from established values in the literature. These parameters, detailed in Table 1, include the discount factor (β) set at 0.98, following Rennert et al. (2022), and the trade elasticity (σ) set at 6.03, following Head and Mayer (2014).[37]

[35] The selection of these economies is motivated by the fact that the US, China, India, and the EU accounted for approximately 60% of global carbon emissions in 2022.

[36] In principle, the model can be extended to an arbitrary number of economies and sectors. In this paper, I focus on a limited number of economies and sectors to balance data availability and clarity of interpretation. For example, individual-level data are not available for all WIOD countries, and the China Family Panel Studies aggregate all manufacturing industries into a single sector. This parsimonious aggregation allows the key mechanisms and policy tradeoffs to be presented more transparently.

[37] Head and Mayer (2014) study 744 estimates of trade elasticity from a sample of 32 papers. The median of estimates is -5.03, i.e., $1 - \sigma = -5.03$.

Table 1: Summary of Imposed Parameters

Parameter	Value	Description	Source
β	0.98	Discount factor	Rennert et al. (2022)
σ	6.03	Trade elasticity	Head and Mayer (2014)
$\delta^C_{t-\tau}$	See Text	Carbon stock depreciation	Cruz and Rossi-Hansberg (2023)
$\mathcal{S}$	2200	Pre-industrial carbon stock	Golosov et al. (2014)

Additionally, the carbon stock depreciation rate (δ^C_τ) is calibrated following the approach of Cruz and Rossi-Hansberg (2023). In particular, I parameterize δ^C_τ as follows: $\delta^C_\tau = 1 - a_0 - \sum_{s=1}^{3} a_s e^{-\tau/b_s}$.[38] This approximation reflects the finding of Stocker et al. (2013) that more than 20% of carbon emissions in a given year will remain in the atmosphere for more than 1000 years. Following Golosov et al. (2014), the pre-industrial carbon stock ($\mathcal{S}$) is set to 2200 Gt of CO_2, with the baseline year for the measurement being 1850 ($T_{\min} = 1850$).

3.2 Parameters Calibrated Outside of the Model

In this section, I describe the calibration procedure and results for the key parameters of the model: the dispersion of capital efficiencies ρ_K, the dynamic agglomeration externality ρ_Φ, and the dispersion of non-pecuniary preferences ρ_L. To keep the exposition focused and concise, the calibration procedures and results for the remaining parameters in Table 2 are reported in Appendix A.3.

Capital Supply Elasticities I derive the following specification based on Equation 12 to estimate the capital supply elasticities ρ_K and ρ_Φ (see Appendix A.13 for the derivation):

$$\Delta \ln\left(\frac{K^j_{n,t}}{K^o_{n,t}}\right) = \Big(\rho_K - 1\Big)\Delta \ln\left(\frac{r^j_{n,t}}{r^o_{n,t}}\right) + \big(\rho_K \cdot \rho_\Phi\big)\Delta \ln\left(\frac{K^j_{n,t-1}}{K^o_{n,t-1}}\right) + \quad \tilde{\eta}^j_{n,t} \tag{20}$$

where $\Delta x^j_{n,t} \equiv x^j_{n,t} - x^j_{n,t-1}$ denotes the first difference, and $\tilde{\eta}^j_{n,t} = \tilde{\phi}^j_n + \hat{\phi}^j_n \times t + \nu^j_{n,t}$ includes economy-sector fixed effects $\tilde{\phi}^j_n$, an economy-sector-specific linear time trend

[38] The coefficients are $a_0 = 0.2173$, $a_1 = 0.2240$, $a_2 = 0.2824$, $a_3 = 0.2763$, $b_1 = 394.4$, $b_2 = 36.54$, and $b_3 = 4.304$.

Table 2: Summary of Parameters Calibrated Outside of the Model

Parameter	Description	Source
ρ_K	Dispersion of capital efficiencies	WIOD
ρ_Φ	Dynamic agglomeration externality	WIOD
ρ_L	Dispersion of non-pecuniary preferences	See Text
$\kappa_n^{j,k}$	Labor mobility costs	See Text
η_1	Energy demand elasticity	See Appendix
η_2	Fossil fuel elasticity	See Appendix
w_n^{NE}	Non-employment income	See Appendix
$d_{nm,0}^j$	Exogenous trade costs	See Appendix
δ_n^K	Capital depreciation	PWT
φ_n^h	Supply cost of fossil fuel by type	IMF
ϕ^h	Carbon emission by fossil fuel	RR2020
E^L	Land-change emissions	RR2020
λ_T	Carbon-temperature conversion	MET
α_n	Non-energy expenditure share	WIOD
α_n^j	Non-energy expenditure share by sector	WIOD
$\gamma_n^{j,V}$	Value added share	WIOD
$\gamma_n^{j,k}$	Material input share	WIOD
$\psi_n^{j,L}$	Labor input share	WIOD
$\psi_n^{j,K}$	Capital input share	WIOD
$\psi_n^{j,E}$	Energy demand share	WIOD
$\psi_n^{B,F}$	Fossil fuel input share	WIOD

Notes: RR2020: Ritchie and Roser (2020), MET: Met Office Hadley Centre (2024), PWT: Penn World Table, WIOD: World Input-Output Database 2016 Release, WIOD 2013: World Input-Output Database 2013 Release, IMF: Black et al. (2023), MacMap: Guimbard et al. (2012).

$\hat{\phi}_n^j \times t$, and an error term $\nu_{n,t}^j$. To run the regression, I obtain data on sector-level capital stock $P_{n,t}K_{n,t}^j$ and capital compensation $r_{n,t}^j P_{n,t} K_{n,t}^j$ from the Socio-Economic Accounts (SEA) of the WIOD 2016 Release for 56 sectors in 43 countries from years 2000 to 2014. Rental rates of capital $r_{n,t}^j$ are computed by dividing sector-level capital compensation by sector-level capital stocks. I include all 56 sectors in the estimation to exploit the full cross-sector variation. I take the petroleum and coke manufacturing sector (P) as the reference sector ($o = P$).[39]

[39]The choice of the reference sector is a normalization and does not affect the estimated parameters.

The specification in Equation (20) may suffer from simultaneity bias, as capital demand and capital supply jointly determine the equilibrium capital allocation and the rental rate. To address this endogeneity concern, I employ an instrumental variable (IV) approach that uses the annual change in sector-level exposure to environmental policies, $\Delta \text{EPS}^j_{n,t}$, as an instrument for changes in the rental rate.

The sector-level exposure to environmental policies, $\text{EPS}^j_{n,t}$, is constructed using the country-level Environmental Policy Stringency (EPS) index developed by the OECD (Kruse et al., 2022), and country-sector-level carbon emissions per unit of output in the year 2000, based on data from the World Input-Output Database Environmental Accounts (Corsatea et al., 2019). The EPS index, available for 40 countries from 1990 to 2023, measures the stringency of 13 policy instruments (including market-based, non-market-based, and technology-support policies) on a scale from 0 to 6, with higher values indicating stricter environmental policies.

Since this paper focuses on market-based climate policies, the scores for market-based instruments are converted into USD per tonne of GHG emissions using conversion tables provided by Kruse et al. (2022), and then averaged. The sector-level policy exposure $\text{EPS}^j_{n,t}$ is defined as the product of this converted EPS measure and the sector-level carbon emissions per unit of output in the year 2000 from Corsatea et al. (2019). Identification of $(\rho_K - 1)$ relies on the assumption that environmental policies influence capital supply only indirectly, by raising the cost of emitting and reducing capital demand disproportionately in more emission-intensive sectors, hence shifting their rental rates.[40]

The specification in Equation (20) may also suffer from autocorrelation bias because capital adjusts gradually in response to past shocks. In particular, the lagged capital differential $\Delta \ln \left(\frac{K^j_{n,t-1}}{K^o_{n,t-1}} \right)$ may be correlated with the error term $\nu^j_{n,t}$ if unobserved shocks to capital supply or demand are serially correlated. To address this concern, I use the five-period lagged capital $\ln \left(\frac{K^j_{n,t-5}}{K^o_{n,t-5}} \right)$ as an instrument similar to the approach of Arellano and Bond (1991). This instrument is correlated with the endogenous lagged capital differential but assumed orthogonal to the current error term under the assumption that

[40]The instrument has a shift-share structure, interacting cross-country, time-varying aggregate policy stringency with fixed year-2000 sector emission intensities. Emission intensities are fixed before most of the sample's EPS variation, so the instrument is plausibly orthogonal to contemporaneous shocks to sector-specific capital efficiency $a^j_{n,t}$.

Table 3: Estimation Result for Capital Supply Elasticities ($\rho_K - 1$, $\rho_K \cdot \rho_\Phi$)

Dependent variable: $\Delta \ln \left(\frac{K^j_{n,t}}{K^P_{n,t}} \right)$	OLS	IV: $\Delta \text{EPS}^j_{n,t}$
$\Delta \ln \left(\frac{r^j_{n,t}}{r^P_{n,t}} \right)$	0.0187***	0.3099***
	(0.0042)	(0.0953)
$\Delta \ln \left(\frac{K^j_{n,t-1}}{K^P_{n,t-1}} \right)$	–0.1582***	0.8127***
	(0.0133)	(0.0973)
Country-Sector FE	YES	YES
Country-Sector Trend	YES	YES
Observations	21,744	12,283
First Stage F-Statistic		56.84
R-squared	0.0379	

Notes: $\Delta \ln(K^j_{n,t}/K^P_{n,t})$ denotes the annual change in (log) ratio of sector-level capital stocks relative to the petroleum and coke manufacturing sector (P). $\Delta \ln(r^j_{n,t}/r^P_{n,t})$ denotes the annual change in (log) ratio of rental rates of capital relative to the petroleum and coke manufacturing sector (P). Standard errors are clustered at the country-sector level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Source: WIOD 2016 Release.

shocks affecting capital allocation do not persist beyond four years.

The results are presented in Table 3. Column 3 provides the preferred IV estimates for $\rho_K - 1$ and $\rho_K \cdot \rho_\Phi$, yielding $\rho_K = 1.310$ and $\rho_\Phi = 0.620$.[41] These results indicate that a 1 percent increase in the rate of return in a given sector, relative to petroleum and coke manufacturing, is associated with a 0.310 percent increase in capital allocated to that sector. This elasticity implies substantial heterogeneity in returns to capital investment even within sectors, which dampens sectoral capital reallocation. Moreover, the estimate for ρ_Φ suggests that a 1 percent increase in past capital allocation to a given sector, relative to the petroleum and coke manufacturing sector, is associated with a 0.813 percent increase in current capital allocation. This implies that the evolution of sector-specific capital efficiency is highly dependent on past capital investment in that sector.

[41]The OLS estimate for the lagged capital coefficient is negative, while the IV estimate is positive. This reversal is consistent with the dynamic-panel bias of Nickell (1981): OLS is downward biased when the lagged dependent variable is correlated with panel-specific residuals. The lagged-capital IV corrects this bias.

Dispersion of Non-pecuniary Preferences For the estimation labor supply parameters, I merge longitudinal labor force surveys from the following eight countries: the United States (Current Population Survey, 2000-2023), China (China Family Panel Studies, 2010-2020), India (Periodic Labour Force Survey, 2018-2020), France (Continuous Labour Force Survey, 2014-2020), Argentina (Permanent Household Survey, 2003-2019), Australia (The Household, Income and Labour Dynamics in Australia, 2002-2022), South Korea (Korean Labor & Income Panel Study, 1999-2022), and the United Kingdom (Labour Force Survey, 2016-2022).[42] These datasets track the flows $L_{n,t}^{j,k}$ of individuals across sectors and employment status over time and report their income as well as individual characteristics such as gender, age, and education. Details on the data construction procedure are provided in Appendix A.2.

To estimate the dispersion of non-pecuniary preferences ρ_L, I use the following specification (see Appendix A.13 for the derivation):

$$y_{n,t}^{j,k} = \omega_n^k + \omega_n^j + \left(\frac{\beta}{\rho_L}\right) \ln\left(\frac{w_{n,t+1}^k}{w_{n,t+1}^j}\right) + \eta_{n,t+1}^{j,k} \tag{21}$$

where $y_{n,t}^{j,k} \equiv \ln\left(\frac{L_{n,t}^{j,k}}{L_{n,t}^{j,j}}\right) - (1-\beta)C_n^{j,k} - \beta \ln\left(\frac{L_{n,t+1}^k L_{n,t+1}^{j,k}}{L_{n,t+1}^j L_{n,t+1}^{k,k}}\right)$, and $C_n^{j,k} \equiv \frac{\ln(1-\kappa_n^{j,k})}{\rho_L}$ denotes the bilateral mobility cost terms estimated in Appendix A.3; ω_n^j and ω_n^k are, respectively, economy-origin and economy-destination fixed effects defined as $\omega_n^j \equiv -\beta \ln(\mu_n^j)$ and $\omega_n^k \equiv \beta \ln(\mu_n^k)$; $\ln(\frac{w_{n,t+1}^k}{w_{n,t+1}^j})$ captures the (log) residualized wage of the destination sector k relative to the residualized wage of the origin sector j at time $t+1$; and, $\eta_{n,t+1}^{j,k}$ is a forecast error. For the estimation, I assume the discount factor β equals 0.98.

To address the potential endogeneity concern that forecast errors may be correlated with the regressors, I estimate the regression specification (21) using an instrumental variable (IV) approach, employing two-period lagged residualized wages, $\ln\left(\frac{w_{n,t-1}^k}{w_{n,t-1}^j}\right)$, as an instrument, following Artuç et al. (2010). This instrument is assumed to be correlated with future wage differentials but uncorrelated with future labor supply shocks, such as preference shocks or migration forecast errors.

[42] France proxies the European Union; the United Kingdom, Australia, South Korea, and Argentina proxy the Rest of the World.

Table 4: Estimation Result for Dispersion of Non-pecuniary Preferences $\left(\frac{\beta}{\rho_L}\right)$ with $\beta = 0.98$

Dependent variable: $y^{j,k}_{n,t}$	OLS	IV: $\ln\left(w^k_{n,t-1}/w^j_{n,t-1}\right)$
$\ln\left(\frac{w^k_{n,t+1}}{w^j_{n,t+1}}\right)$	0.4327***	0.6903***
	(0.0770)	(0.1812)
Country-Origin FE	YES	YES
Country-Destination FE	YES	YES
Observations	3,597	2,995
First Stage F-Statistic		41.84
R-squared	0.0676	

Notes: $y^{j,k}_{n,t}$ denotes the (log) difference of labor flows adjusted for mobility costs and option values. $\ln\left(w^k_{n,t+1}/w^j_{n,t+1}\right)$ denotes the (log) residualized wage ratio between sectors k and j at time $t+1$. Standard errors are clustered at the country-origin-destination level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Sources: Panel labor force surveys of the US (CPS, 2000-2023), China (CFPS, 2010-2020), India (2018-2020), France (2014-2020), Argentina (2003-2019), Australia (2002-2022), South Korea (KLIPS, 1999-2022), and the UK (LFS, 2016-2022).

The results are presented in Table 4. Column 3 provides the preferred IV estimate for $\frac{\beta}{\rho_L}$, yielding $\rho_L = 1.420$. This estimate suggests that a 1 percent increase in the next-period wage of the destination sector relative to the next-period wage of the origin sector increases labor flows to the destination sector by 0.690 percent. Hence, labor flows are not highly responsive to changes in future wage differentials, and the dispersion of non-pecuniary motives for worker mobility is significant.

3.3 Parameters Calibrated by Method of Simulated Moments

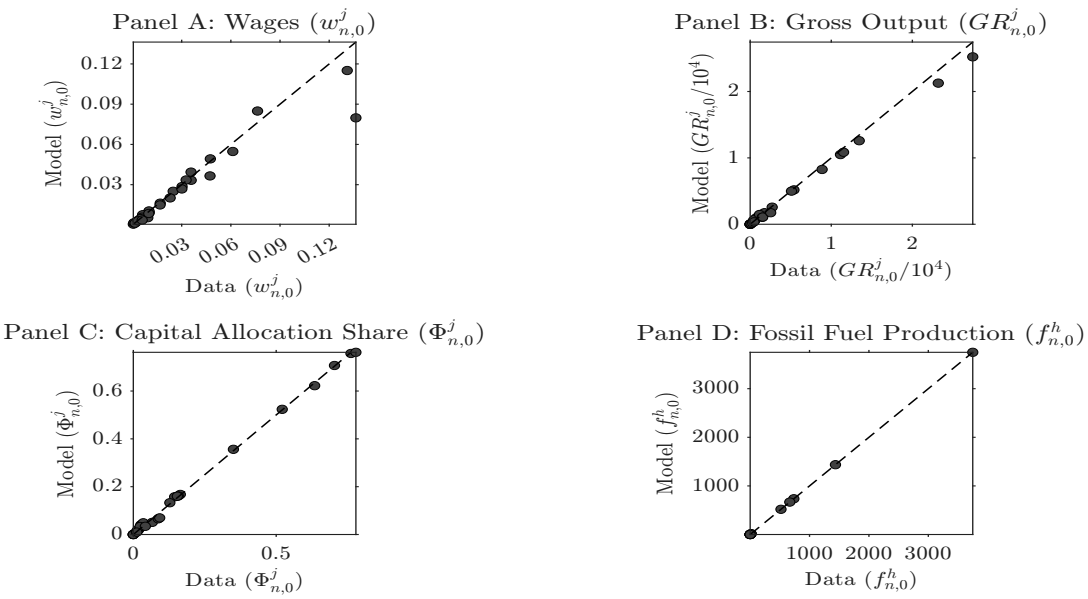
The parameters listed in Table 5 are jointly calibrated by the Method of Simulated Moments (MSM), which iteratively minimizes the distance between model-simulated and observed data moments, conditional on parameters that are imposed or calibrated outside the model. For a detailed description of the targeted moments and the MSM procedure, refer to Appendix A.4.

Table 5: Summary of Parameters Calibrated by Method of Simulated Moments

Parameter	Description	Targeted Moments
a_n^j	Average capital efficiency	Real interest rate, capital allocation, capital compensation
μ_n^j	Non-pecuniary benefits	Sector-level wages
$z_{n,0}^j$	Sector-specific TFP	Sector-level gross output, bilateral trade shares
ω_n^h	Weights on fossil fuels	Fossil fuel production by type
α_n^{CL}	Weight on clean energy	Expenditure on the clean energy
ξ_n	Environmental damage	GDP response to 1°C increase in global temperature

Notes: Parameters are estimated by matching model moments to observed data moments using the Method of Simulated Moments.
Sources: IMF IFS, Timmer et al. (2015), Ritchie and Rosado (2017), Bilal and Känzig (2024).

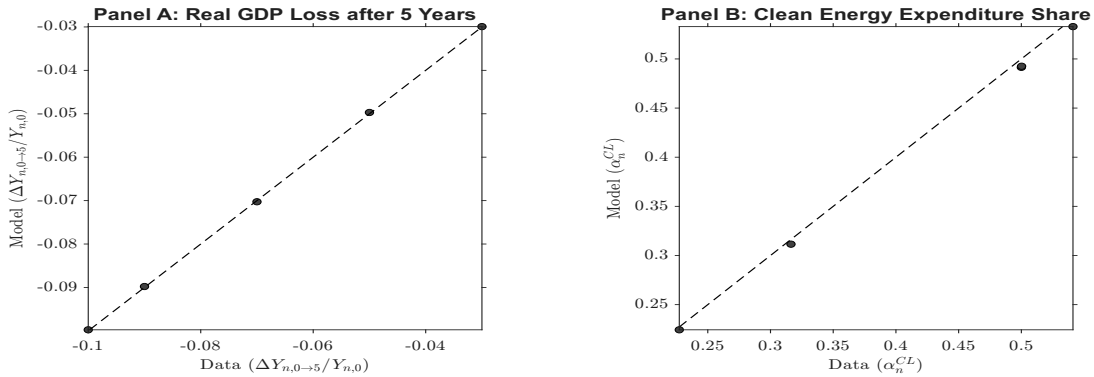
Figure 1: Model Fit: Model vs. Data Moments



Notes: Comparison between the model-generated moments and the observed data moments for wages, gross output, capital allocation share, and fossil fuel production.

Model Fit Figures 1 and 2 compare the data moments targeted during the MSM calibration procedure with the model-generated moments. Overall, the calibrated model

Figure 2: Model Fit: Model vs. Data Moments



Notes: Comparison between the model-generated moments and the observed data moments for real GDP loss after 5 years in response to a 1 °C increase in the global temperature and final consumption expenditure on the clean energy sector.

provides a good fit to the data. The parameter estimates obtained through the MSM procedure are presented in Appendix A.8.

Table 6: Summary of Initial Conditions

Variable	Description	Source
$W_{n,0}$	Aggregate capital	WIOD
$L^j_{n,0}$	Labor allocation	WIOD
$\mathcal{C}_0$	Carbon stock	RR2020
$\mathcal{D}_{n,0}$	Environmental quality	BK, MET
$D^h_{n,0}$	Fossil fuel reserves	RR2017
$\tau^{Y,B}_{n,0}$, $\tau^{Y,CL}_{n,0}$	Sector-specific consumption taxes	–
$\tau^{K,B}_{n,0}$, $\tau^{K,CL}_{n,0}$	Sector-specific capital taxes	–
$\tau^{L,B}_{n,0}$, $\tau^{L,CL}_{n,0}$	Sector-specific labor taxes	–
$\tau^{f,h}_{n,0}$	Fuel-specific taxes	IMF
φ^h_n	Fossil fuel extraction costs	IMF

Notes: RR2020: Ritchie and Roser (2020), BK: Bilal and Känzig (2024), MET: Met Office Hadley Centre (2024), WIOD: World Input-Output Database 2016 Release, IMF: Black et al. (2023), RR2017: Ritchie and Rosado (2017).

3.4 Initial Conditions

Table 6 summarizes the variables that characterize the initial conditions of the model. The initial conditions are computed for the year 2014, or the most recent available year. For a detailed description of the variables and data sets used to compute the initial conditions, refer to the Appendix A.5.

4 Quantitative Analysis

4.1 Mechanisms

In this section, I assess the quantitative importance of two channels that govern the impact of climate policies on global emissions: (i) the substitution effect, and (ii) the scale effect. As described in Section 2.2, the substitution effect refers to the change in emissions per unit of output due to shifts in production and demand away from or toward the brown energy sector; in contrast, the scale effect refers to the change in emissions driven by changes in aggregate demand and production through the energy price index, investment, and input-output linkages.

To assess the magnitude of these two channels, I focus on a clean energy subsidy, which generates substitution and scale effects that operate in opposing directions: a subsidy lowers the price of clean energy relative to brown energy, reducing emissions per unit of output, but also lowers the energy price index, increasing overall energy demand and output.[43]

Substitution Effect Imperfect factor mobility limits production reallocation across sectors in response to climate policies, weakening the substitution effect. To investigate the role of imperfect factor mobility, I compare two models: (i) the baseline model with the estimated levels of factor mobility and (ii) a counterfactual model featuring perfect factor mobility (hereafter, PM).[44] The PM model features perfectly elastic capital and labor supply such that factors can immediately move across sectors in response to climate

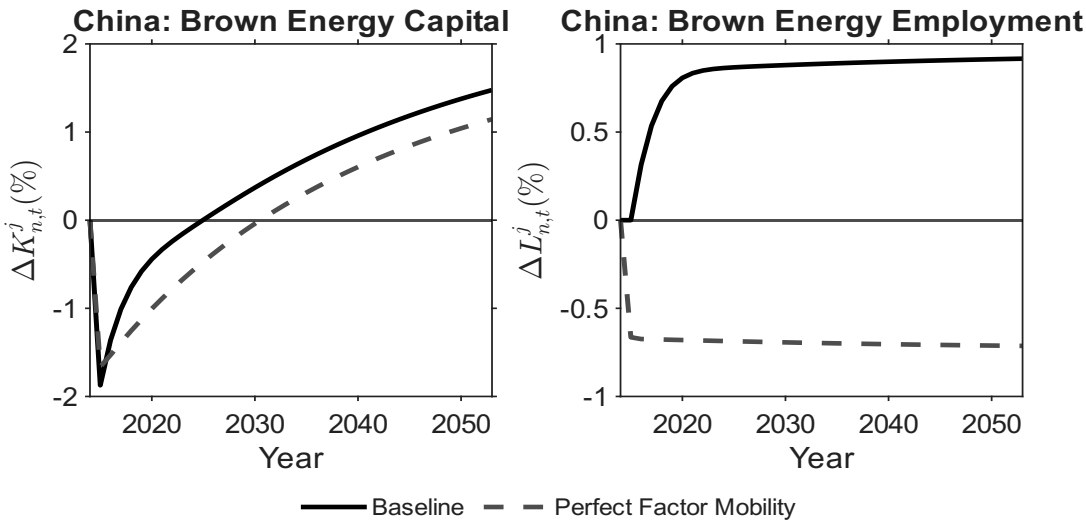
[43]For a carbon tax, the substitution and scale effects go in the same direction by making brown energy relatively more expensive and increasing the energy price index.

[44]The literature typically emphasizes demand-driven energy substitution governed by the elasticity parameter η_1. Since this paper focuses on factor mobility, I keep η_1 fixed at its estimated value of 2.28.

policies and equalize factor prices, similar to standard models used in climate policy analysis (Golosov et al., 2014; Barrage and Nordhaus, 2024).[45]

I simulate an unanticipated and permanent 10-percentage-point increase in the clean energy subsidy in 2015 in both models.[46] I then compare changes in key variables relative to business-as-usual (BAU) levels where the policy is not implemented. To evaluate cross-country heterogeneity, I conduct two experiments: one for China and one for the United States.[47]

Figure 3: Model Comparison:
The Impact of a Clean Energy Subsidy in China on the Brown Energy Sector in China



Notes: The impact of China's unanticipated and permanent 10-percentage-point reduction in the consumption tax on the clean energy sector in 2015 on capital and employment in the brown energy sector. The figure compares the baseline model with the model with perfect factor mobility. All values are reported relative to the business-as-usual (BAU) scenario without the policy.

Figure 3 shows the impact of China's clean energy subsidy on capital and employment in its brown energy sector. In both models, capital falls initially relative to BAU due to the substitution effect, then rises over time as the scale effect, driven by aggregate

[45]The PM model is calibrated using the MSM procedure to match the same moments used to calibrate the baseline model, except for the moments used to calibrate sector-specific capital efficiencies and non-pecuniary benefits.

[46]That is, the experiment involves permanently reducing the consumption tax ($\tau^{Y,CL}_{n,t}$) on the clean energy sector by 10 percentage points.

[47]China and the United States are selected because they are the largest emitters of CO_2, accounting for 31.2% and 13.5% of global emissions (excluding land-use change) in 2022, respectively.

capital accumulation, dominates. On impact, capital falls more in the baseline model than in the PM model because the policy shock is unanticipated: workers chose their sector allocation in 2014 before the subsidy was announced, so no reallocation occurs in 2015 and the entire reallocation burden falls on capital. In China, the subsidy induces a substantial absolute expansion in the clean energy sector, generating capital reallocation large enough to outweigh both the capital-labor complementarity and imperfect capital mobility channels that would otherwise dampen the capital response. In the PM model, by contrast, the reallocation burden is shared with labor, so capital falls by less on impact.

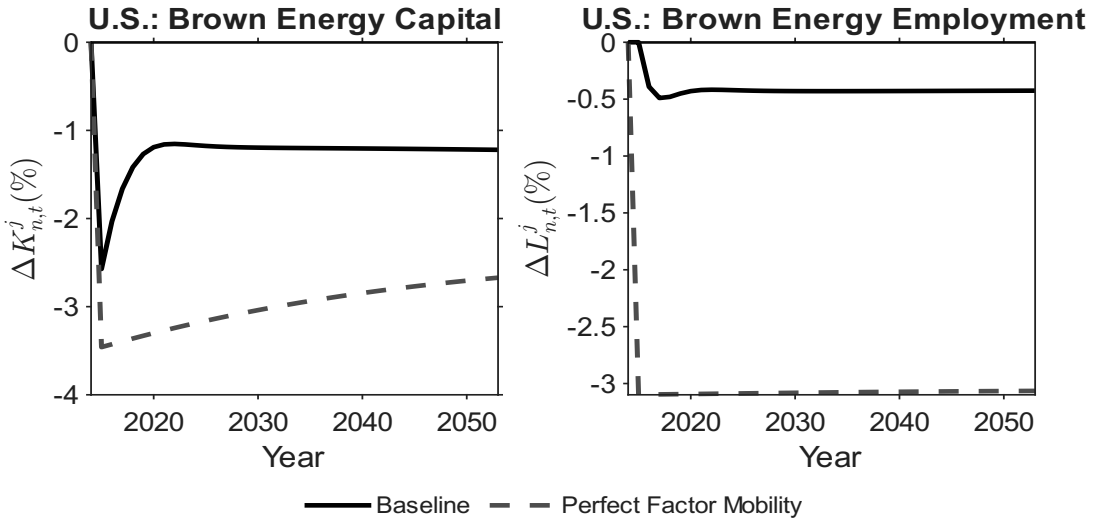
However, from 2016 onward, the role of imperfect capital mobility becomes evident. Capital in the brown energy sector returns to the BAU level by 2025, six years faster than in the PM model. This faster recovery directly reflects the weaker substitution effect under imperfect capital mobility that limits the reallocation of capital from the brown energy sector to the clean energy sector, allowing the scale effect to dominate sooner despite the larger initial decline. The employment responses reinforce this interpretation; in the baseline model employment rises gradually and monotonically, while in the PM model, it drops sharply on impact and remains permanently below the BAU level.

Figure 4 shows the impact of a clean energy subsidy in the United States on capital and employment in its brown energy sector. As in China, capital initially falls relative to the BAU level but rises over time due to the scale effect in both models. The initial capital drop in the baseline model is 74.4% of the decline in the PM model. Employment in both models falls relative to BAU. In the baseline model, there is no employment response in 2015 since workers cannot adjust immediately; employment declines by 12.6% of the PM model's drop in 2016 and by 13.9% in 2050.

In both China and the United States, imperfect factor mobility substantially weakens the substitution effect of clean energy subsidies over time. The results also highlight differences in capital and employment responses across the two economies, driven by heterogeneous scale effects. The scale effect appears weaker in the United States, causing capital in the brown energy sector to stay below BAU and employment to remain permanently below BAU, a point to which I return below.

Figure 5 shows the impact of clean energy subsidies in China and the United States on global emissions. The magnitude of scale effects differs substantially across the two economies. When China implements the subsidy, global emissions rise above BAU in

Figure 4: Model Comparison:
The Impact of a U.S. Clean Energy Subsidy on the U.S. Brown Energy Sector



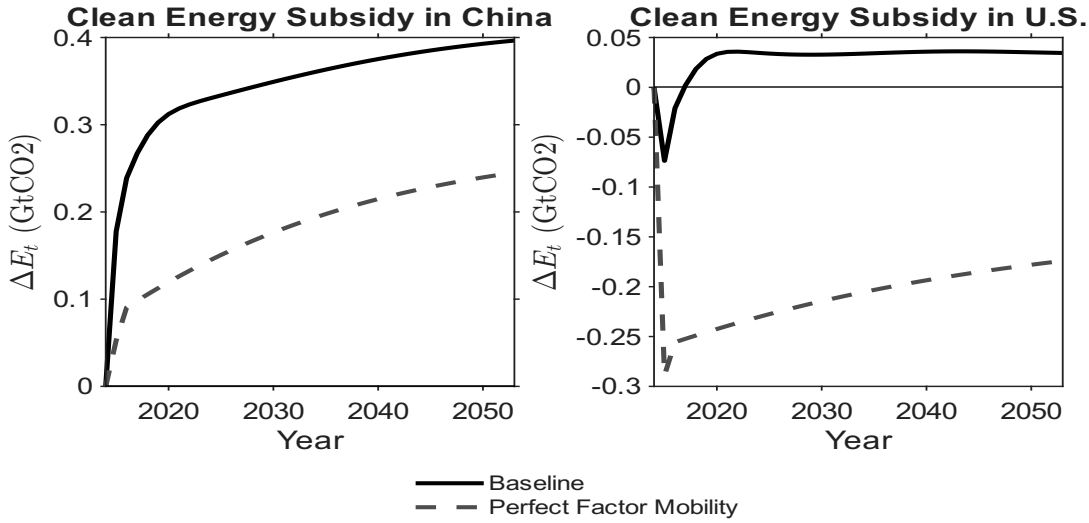
Notes: The impact of an unanticipated and permanent 10-percentage-point reduction in the U.S. consumption tax on the clean energy sector in 2015 on capital and employment in the brown energy sector. The figure compares the baseline model with the model with perfect factor mobility. All values are reported relative to the business-as-usual (BAU) scenario without the policy.

both models. When the United States implements the subsidy, global emissions initially fall below BAU in both models, but subsequently rise above BAU only in the baseline model; in the PM model, emissions remain below BAU permanently. Appendix Figure A2 shows qualitatively similar patterns for India and the European Union.

Imperfect factor mobility also substantially weakens the effectiveness of climate policies. For China, the emission increase in the baseline model is 336.6% of the increase in the PM model in 2015 and 163.9% in 2050. For the United States, the emission decline in the baseline model is 25.5% of the decline in the PM model in 2015.

Scale Effects As shown in Figure 5, the clean energy subsidy raises global emissions substantially more when implemented by China compared to when implemented by the United States. This pattern indicates that the scale effects associated with China are substantially larger than those associated with the United States. To understand the sources of these scale effects, I consider two additional counterfactual models. First, I consider a version of the PM model with constant aggregate capital (hereafter PMK) to

Figure 5: Model Comparison:
The Impact of Clean Energy Subsidies on Global Emissions



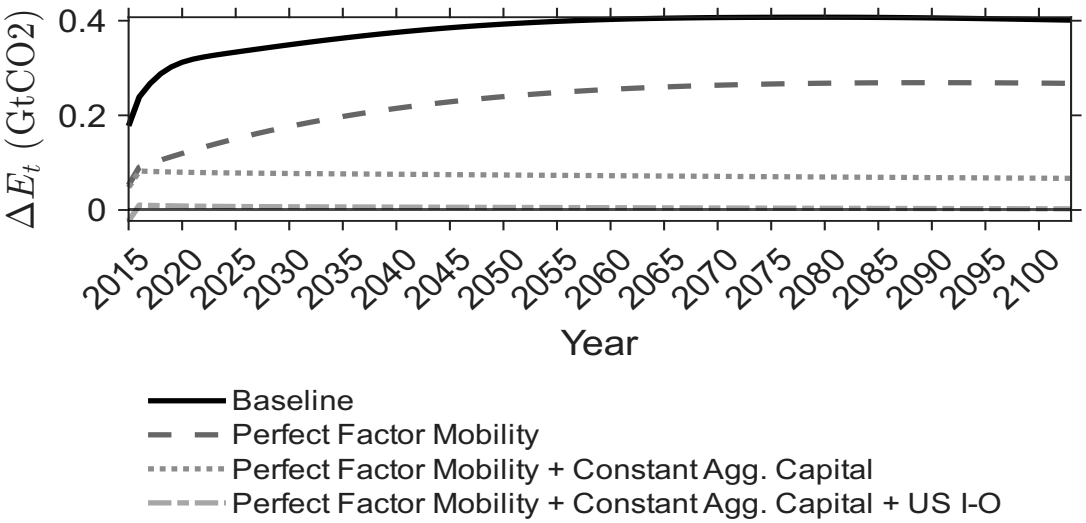
Notes: The left (right) panel displays the impact of an unanticipated and permanent 10 percentage point reduction in the consumption tax on the clean energy sector in China (the United States) in 2015 on global emissions. The figure compares the baseline model with the model with perfect factor mobility. All values are relative to the BAU case where the policy is not implemented.

remove the scale effect driven by capital accumulation. Second, motivated by the decline in global emissions when the subsidy is implemented by the United States, I consider a version of the PMK model where all economies have the same input-output structure as the United States (hereafter PMK-IO) by setting $\gamma_n^{j,k} = \gamma_{\text{US}}^{j,k}$ for all $n \in \mathcal{N}$, $j,k \in \mathcal{J}$; this model removes the scale effect driven by economy-specific production linkages.[48]

Figure 6 reports the response of global emissions to China's clean energy subsidy across the four models. Removing the capital accumulation channel leaves the initial increase in emissions in 2015 unchanged but prevents emissions from rising further over time. By 2100, the increase in global emissions in the PMK model amounts to 30.9 percent of the increase in the PM model. In contrast, in the PMK-IO model, the subsidy has a minimal impact on global emissions, indicating that China's production structure, which relies heavily on energy-intensive industries such as manufacturing, drives the larger

[48]The PMK and PMK-IO models are calibrated using the MSM procedure to match the same moments used to calibrate the baseline model, except for the moments used to calibrate sector-specific capital efficiencies and non-pecuniary benefits.

Figure 6: Model Comparison:
Impact of China's Clean Energy Subsidy on Global Emissions



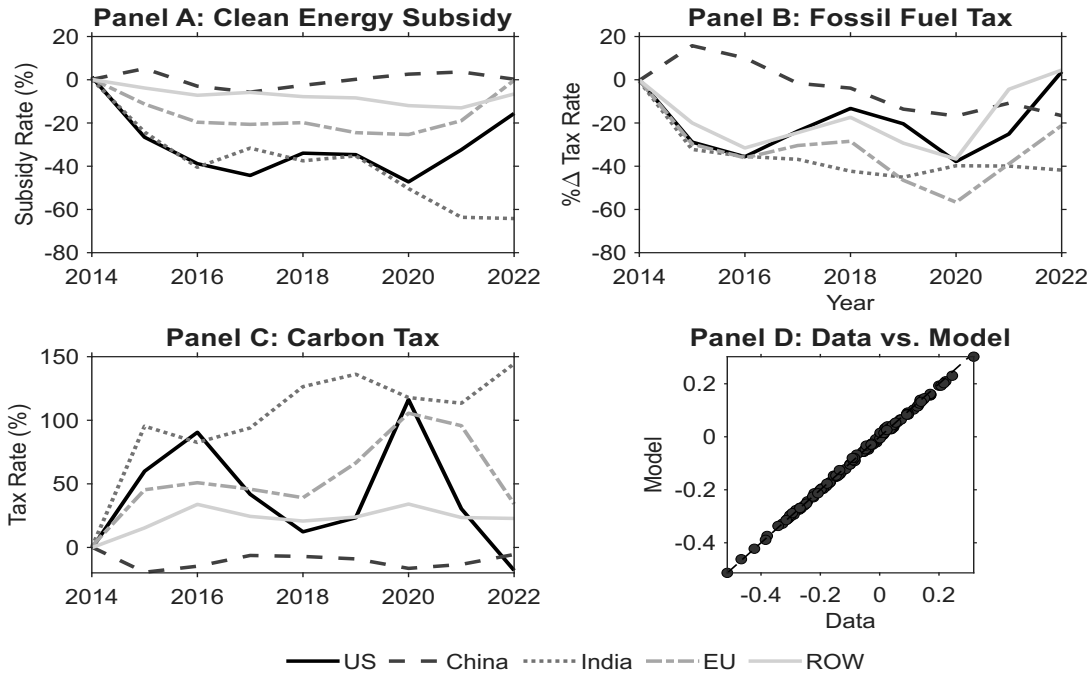
Notes: This figure shows the impact of an unanticipated and permanent 10 percentage point reduction in China's consumption tax on the clean energy sector in 2015 on global emissions across four model specifications. The lines report the baseline model with imperfect factor mobility, a model with perfect factor mobility (PM), a model with perfect factor mobility and constant aggregate capital (PMK), and a model with perfect factor mobility, constant aggregate capital, and the United States input-output structure for all economies (PMK-IO). All values are relative to the BAU case where the policy is not implemented.

scale effects. This result highlights that production linkages, which are absent from most climate policy models, are a quantitatively important driver of the scale effect.

4.2 Policy Evaluation

Model-Implied Policy Instruments In this section, I evaluate the economic and environmental implications of alternative policy designs. I begin by computing model-implied energy-related policy instruments that allow the model to reproduce the observed paths of clean energy production, brown energy production, and carbon emissions from 2015 to 2022. These model-implied instruments discipline the magnitude and evolution of policy interventions and serve as a realistic baseline for the counterfactual experiments that follow. These instruments should be interpreted as structural distortions that rationalize observed outcomes in the model, rather than literal statutory rates (Chari et al., 2007;

Figure 7: Model-Implied Policy Measures and Model Fit



Notes: Panels report model-implied consumption subsidies on the clean energy sector, $-\tau_{n,t}^{Y,CL}$ (Panel A), percentage deviations in fuel-specific gross tax rates $\{1+\tau_{n,t}^{f,h}\}_{h\in\{o,g,c\}}$ from the values reported in Black et al. (2023) (Panel B), and consumption taxes on the brown energy sector, $\tau_{n,t}^{Y,B}$ (Panel C), required to reproduce observed growth in energy production and emissions between 2015 and 2022 computed using the OECD Inter-Country Input-Output Tables (Yamano et al., 2023) and Ritchie and Roser (2020). Panel D presents the model fit. Policy instruments are held fixed at their 2022 levels after 2022.

Hsieh and Klenow, 2009).[49]

Specifically, I obtain model-implied clean and brown energy consumption taxes $\{\tau_{n,t}^{Y,CL}, \tau_{n,t}^{Y,B}\}$ by adjusting them each year so that the model reproduces the observed growth rates of gross output in the clean and brown energy sectors between 2015 and 2022.[50] I further obtain fuel-specific gross tax rates $\{1+\tau_{n,t}^{f,h}\}_{h\in\{o,g,c\}}$ using the IMF Fossil

[49]They capture the combined effects of explicit policies and unmeasured factors such as regulatory barriers, environmental regulations, and political economy considerations.

[50]Gross output growth rates in the clean and brown energy sectors are computed using the OECD Inter-Country Input-Output (ICIO) Tables (Yamano et al., 2023) for 2015–2022. ICIO sectors are aggregated into the model's sector classification as follows: the clean energy sector corresponds to ICIO sector D (electricity, gas, steam, and air conditioning supply); the brown energy sector corresponds to ICIO sectors B05–B09 (mining of coal, crude petroleum, natural gas, metal ores, and other mining) and C19 (petroleum refining); and remaining sectors are aggregated into agriculture (A01–A03), manufacturing (C10–C31), and services (E–T).

Fuel Subsidies Data (Black et al., 2023) as a baseline and scale them uniformly for each fuel type each year to match the observed growth in economy-level carbon emissions from 2015 to 2022 reported by Ritchie and Roser (2020). After 2022, all policy instruments are held fixed at their 2022 levels.

Figure 7 presents the model-implied policy instruments across economies and over time. Panel D shows that the model fits the targeted moments well. The model-implied instruments reveal several patterns. Panel A shows that China is the only economy that subsidized clean energy production over this period, while all other economies imposed substantial implicit taxes on clean energy throughout.[51] These implicit taxes likely reflect broad taxes on energy consumption rather than policies specifically targeting clean energy.

Panel C shows substantial heterogeneity in taxes on brown energy consumption. India and the European Union imposed large and increasing implicit taxes, while the United States shifted from a tax to a subsidy by 2022. China subsidized brown energy consumption throughout the period. Panel B reveals that all economies increasingly subsidized fossil fuel extraction, consistent with Black et al. (2023), suggesting that economies levying taxes on brown energy consumption simultaneously subsidized fossil fuel extraction, partly offsetting their own climate efforts.[52]

Alternative Policy Designs I next examine whether alternative clean energy subsidy instruments outperform the model-implied consumption-based instruments under imperfect factor mobility. Specifically, I compare consumption subsidies with input-based subsidies to capital and labor used in the clean energy sector. To isolate the role of instrument choice, I adopt a unilateral design: each economy independently reallocates its own model-implied energy-related subsidy expenditure toward a single clean energy instrument, while all other economies retain their model-implied instruments. This design isolates the effect of a single economy's policy reform and directly addresses the question of how an economy should use its own fiscal resources.

I consider four scenarios, each implemented as an unanticipated policy change at the beginning of 2015, with all state variables initialized at their BAU levels at the end of

[51] In particular, the subsidy rate in China peaked at 5.14 percent in 2015, turned negative between 2016 and 2018, recovered to a second peak of 3.62 percent in 2021, and declined to near zero by 2022.

[52] This pattern could reflect the influence of political economy and energy security considerations on policy design.

2014. Scenario 1 is the baseline and implements the model-implied policy instruments $\{\tau_{n,t}^{Y,B}, \tau_{n,t}^{Y,CL}, \tau_{n,t}^{f,h}\}$ for all economies.[53] In Scenarios 2–4, the implementing economy reallocates its total energy subsidy expenditure $\widetilde{\Omega}_{n,t}^{total}$ exclusively toward clean energy subsidies, varying only the instrument type, while all other economies retain their model-implied instruments.[54] Scenario 2 implements a clean energy consumption subsidy ($\tau_{n,t}^{Y,CL}$), Scenario 3 a clean energy capital subsidy ($\tau_{n,t}^{K,CL}$), and Scenario 4 a clean energy labor subsidy ($\tau_{n,t}^{L,CL}$).[55] In all scenarios, policy instruments are held fixed at their 2022 values after 2022.

I measure domestic and global cumulative emission changes relative to BAU, $\widehat{\mathcal{E}}_{n,y}$ and $\widehat{\mathcal{E}}_{y}$ respectively, and welfare as cumulative real consumption across all agents with equal welfare weights, $\widehat{\mathcal{W}}_{n,y}$.[56]

Tables 7 and 8 show substantial heterogeneity in the effects of unilateral reform across economies. Relative to the model-implied instruments baseline, reallocating subsidy resources lowers global cumulative emissions when the United States, China, or India acts alone, but raises them when the European Union or the Rest of World acts alone. The global effect of unilateral reform therefore depends on each economy's production structure and the strength of the cross-border scale effects it generates.

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[53] Government revenue from consumption-based instruments in units of final good is $\widetilde{\Omega}_{n,t}^{Y,j} \equiv \frac{\tau_{n,t}^{Y,j} p_{n,t}^{j} Y_{n,t}^{j}}{P_{n,t}}$ for $j \in \{B, CL\}$. Revenue from fuel-specific instruments is $\widetilde{\Omega}_{n,t}^{f,h} \equiv \frac{\tau_{n,t}^{f,h} s_{n,t}^{h} f_{n,t}^{h}}{P_{n,t}}$ for $h \in \{o, g, c\}$. Total energy-related subsidy expenditure is $\widetilde{\Omega}_{n,t}^{total} \equiv -\sum_{j\in\{B,CL\}} \widetilde{\Omega}_{n,t}^{Y,j} \mathbb{1}\{\widetilde{\Omega}_{n,t}^{Y,j} < 0\} - \sum_{h\in\{o,g,c\}} \widetilde{\Omega}_{n,t}^{f,h} \mathbb{1}\{\widetilde{\Omega}_{n,t}^{f,h} < 0\}$.

[54] Clean energy subsidies are often more politically feasible than carbon taxes, making this a realistic reform margin consistent with recent policies such as the U.S. Inflation Reduction Act.

[55] Fiscal equivalence requires subsidy expenditure under each alternative instrument to equal total baseline subsidy expenditure for all $n \in \mathcal{N}$ and $t \in \{2015, \ldots, 2022\}$. Specifically, Scenario 2 sets $\widetilde{\Omega}_{n,t}^{Y,CL} = -\widetilde{\Omega}_{n,t}^{total}$, Scenario 3 sets $\widetilde{\Omega}_{n,t}^{K,CL} \equiv \tau_{n,t}^{K,CL} r_{n,t}^{CL} K_{n,t}^{CL} = -\widetilde{\Omega}_{n,t}^{total}$, and Scenario 4 sets $\widetilde{\Omega}_{n,t}^{L,CL} \equiv \frac{\tau_{n,t}^{L,CL} w_{n,t}^{CL} L_{n,t}^{CL}}{P_{n,t}} = -\widetilde{\Omega}_{n,t}^{total}$.

[56] Formally, $\widehat{\mathcal{E}}_{y} = \sum_{n\in\mathcal{N}} \widehat{\mathcal{E}}_{n,y}$ and $\mathcal{W}_{n,y} \equiv \sum_{t=2015}^{y} \left(C_{n,t}^{K} + \sum_{j\in\mathcal{J}} c_{n,t}^{j} L_{n,t}^{j} \right)$. $\widehat{\mathcal{W}}_{n,y}$ denotes the percentage change in welfare relative to BAU. In the model, net government revenue is rebated as lump-sum transfers to capitalists only. This assumption is standard but consequential for welfare comparisons: the fiscal cost of any subsidy is borne entirely by capitalists through reduced transfers. Although the model features infinitely-lived agents, consumption at different dates can represent different generations. Following Ramsey (1928), Heal (2005), Stern (2007), and Stern (2008), this welfare measure does not discount future consumption, reflecting the ethical position that discounting future utility is "ethically indefensible and arises merely from the weakness of the imagination" (Ramsey, 1928).

Table 7: Unilateral Design: Domestic and Global Cumulative Emissions

	Scenario 1 Model-Implied		Scenario 2 Consumption		Scenario 3 Capital		Scenario 4 Labor	
	2030	2100	2030	2100	2030	2100	2030	2100
Panel A: Domestic Cumulative Emissions ($\widehat{\mathcal{E}}_{n,y}$, $GtCO_2$ change from BAU)								
United States	5.7	37.7	-19.1	-95.9	-17.2	-82.9	-17.4	-87.0
China	16.1	120.2	-12.6	-66.7	-23.0	-125.3	-13.9	-66.5
India	11.2	60.9	-5.8	-24.0	-5.8	-22.8	-5.6	-24.1
European Union	-2.6	-22.8	-7.3	-26.0	-7.1	-24.9	-7.4	-26.9
Rest of World	0.5	-14.0	-10.7	-51.5	-10.8	-51.9	-10.8	-52.0
Panel B: Global Cumulative Emissions								
($\widehat{\mathcal{E}}_y$, $GtCO_2$ change from BAU, economy acts alone)								
United States	30.9	182.1	12.0	42.9	13.3	48.9	10.8	35.4
China	30.9	182.1	9.5	56.1	-1.1	-0.3	5.9	44.2
India	30.9	182.1	24.0	142.3	24.0	143.2	23.4	138.5
European Union	30.9	182.1	40.8	238.3	41.2	240.5	40.0	234.2
Rest of World	30.9	182.1	56.9	287.7	57.0	287.6	56.7	286.4

Notes: Panel A reports domestic cumulative emission changes from 2015 to the specified year relative to BAU. Panel B reports global cumulative emission changes when only the row economy implements the alternative instrument, holding all other economies at model-implied instruments. Scenario 1 implements the model-implied energy-related policy instruments for all economies. Scenarios 2–4 are fiscally equivalent for the implementing economy, reallocating its total model-implied subsidy expenditure toward clean energy subsidies only, while all other economies retain model-implied instruments. Scenario 2 implements clean energy consumption subsidies ($\tau_{n,t}^{Y,CL}$), Scenario 3 clean energy capital subsidies ($\tau_{n,t}^{K,CL}$), and Scenario 4 clean energy labor subsidies ($\tau_{n,t}^{L,CL}$).

United States and China By 2100, the model-implied instruments increase domestic cumulative emissions in both economies, especially in China. All three alternative instruments reduce domestic emissions, but the ranking differs across the two cases. In the United States, the consumption subsidy performs best because it subsidizes clean energy use regardless of whether the energy is produced domestically or imported, inducing the strongest shift away from brown energy. In China, the capital subsidy performs best because the domestic production margin is more powerful: subsidizing capital used in clean energy directly expands domestic clean energy production and pulls resources away from brown energy and other energy-intensive sectors.

All three alternative instruments also reduce global cumulative emissions relative to

Table 8: Unilateral Design: Domestic Cumulative Welfare Changes

	Scenario 1 Model-Implied		Scenario 2 Consumption		Scenario 3 Capital		Scenario 4 Labor	
	2030	2100	2030	2100	2030	2100	2030	2100
Panel C: Domestic Cumulative Welfare ($\widehat{\mathcal{W}}_{n,y}$, % change from BAU)								
United States	-1.16	-0.82	-4.21	-4.83	-5.01	-5.48	-3.11	-4.00
China	-1.93	-1.82	-4.61	-5.33	-9.11	-10.18	0.54	0.26
India	1.32	-2.38	-0.82	-1.07	-2.05	-2.00	0.47	-0.19
European Union	0.32	0.23	-0.35	-0.86	-0.59	-1.09	-0.11	-0.64
Rest of World	0.37	-1.39	-1.24	-2.55	-1.29	-2.58	-1.18	-2.50

Notes: Domestic cumulative welfare changes relative to BAU, measured as the percentage change in cumulative real consumption across all agents from 2015 to the specified year. See notes to Table 7 for scenario descriptions.

the model-implied instruments baseline in both economies. China stands out because its capital subsidy is the only case that brings global cumulative emissions approximately back to BAU by 2100. This result is consistent with a particularly large domestic emission reduction combined with weaker spillovers abroad than under the other two instruments.

The welfare ranking differs sharply from the emission ranking in both economies. Capital subsidies impose the largest welfare costs in the United States and China. These losses are shared by both capitalists and workers in the United States and are especially large for workers in China (see Appendix Table A10). This pattern is consistent with the capital subsidy distorting capital allocation toward clean energy, reducing aggregate efficiency and aggregate capital. In China, the effect is amplified by the central role of manufacturing in production and employment, as documented in Appendix Table A11.

Labor subsidies perform best on the welfare margin. In the United States, they minimize welfare losses because they generate the smallest worker losses among the three alternative instruments. In China, they raise welfare because worker gains more than offset capitalist losses. Appendix Table A10 shows that the labor subsidy increases worker welfare while reducing capitalist welfare, consistent with the labor subsidy relaxing mobility frictions, shifting workers toward clean energy, and inducing capital reallocation indirectly through capital-labor complementarity.

India and European Union The model-implied instruments increase domestic cumulative emissions in India but reduce them in the European Union, consistent with a more stringent baseline policy stance in the latter. All three alternative instruments reduce domestic emissions in both economies, and the differences across instruments are small relative to those in the United States and China.

In contrast to the United States and China, unilateral reform in India and the European Union raises global cumulative emissions relative to the model-implied instruments baseline. In both economies, the subsidy lowers the domestic energy price and expands non-energy production. Through international trade linkages, this expansion raises output and energy demand abroad enough to offset the domestic emission reductions.

Capital subsidies again impose the largest welfare costs, but for different reasons. In India, they direct resources toward clean energy despite India's weaker specialization in that sector. In the European Union, they redirect resources away from services, which account for a large share of domestic production, toward a relatively small clean energy sector (see Appendix Table A11). Labor subsidies perform best on the welfare margin in both economies. In India, they improve welfare at shorter horizons because worker gains outweigh capitalist losses, although the net welfare gain disappears by 2100. In the European Union, they minimize welfare losses because they generate the smallest worker losses among the three alternative instruments (see Appendix Table A10).

Summary Two conclusions emerge. First, the global effect of unilateral reform differs sharply across economies: reallocating subsidy resources lowers global emissions relative to the model-implied instruments baseline for the United States, China, and India, but raises them for the European Union and the Rest of World. Second, there is a clear tradeoff between emissions and welfare. Capital subsidies can generate large domestic emission reductions but often impose the largest welfare costs because they distort capital allocation and reduce aggregate capital. By contrast, labor subsidies perform best on the welfare margin in most cases and even raise welfare in China, while still delivering substantial domestic emission reductions. Overall, effective climate policy design depends on economy-specific production structure, international spillovers, and the nature of factor adjustment frictions.

Simultaneous Design Appendix A.9 presents results for the simultaneous policy design, in which all five economies reallocate their model-implied subsidy expenditure toward the same alternative instrument at the same time. The qualitative findings are broadly consistent with the unilateral results. Capital subsidies tend to deliver the largest global emission reductions but also the largest welfare costs, particularly in China, while labor subsidies minimize welfare losses and improve welfare in China and India. The main difference is that simultaneous reallocation pools fiscal effort across economies, produces larger global emission reductions, and eliminates the adverse spillovers that arise when economies act alone. The ranking of instruments by emission reduction and welfare nevertheless continues to depend on economy-specific economic structure.

5 Conclusion

This paper develops a dynamic general equilibrium model to study climate policies under imperfect capital and labor mobility. When factor mobility is imperfect, clean energy subsidies can raise global emissions: the expansion in aggregate output through production linkages and capital accumulation outweighs the decline in emissions per unit of output.

Two conclusions emerge. First, the global effect of unilateral climate policy differs sharply across economies: clean energy subsidies are more likely to raise global emissions when factor reallocation is sluggish, input-output linkages are strong, and the implementing economy is large enough to affect world prices. Conversely, in economies with more mobile capital and labor, clean energy subsidies are more likely to reduce emissions. Second, instruments that deliver larger emission reductions tend to impose larger welfare costs, with labor subsidies alleviating this tension most effectively. These findings suggest that factor adjustment frictions should be an explicit input to international climate policy design.

Several natural extensions remain. One is to characterize dynamically optimal climate policy under adjustment frictions and international linkages. Others include incorporating strategic interaction across governments and introducing heterogeneity in household asset ownership and intergenerational tradeoffs.

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A Appendix

A.1 Description of Equilibrium Variables

In this section, I provide detailed descriptions of the sets of parameters Θ, prices $\mathbb{P}$, and endogenous variables $\mathbb{Y}$ shown in Section 2.8. The variables in the set of prices $\mathbb{P}$ are the shown in Table A1. The variables in the set of parameters Θ are shown in Table A2. The variables in the set of endogenous variables $\mathbb{Y}$ are shown in Table A3.

Table A1: List of Prices and Taxes

Variable	Description
$p^{j}_{mn,t}$	Price in economy m for variety produced in sector j of economy n at time t
$P^{M,j}_{n,t}$	Price of material in sector j of economy n at time t
$P^{E}_{n,t}$	Energy price index in economy n at time t
$s^{h}_{n,t}$	Before-tax price of fossil fuel type h in economy n at time t
$s_{n,t}$	Price of a fossil fuel input in economy n at time t
$P_{n,t}$	Price of the final good in economy n at time t
$P^{j}_{n,t}$	Sector-level price index for sector j in economy n at time t
$w^{j}_{n,t}$	Wage in sector j of economy n at time t
$r^{j}_{n,t}$	Rental rate of capital in sector j of economy n at time t
$R_{n,t+1}$	Gross interest rate in economy n at time $t+1$
$\tau^{K,B}_{n,t}$	Capital tax on the brown energy sector in economy n at time t
$\tau^{K,CL}_{n,t}$	Capital tax on the clean energy sector in economy n at time t
$\tau^{L,B}_{n,t}$	Labor tax on the brown energy sector in economy n at time t
$\tau^{L,CL}_{n,t}$	Labor tax on the clean energy sector in economy n at time t
$\tau^{Y,B}_{n,t}$	Consumption tax on the brown energy sector in economy n at time t
$\tau^{Y,CL}_{n,t}$	Consumption tax on the clean energy sector in economy n at time t
$\tau^{f,h}_{n,t}$	Consumption tax specific to fossil fuel type h in economy n at time t

Table A2: List of Model Parameters

Variable	Description
$\kappa_n^{j,k}$	Bilateral labor mobility costs
ρ_L	Dispersion of idiosyncratic non-pecuniary preferences
ρ_K	Dispersion of capital efficiencies
ρ_Φ	Dynamic capital allocation externality
η_1 (η_2)	Elasticity of substitution between energy goods (fossil fuels)
β (σ)	Discount factor (Elasticity of substitution across varieties)
δ_n^K ($\delta_{t-\tau}^C$)	Economy-specific capital depreciation (Carbon stock depreciation rate)
ξ_n	Environmental damage in economy n
λ_T	Carbon-temperature conversion
α_n	Non-energy consumption share in economy n
α_n^j	Non-energy consumption share by sector j in economy n
$\gamma_n^{j,V}$	Value added share for sector j in economy n
$\gamma_n^{j,k}$	Material input share for sector j in economy n
$\psi_n^{j,L}$	Labor input share for sector j in economy n
$\psi_n^{j,K}$	Capital input share for sector j in economy n
$\psi_n^{j,E}$	Energy input share for sector j in economy n
$\psi_n^{B,F}$	Fossil fuel input share in economy n
$d_{nm,t}^j$	Exogenous trade costs between economies n and m for sector j at time t
$z_{n,0}^j$	Sector-specific TFP for sector j in economy n at baseline
μ_n^j	Non-pecuniary benefits for sector j in economy n
$a_{n,t}^j$	Capital efficiency for sector j in economy n
a_n^j	Time-invariant component of capital efficiency for sector j in economy n
φ_n^h	Supply cost of fossil fuel by type h
ϕ^h	Carbon emission by fossil fuel type h
E^L	Land-change emissions
ω_n^h	Weight on fossil fuels for fuel type h in economy n
$\alpha_n^B(\alpha_n^{CL})$	Weights on Brown (Clean) Energy in economy n
w_n^{NE}	Non-employment income in economy n

Table A3: List of Endogenous Variables

Variable	Description
$C^K_{n,t}$	Consumption of the representative capitalist in economy n at time t
$c^j_{n,t}$	Consumption of a worker in sector j of economy n at time t
$W_{n,t+1}$	Aggregate capital of the representative capitalist in economy n at time $t+1$
$A_{n,t+1}$	Risk-free assets held by capitalists in economy n at time $t+1$
$Y_{n,t}$	Output of final goods in economy n at time t
$L_{n,t}$	Aggregate supply of labor in economy n at time t
$L^j_{n,t}$	Supply of labor in sector j of economy n at time t
$K_{n,t}$	Aggregate stock of capital in economy n at time t
$K^j_{n,t}$	Stock of capital in sector j of economy n at time t
$F_{n,t}$	Aggregate supply of fossil fuels in economy n at time t
$F^h_{n,t}$	Supply of fossil fuel type h in economy n at time t
$y^j_{mn,t}$	Output produced in sector j of economy n and consumed in economy m
$Y^j_{n,t}$	Sectoral output in sector j of economy n at time t
$C^j_{n,t}$	Consumption of sectoral goods from sector j in economy n at time t
$Q^{j,E}_{n,t}$	Energy input in sector j of economy n at time t
$M^j_{n,t}$	Material input in sector j of economy n at time t
$M^{j,k}_{n,t}$	Material input from sector k used in sector j of economy n at time t
$\pi^j_{nm,t}$	Share of sector j expenditure of economy n spent on sector j of economy m
$E_{n,t}$	Total carbon emissions due to fossil fuel production in economy n at time t
E_t	Global CO_2 emissions at time t
$\mathcal{C}_t$	Global stock of carbon dioxide in the atmosphere at time t
$\mathbb{T}(\mathcal{C}_t)$	Global temperature at time t relative to the global temperature at time T_{min}
$\mathcal{D}_n(\mathcal{C}_t)$	Environmental quality in economy n at time t
$\Phi^j_{n,t}$	Share of aggregate capital allocated to sector j in economy n
$V^j_{n,t}$	Expected lifetime utility of a worker in sector j of economy n at time t
$V^K_{n,t}$	Lifetime utility of the representative capitalist in economy n at time t
$m^{j,k}_{n,t}$	Share of workers moving from sector j to sector k in economy n at time t
$\Omega_{n,t}$	Lump-sum government transfer in economy n at time t

A.2 Description of Labor Force Surveys and Variable Construction

In this section, I describe the labor force surveys used in the analysis which track the movement of individuals across sectors and employment status over time and report individuals' income as well as individual characteristics such as gender, age, and education.

United States The data for the United States is sourced from the IPUMS Current Population Survey (CPS) Annual Social and Economic Supplement (ASEC) for the years 2000 to 2023. The dataset is annual and includes variables such as individual panel ID, year, sex, age, marital status, individual weight, employment status, sector of employment, sector of employment last year, education, income, and unemployment benefits. The sample includes individuals aged 25 to 65, excluding retirees and individuals unable to work. Income values exceeding 99990 or less than zero are treated as missing. The dataset tracks employment status (unemployed, employed) and movements across sectors.

China The data for China comes from the China Family Panel Studies (CFPS), covering the years 2010 to 2020. The survey is biennial, and the data includes variables such as sector of origin (sector of employment last year) and sector of destination (sector of employment), with sectors categorized into Agriculture, Brown, Manufacturing, Clean Energy, and Services. The labor mobility shares are computed by annualizing the labor flows using the method of Artuç et al. (2010). The dataset also includes individual-level data such as income and employment status (unemployed, employed).

India The data for India is sourced from the Periodic Labour Force Survey (PLFS) for the years 2018 to 2020. The survey is quarterly, and the data includes variables such as quarter, visit, stratum, subsample, individual panel ID, years of education, sector, income, individual weight, employment status, age, hours worked per week, wage, and sex. The sample includes individuals aged 25 to 65. Employment status is derived based on a combination of variables indicating self-employment, wage employment, unemployment, and inactivity. The labor mobility shares are computed by annualizing the labor flows using the method of Artuç et al. (2010).

France The data for France is sourced from the Continuous Labour Force Survey (EEEC) for the years 2014 to 2020. The survey is quarterly, and the data includes variables such as individual panel ID, year, quarter, age, individual weight, birth year, birth month, start year, sector, employment status, sex, education, income, and unemployment benefits. The sample includes individuals aged 25 to 65 who are either employed or actively seeking employment. Sector classification follows NAF Rev. 2, which aligns with ISIC Rev. 4. The labor mobility shares are computed by annualizing the labor flows using the method of Artuç et al. (2010).

Argentina The data for Argentina is derived from the Permanent Household Survey (EPH) for the years 2003 to 2019. The survey is quarterly, and the data includes variables such as year, quarter, age, individual weight, income, wage, sector, education, employment status inactivity status, sex, job search, and individual panel ID. The sample includes individuals aged 25 to 65 who are either employed or actively seeking employment. Sector classification follows CAES-Mercosur, which aligns with ISIC Rev. 4. The labor mobility shares are computed by annualizing the labor flows using the method of Artuç et al. (2010).

Australia The dataset for Australia is the the Household, Income and Labour Dynamics in Australia (HILDA) Survey, covering 2002 to 2022. The survey is annual and provides longitudinal information on individual movements across sectors and employment status, including income and personal characteristics. Variables include individual panel ID, age, sex, marital status, individual weight, sector, education, income, employment status, school years, and annual work hours. The sample includes individuals aged 25 to 65 who are either employed or actively seeking employment.

South Korea The data for South Korea is from the Korean Labor & Income Panel Study (KLIPS) for the years 1999 to 2022. The survey is annual, and the data includes variables such as individual panel ID, year, sector, employment status, income, education, and weekly work hours. The sample includes individuals aged 25 to 65 who are either employed or actively seeking employment. Sector classification follows the 3-digit Korean Standard Industry Classification (KSIC), which aligns with the 3-digit ISIC Rev. 4.

United Kingdom The data for the United Kingdom comes from the Labour Force Survey (LFS) for the years 2016 to 2022. The survey is both quarterly and annual, and the data includes variables such as individual panel ID, year, quarter, age, individual weight, employment status, sector, and income. The sample includes individuals aged 25 to 65 who are either employed or actively seeking employment. Sector classification follows the 2-digit UK Standard Industry Classification (SIC), which aligns with the 2-digit ISIC Rev. 4.

A.3 Detailed Estimation Procedure for Parameters

This section discusses the estimation procedure for parameters not discussed in the main text. Throughout the appendix, I use $\tau^{j}_{n,t}$ as shorthand for the consumption tax $\tau^{Y,j}_{n,t}$ and $\tau^{h}_{n,t}$ as shorthand for the fossil-fuel tax $\tau^{f,h}_{n,t}$, when the tax type is unambiguous from context.

Energy Demand Elasticity I estimate the energy demand elasticity using the following specification (see Appendix A.13.3 for the derivation):

$$\ln\left(\frac{Y^{CL}_{n,t}}{Y^{CL}_{n,t-1}}\right) - \ln\left(\frac{Y^{B}_{n,t}}{Y^{B}_{n,t-1}}\right) = -\eta_1\left(\ln\left(\frac{P^{CL}_{n,t}}{P^{CL}_{n,t-1}}\right) - \ln\left(\frac{P^{B}_{n,t}}{P^{B}_{n,t-1}}\right)\right) + \delta_t + \iota_{n,t} \tag{22}$$

where δ_t is the year fixed effects, and $\iota_{n,t}$ is the residual term.[57] In words, I regress the difference in output growth between the brown and clean energy sectors on the difference in price growth between these two sectors.

[57] In particular, $\delta_t \equiv \frac{\eta_1}{|\mathcal{N}|}\sum_{m\in\mathcal{N}} \ln\left(\frac{(1+\tau^{CL}_{m,t})(1+\tau^{B}_{m,t-1})}{(1+\tau^{B}_{m,s})(1+\tau^{CL}_{m,t-1})}\right)$ and $\iota_{n,t} \equiv -\eta_1 \ln\left(\frac{(1+\tau^{CL}_{m,t})(1+\tau^{B}_{m,t-1})}{(1+\tau^{B}_{m,s})(1+\tau^{CL}_{m,t-1})}\right) - \frac{\eta_1}{|\mathcal{N}|}\sum_{m\in\mathcal{N}} \ln\left(\frac{(1+\tau^{CL}_{m,t})(1+\tau^{B}_{m,t-1})}{(1+\tau^{B}_{m,s})(1+\tau^{CL}_{m,t-1})}\right)$.

Table A4: Estimation Result for Energy Demand Elasticity ($-\eta_1$)

Dependent: $\ln\left(\frac{Y^{CL}_{n,t}}{Y^{CL}_{n,t-1}}\right) - \ln\left(\frac{Y^{B}_{n,t}}{Y^{B}_{n,t-1}}\right)$	(1) OLS	(2) IV: $\ln\left(\frac{r^{CL}_{n,t-1}}{r^{CL}_{n,t-2}}\right) - \ln\left(\frac{r^{B}_{n,t-1}}{r^{B}_{n,t-2}}\right)$
$\ln\left(\frac{P^{CL}_{n,t}}{P^{CL}_{n,t-1}}\right) - \ln\left(\frac{P^{B}_{n,t}}{P^{B}_{n,t-1}}\right)$	-0.37**	-2.28***
	(0.19)	(0.77)
Year FE	YES	YES
Observations	630	574
First Stage F-Statistic		9.89
R-squared	0.262	

Notes: The dependent variable is the one-period lagged (log) difference in the growth rates of clean energy sector output and brown energy sector output. Column 2 uses the difference in the rental rates of capital as the instrument. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To estimate the specification (22), I use the Socio-Economic Accounts (SEA) of the WIOD 2016 release, which provides sector-level output and output price indices normalized to 2010 levels. To address the potential endogeneity issue where prices may be correlated with the demand shifters in the residuals, I use the one-period lagged (log) difference between the growth rates of rental rates of capital in the clean energy sector and the brown energy sector, $\ln\left(\frac{r^{CL}_{n,t-1}}{r^{CL}_{n,t-2}}\right) - \ln\left(\frac{r^{B}_{n,t-1}}{r^{B}_{n,t-2}}\right)$, as an instrument. The parameter η_1 is identified under the assumption that variations in growth rates of rental rates of capital influence the relative output of the energy sectors only by affecting the production costs across these sectors and not the demand for energy. The results are presented in Table A4. Column 2 displays the preferred estimate for $-\eta_1$, yielding $\eta_1 = 2.28$, which suggests that brown and clean energy goods are substitutes.[58]

[58]The first-stage F-statistic of 9.89 falls just below the conventional threshold of 10. The substantial gap between the OLS estimate (-0.37) and the IV estimate (-2.28) indicates that the instrument moves the estimate away from the biased OLS value, and the IV point estimate falls within standard literature ranges for the clean-brown energy substitution elasticity.

Fossil Fuel Elasticity I estimate the elasticity of substitution across fossil fuels using the following specification (see Appendix A.13.4 for the derivation):

$$\ln\left(\frac{f^k_{n,t}}{f^h_{n,t}}\right) = -\eta_2 \ln\left(\frac{s^k_{n,t}}{s^h_{n,t}}\right) - \eta_2 \ln\left(\frac{1+\tau^k_{n,t}}{1+\tau^h_{n,t}}\right) + \delta^k_n + \tilde{\iota}^k_{n,t} \tag{23}$$

where δ^k_n are country-fuel fixed effects defined as $\delta^k_n \equiv \ln\left(\frac{\omega^k_n}{\omega^h_n}\right)$ for some base fuel type h; and, $\tilde{\iota}^k_{n,t}$ is an error term. In words, I regress the relative fossil fuel production on the relative supply costs and fuel-specific taxes. For the estimation, I consider crude oil (o) as the base fuel type.

Table A5: Estimation Result for Fossil Fuel Elasticity ($-\eta_2$)

Dependent: $\ln\left(\frac{f^k_{n,t}}{f^o_{n,t}}\right)$	(1)	(2)
$\ln\left(\frac{s^k_{n,t}}{s^o_{n,t}}\right)$	-2.56***	-2.67***
	(0.41)	(0.31)
Country-Fuel FE	YES	YES
Year FE	NO	YES
Observations	1,219	1,219
R-squared	0.985	0.985

Notes: The dependent variable is the (log) difference in the production of fuel type h relative to the production of crude oil (o). Clustered standard errors (country-fuel) are in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To estimate the specification (23), I use the IMF Fossil Fuel Subsidies Data (Black et al., 2023) which provides information on supply costs ($s^h_{n,t}$) and retail prices ($(1+\tau^h_{n,t})s^h_{n,t}$) for 168 countries from 2015 to 2019. The extraction costs ($\{s^h_{n,t}\}$) for oil, natural gas, and coal are set equal to the supply costs of gasoline (bil. USD/bil. m^3), natural gas (bil. USD/bil. m^3), and coal (bil. USD/Gt). The fuel-specific taxes ($\{1+\tau^h_{n,t}\}$) are computed by dividing the retail prices by the supply costs. Production levels of crude oil, natural gas, and coal ($\{f^h_{n,t}\}$) are sourced from Ritchie and Rosado (2017) for 169 countries from 2015 to 2019, with units of production for crude oil, natural gas, and coal provided in billion m^3, billion m^3, and billion tons, respectively.

The results are presented in Table A5. Column 1 shows the baseline estimate for $-\eta_2$,

yielding $\eta_2 = 2.56$. Column 2 presents the preferred estimate for $-\eta_2$ when I additionally control for year fixed effects. The estimate is 2.67, showing minimal change from the estimate in Column 1. These estimates suggest that fossil fuel inputs are substitutes.

Non-Employment Income To estimate the non-employment income w_n^{NE} as a share of average labor income, I employ the panel labor force surveys described above which contain information on income of non-employed individuals. I begin by estimating the Mincerian regression of (log) wages on sector-year fixed effects, along with education, age, and gender fixed effects, weighted by individual weights to obtain residualized sector-year-level wages δ_{jt} (Mincer, 1974):

$$\underbrace{\omega_{n,t}^{j,i}\left(\log(w_{n,t}^{j,i})\right)}_{\text{Weighted (log) wage in sector } j} = \underbrace{\omega_{n,t}^{j,i}}_{\text{Weight}}\left(\underbrace{\delta_{jt}}_{\text{Sector-Year FE}} + \underbrace{\delta_e}_{\text{Education FE}} + \underbrace{\delta_a}_{\text{Age FE}} + \underbrace{\delta_g}_{\text{Gender FE}} + \underbrace{\varepsilon_{n,t}^{j,i}}_{\text{Residual}}\right)$$

Next, I compute the average residualized wage of employed workers, weighted by employment:

$$\underbrace{L_{n,t}^{j} \cdot \delta_{jt}}_{\text{Weighted (log) residualized wage}} = \underbrace{L_{n,t}^{j}}_{\text{Weight}}\left(\underbrace{\delta_t}_{\text{Year FE}} \mathbb{1}[j \neq \text{NE}] + \underbrace{\delta_t}_{\text{Year FE}} \mathbb{1}[j = \text{NE}] + \underbrace{\varepsilon_{n,t}^{j}}_{\text{Residual}}\right)$$

Finally, I estimate the non-employment income parameter w_n^{NE} by dividing the residualized non-employment income by the average residualized wage of employed workers and averaging over time.

Bilateral Trade Costs The gravity equation leads to the following regression specification:

$$\ln\left((\pi_{nm,t}^{j})^{\frac{1}{1-\sigma}}\right) = \ln(d_{nm,t}^{j}) + \delta_{m,t}^{j} + \delta_{n,t}^{j} + \varepsilon_{nm,t}^{j}$$

where $\delta_{m,t}^{j}$ and $\delta_{n,t}^{j}$ are, respectively, exporter-sector-year fixed effects and importer-sector-year fixed effects;[59] $\varepsilon_{nm,t+1}^{j}$ is the residual term. The bilateral trade shares ($\pi_{nm,t}^{j}$) are

[59]Specifically, $\delta_{m,t}^{j} \equiv \ln(mc_{m,t}^{j}) - \ln(z_{m,t}^{j})$ and $\delta_{n,t}^{j} \equiv \frac{1}{\sigma-1}\ln\left(\sum_{i\in\mathcal{N}}(z_{i,t}^{j})^{\sigma-1}(d_{ni,t}^{j}mc_{i,t}^{j})^{1-\sigma}\right)$.

sourced from the WIOD 2016 Release. Refer to the Appendix A.8 for the detailed estimates.

Other Parameters The capital depreciation rates (δ_n^K) are taken from the Penn World Table 10.01 (Feenstra et al., 2015) for the year 2019, which is the most recent year available. For country aggregates, I take the GDP-weighted average; the values range from 0.042 to 0.058. For detailed estimates, refer to the Appendix A.8.

Supply costs of fossil fuels (φ_n^h) are estimated based on the IMF Fossil Fuel Subsidies Data (Black et al., 2023) for the year 2015. The extraction costs for oil, natural gas, and coal are set equal to the supply costs of gasoline (bil. USD/bil. m^3), natural gas (bil. USD/bil. m^3), and coal (bil. USD/Gt). See Appendix A.8 for detailed estimates.

Carbon emissions data from years 1850 to 2022 are sourced from Ritchie and Roser (2020), with non-industrial and land-change emissions (E^L) computed as the average global non-industrial and land-change emissions over this period. The estimate for E^L is 7.4249 $GtCO_2$ in year 2014. The carbon-temperature conversion parameter (λ_T) is obtained by estimating equation $\mathbb{T}(\mathcal{C}_t) = \lambda_T \log_2\left(\frac{\mathcal{C}_t}{\mathcal{S}}\right)$ using data on global temperature relative to the year 1850 obtained from Met Office Hadley Centre (2024) and global carbon stock computed based on Equation $\mathcal{C}_t = \mathcal{S} + \sum_{\tau=T_{min}}^{t}(1-\delta_{t-\tau}^{C})\left(E_\tau + E^L\right)$. Since the model is calibrated to the 2014 baseline and the quantitative exercises begin in 2015, I use the estimate from the 2014–2022 sample as the preferred calibration target. This yields $\lambda_T = 1.836$ with a robust standard error of 0.043. The model fit and the regression results are presented in Figure A1 and Table A16, respectively.

Consumption expenditure shares (α_n, α_n^j) and input shares (value added ($\gamma_n^{j,V}$), material ($\gamma_n^{j,k}$), labor ($\psi_n^{j,L}$), capital ($\psi_n^{j,K}$), energy ($\psi_n^{j,E}$), fossil fuel ($\psi_n^{B,F}$)) are directly computed using the World Input-Output Table (WIOT) and Socio-Economic Accounts (SEA) of the WIOD 2016 Release for the year 2014, which is the most recent year available. Refer to the Appendix A.8 for detailed estimates.

A.4 Method of Simulated Moments (MSM) Procedure

This section discusses the targeted moments and estimation procedure for parameters estimated using the method of simulated moments.

A.4.1 Targeted Moments

Capital Efficiencies and Non-pecuniary Benefits Capital efficiencies (a_n^j) target real interest rates, capital allocation shares, and economy-level capital compensation in the year 2014. Real interest rates are sourced from International Financial Statistics (IFS), and the capital allocation shares and economy-level capital compensation are calculated using the WIOD 2016 Release. Non-pecuniary benefits (μ_n^j) are matched to sector-level wages in the year 2014 computed using the WIOD 2016 Release.

Productivity Sector-specific total factor productivity ($z_{n,0}^j$) is estimated by targeting bilateral trade shares and sector-level output in 2014, which is the most recent year available in the WIOD 2016 Release. Bilateral trade shares pin down relative sector-level productivity levels across economies for a given sector. To pin down the relative productivities across sectors within each economy, I exploit the positive and monotonic relationship between sector-level output and sector-level productivity. In the estimation, I assume that relative productivity levels across sectors and economies remain constant over time, although I do not assume the global economy is initially in steady state.

Weights on Fossil Fuels and Clean Energy Weights on fossil fuels (ω_n^h) target the production of fossil fuels at the economy level by fuel type sourced from Ritchie and Rosado (2017). The relative fossil fuel demand is given by: $\frac{\omega_n^h}{\omega_n^k} = \left(\frac{\varphi_n^h}{\varphi_n^k}\right)^{\eta_2}\left(\frac{f_{n,t}^h}{f_{n,t}^k}\right)$ which shows that weights are a function of fossil fuel production ($f_{n,t}^k$,$f_{n,t}^h$), elasticity η_2, and the efficiencies (φ_n^k, φ_n^h). Similarly, the energy demand can be re-arranged as follows: $\frac{\alpha_n^{CL}}{\alpha_n^B} = \left(\frac{(1+\tau_{n,t}^{CL})P_{n,t}^{CL}}{(1+\tau_{n,t}^B)P_{n,t}^B}\right)^{\eta_1-1}\left(\frac{P_{n,t}^{CL}Y_{n,t}^{CL}}{P_{n,t}^{CL}Y_{n,t}^B}\right)$. Th equation implies that the weight on the clean energy sector (α_n^{CL}) can be identified by the share of final energy expenditure spent on the clean energy relative to the brown energy sector calculated using the WIOD 2016 Release.

Environmental Damage The environmental damage parameter (ξ_n) is estimated to match the impulse response of real GDP per capita of an economy to a 1 °C increase in global temperature, as estimated by Bilal and Känzig (2024). Specifically, I simulate an unexpected increase in exogenous carbon emissions E_0^X in year 2015 that increases the global temperature by 1 °C in 2015. The shock can be computed as follows: $E_0^X =$

$\mathcal{S} \cdot 2^{\frac{\mathbb{T}(\mathcal{C}_0)}{\lambda_T}} \left(2^{\frac{1}{\lambda_T}} - 1\right)$. I then match the drop in real GDP per capita of each economy, relative to the baseline simulation without the shock, for each economy after 5 years with the estimates of Bilal and Känzig (2024).

A.4.2 Calibration Procedure

The MSM calibration procedure involves the following steps:

1. Initialize Parameters: Start with initial guesses for the parameters Θ^{MSM}.
2. Simulate Model: Run the dynamic general equilibrium model with the initial parameter guesses to generate simulated moments $\mathbb{M}^{\text{Model}}$.
3. Compute Model Moments: Calculate the model moments corresponding to the observed data moments.
4. Minimize Error Function: Adjust the parameters to minimize the distance between the simulated model moments and the observed data moments using the error function:

$$Q_T(\Theta) = \left[\frac{\mathbb{M}_T^{\text{Data}} - \mathbb{M}_T^{\text{Model}}(\Theta)}{\mathbb{M}_T^{\text{Data}}}\right]' I \left[\frac{\mathbb{M}_T^{\text{Data}} - \mathbb{M}_T^{\text{Model}}(\Theta)}{\mathbb{M}_T^{\text{Data}}}\right],$$

   where I is the identity matrix.
5. Iterate: Repeat the simulation and adjustment steps until the parameters converge to values that minimize the error function $Q_T(\Theta)$.
6. Sequential Procedure: The estimation of environmental damage parameters (ξ_n) is performed after the estimation of other parameters. The estimated environmental damage parameter is then fed back into the first step, and the entire procedure is repeated until convergence.

A.5 Initial Conditions

This section describes the initial conditions of the models. The initial conditions of the model are computed for the year 2014 or the most recent year available. The initial

distribution of aggregate capital ($W_{n,0}$) and labor allocations ($L^j_{n,0}$) across sectors and economies are derived from the Socio-Economic Accounts (SEA) of the World Input-Output Database (WIOD) 2016 Release. I assume the labor force ($L_{n,t}$) in each economy remains constant over time.

The global carbon stock ($\mathcal{C}_0$) in the year 2014 is computed based on pre-industrial stock ($\mathcal{S}$), global carbon emissions from years 1850 to 2022 sourced from Ritchie and Roser (2020). The initial environmental quality ($\mathcal{D}_{n,0}$) for each economy is determined by the initial carbon stock and the estimated environmental damage parameter (ξ_n). Given the assumption that only domestic financial asset markets exist in each economy, the initial financial asset position of the representative capitalist in each economy is set to zero, i.e., $A_{n,0} = 0$ for all $n \in \mathcal{N}$.

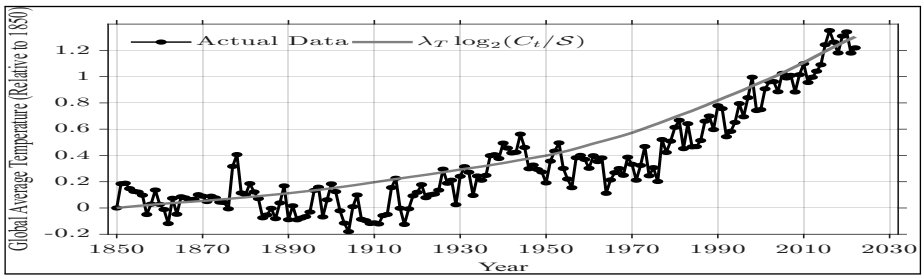
Fossil fuel reserves ($D^h_{n,0}$) are estimated based on Ritchie and Rosado (2017), with units of reserves for crude oil, natural gas, and coal provided in billion m^3, billion m^3, and billion tons, respectively, for the year 2014. Using the same dataset, I compute the average annual discovery from 2010 to 2019 and calculate the reserves assuming new reserves will continue to be discovered at this average annual rate for the next 500 years. This assumption is primarily made to focus on the impacts of environmental policies rather than resource exhaustion; this assumption amplifies the environmental benefits of environmental policies, as larger reserves imply that emissions will continue unless the global economy achieves a clean transition. Estimates are presented in Appendix A.

In the baseline calibration, sector-specific consumption taxes ($\tau^j_{n,0}$) are set to zero for two main reasons. First, countries primarily report value-added taxes (VAT) or sales taxes, which are applied uniformly across all or most products. Therefore, incorporating these taxes into the model would not meaningfully alter the distribution of economic activities across sectors. Second, taxes introduce distortions in the model. Removing uniform consumption taxes and redistributing through government transfers would enhance welfare by reducing these distortions, which is not the primary focus of this paper.

Fuel-specific tax rates are estimated based on the measures of supply costs of fossil fuels (φ^h_n) and retail prices of fossil fuels sourced from IMF Fossil Fuel Subsidies Data (Black et al., 2023) for the year 2014. Fuel-specific tax rates are computed as the difference between the retail price after deducting value-added tax (VAT) of fossil fuels (gasoline, natural gas, coal) and the supply costs of fossil fuels.

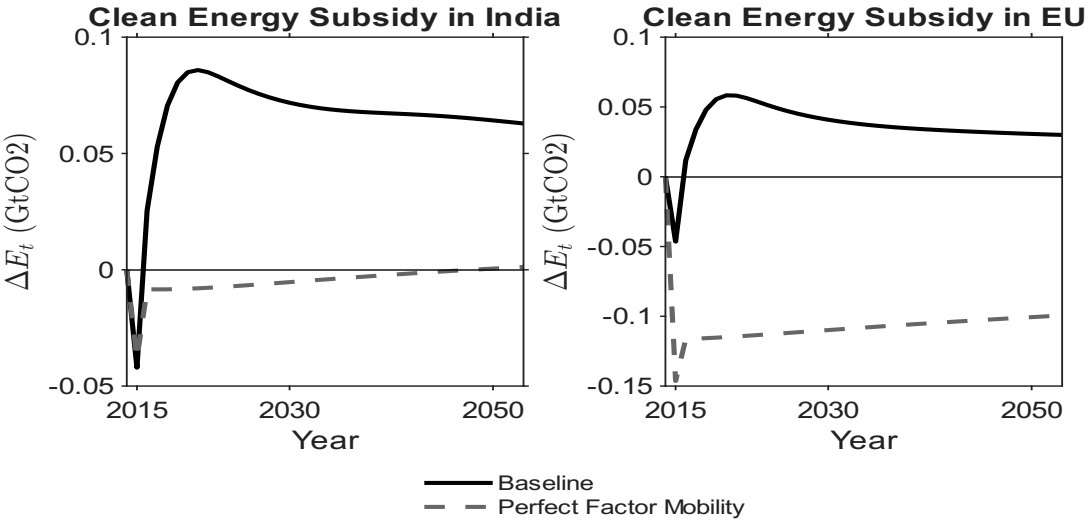
A.6 Figures

Figure A1: Global Average Temperature (°C) relative to Year 1850



Notes: The figure compares actual temperature (°C) data with the predicted temperature (°C) implied by the carbon-temperature conversion parameter (λ_T). The prediction is based on the equation $\mathbb{T}(\mathcal{C}_t) = \lambda_T \log_2(\mathcal{C}_t/\mathcal{S})$, where $\mathbb{T}(\mathcal{C}_t)$ is the global temperature, $\mathcal{C}_t$ is the carbon stock and $\mathcal{S}$ is the pre-industrial carbon stock. The fitted series uses the preferred 2014–2022 estimate, $\lambda_T = 1.836$, with a robust standard error of 0.043.
Sources: Ritchie and Roser (2020) (carbon emissions), Met Office Hadley Centre (2024) (temperature).

Figure A2: Model Comparison:
The Impact of Clean Energy Subsidies on Global Emissions



Notes: The left (right) panel displays the impact of an unanticipated and permanent 10 percentage point reduction in the consumption tax on the clean energy sector in India (European Union) in 2015 on global emissions. The figure compares the baseline model with the model with perfect factor mobility. All values are relative to the BAU case where the policy is not implemented.

A.7 Tables

Table A6: List of Economies and Countries in Country Aggregates

Economy	Country
United States (US)	United States
China (CHN)	China
India (IND)	India
European Union (EU)	Austria, Belgium, Bulgaria, Croatia, Czech Republic, Denmark, Germany, Estonia, Greece, Spain, France, Ireland, Italy, Cyprus, Latvia, Lithuania, Luxembourg, Hungary, Malta, Netherlands, Poland, Portugal, Romania, Slovenia, Slovakia, Finland, Sweden
Rest of the World (ROW)	Australia, Brazil, Canada, Switzerland, United Kingdom, Indonesia, Mexico, Malta, Norway, Russia, Turkey, Japan, South Korea, Taiwan

Notes: The European Union includes 27 member countries, and the Rest of the World (ROW) includes selected major economies and a composite of other countries.
Source: World Input-Output Database (WIOD) 2016 Release.

Table A7: Industry Classification - Clean Energy and Brown Energy

Sector	Industry
Clean Energy (CL)	Electricity, gas supply
Brown Energy (B)	Mining and quarrying Manufacture of coke, petroleum

Notes: Industries are categorized into sectors based on the World Input-Output Database 2016 Release classification. The clean energy sector includes industries related to renewable energy and technology, while the brown energy sector includes industries related to fossil fuels.

Table A8: Industry Classification - Agriculture, Manufacturing, Services

Sector	Industries
Agriculture (AG)	Crop and animal production; Forestry and logging; Fishing and aquaculture
Manufacturing (MN)	Manufacture of food, beverages, tobacco; Textiles, apparel, leather; Wood and paper products; Printing and recorded media; Chemicals and pharmaceuticals; Rubber and plastic products; Non-metallic, basic, and fabricated minerals; Computer and electronics; Electrical equipment; Motor vehicles
Services (SR)	Motor vehicle trade and repair; Wholesale and retail trade; Land, water, and air transport; Warehousing and transport support; Postal and courier activities; Accommodation and food services; Publishing, motion picture, and video production; Telecommunications; Computer programming and consultancy; Financial services, insurance, and pension funding; Real estate; Legal, accounting, and management consultancy; Architectural and engineering activities; Scientific research and development; Advertising and market research; Other professional and technical activities; Administrative and support services; Public administration and defence; Education, health, and social work; Other service activities; Activities of extraterritorial organizations

Notes: Industries are categorized into sectors based on the World Input-Output Database (WIOD) 2016 Release classification.

Table A9: Threshold for Market-Based EPS

Score	Market based policies				
	CO_2 certificate	Renewable energy certificates (%)	CO_2 taxes (USD/tonne CO_2)	NO_x taxes (USD/tonne NO_x)	SO_x taxes (USD/tonne SO_x)
0	0	0	0	0	0
1	5	0.025	5	45	58
2	15	0.065	15	113.5	148
3	25	0.095	25	160.5	212
4	35	0.125	35	207.5	276
5	45	0.155	45	254.5	340
6	50	0.17	50	278	372

Source: Kruse et al. (2022).

Table A10: Welfare Decomposition: Capitalist vs. Worker

	Recovered		Consumption		Capital		Labor	
	2030	2100	2030	2100	2030	2100	2030	2100
Panel A: Capitalist Welfare ($\widehat{\mathcal{W}}^K_{n,y}$, % of total BAU welfare)								
United States	-0.68	-0.77	-2.18	-2.40	-2.20	-2.26	-2.19	-2.60
China	-2.88	-2.91	-2.86	-3.11	-2.96	-3.42	-2.84	-2.88
India	6.13	5.50	-1.76	-1.91	-1.52	-1.55	-2.01	-2.27
European Union	1.11	1.04	-0.29	-0.36	-0.28	-0.34	-0.31	-0.37
Rest of World	1.93	1.44	-0.30	-0.60	-0.30	-0.59	-0.31	-0.61
Panel B: Worker Welfare ($\widehat{\mathcal{W}}^L_{n,y}$, % of total BAU welfare)								
United States	-0.48	-0.05	-2.02	-2.43	-2.80	-3.21	-0.92	-1.40
China	0.95	1.09	-1.75	-2.22	-6.15	-6.75	3.38	3.13
India	-4.81	-7.88	0.95	0.84	-0.53	-0.44	2.47	2.08
European Union	-0.80	-0.81	-0.05	-0.51	-0.31	-0.75	0.20	-0.27
Rest of World	-1.55	-2.82	-0.94	-1.95	-0.99	-1.99	-0.87	-1.89

Notes: Panel A reports the capitalist component of welfare change, expressed as a percentage of total BAU welfare. Panel B reports the worker component, expressed in the same units. Since both panels use the same denominator as total welfare in Table 8, their entries sum to the total welfare change.

Table A11: Sector Structure and Specialization Measures in 2014

		US	China	India	EU	ROW
Panel A: Production Structure						
Manufacturing	Domestic output (%)	18.1	47.5	33.1	26.2	26.6
	Employment (%)	8.3	19.5	12.7	14.9	12.4
	World output (%)	4.5	9.1	1.5	4.3	9.4
Services	Domestic output (%)	74.5	37.4	49.1	65.9	61.9
Clean Energy	Domestic output (%)	2.0	4.4	2.2	3.9	3.6
Panel B: Specialization and Capital Efficiency						
Clean Energy	RCA_n^{CL}	0.60	1.32	0.67	1.15	1.08
Capital efficiency	Services	1.43	1.19	1.10	1.46	1.22
	Clean Energy	0.99	0.92	1.18	0.86	0.97

Notes: Values are computed from sectoral allocations in 2014 as percentages of domestic output, employment, and world output. Relative comparative advantage is computed as $\text{RCA}_n^{CL} = (GR_n^{CL}/GR_n)/(GR_{\text{world}}^{CL}/GR_{\text{world}})$, where GR_n^{CL} is the gross output of the clean energy sector; values above (below) one indicate comparative advantage (disadvantage). Capital efficiency is a_n^j normalized by its economy-specific mean.
Source: World Input-Output Database 2016 Release.

A.8 Parameter Estimates

In this section, I provide the estimates of parameters described in Section 3.

Table A12: Capital Depreciation Rates (δ_n^K) for 2014

Country/Aggregate	Depreciation Rate (δ_n^K)
United States (US)	0.043
China	0.056
India	0.055
European Union (EU)	0.040
Rest of the World (ROW)	0.046

Source: Penn World Table 10.01 (Feenstra et al., 2015).

Table A13: Supply Costs of Fossil Fuels (2015)

Economy	Crude Oil (φ_n^o)	Natural Gas (φ_n^{CL})	Coal (φ_n^c)
United States (US)	1489.4	0.188	61.7
China	1358.3	0.443	78.0
India	1358.3	0.407	97.2
European Union (EU)	1329.8	0.453	104.2
Rest of the World (ROW)	1383.9	0.329	97.0

Notes: The supply costs for crude oil, natural gas, and coal are in units of billion USD per billion m^3 for gasoline, billion USD per billion m^3 for natural gas, and billion USD per gigaton for coal, respectively. Source: IMF Fossil Fuel Subsidies Data (Black et al., 2023).

Table A14: Fuel-Specific Taxes for Fossil Fuels (2015)

Economy	Crude Oil ($\tau_{n,0}^{f,o}$)	Natural Gas ($\tau_{n,0}^{f,g}$)	Coal ($\tau_{n,0}^{f,c}$)
United States (US)	0.0708	0.0873	-0.0446
China	0.5110	0.2792	-0.1077
India	0.6845	-0.0978	-0.0343
European Union (EU)	1.4097	0.2991	-0.0613
Rest of the World (ROW)	0.6755	0.2022	-0.0162

Notes: Fuel-specific tax rates are computed as the difference between the retail price after deducting value-added tax (VAT) of fossil fuels (gasoline, natural gas, coal) and the supply costs of fossil fuels. Source: IMF Fossil Fuel Subsidies Data (Black et al., 2023).

Table A15: Fossil Fuel Reserves ($D_{n,0}^h$) for 2019

Economy	Crude Oil ($D_{n,0}^o$)	Natural Gas ($D_{n,0}^g$)	Coal ($D_{n,0}^c$)
United States (US)	469.52	746518.90	282.52
China	170.68	233563.33	3490.82
India	18.10	42684.44	1851.97
European Union (EU)	10.01	19026.53	712.77
Rest of the World (ROW)	4561.45	1836393.44	6400.67

Notes: The units of reserves for crude oil, natural gas, and coal are in billion m^3, billion m^3, and billion tons, respectively. Estimates are based on Ritchie and Rosado (2017) for the baseline year 2019. I compute the average annual discovery from 2010 to 2019; then, I calculate the reserves assuming new reserves will continue to be discovered at this average annual rate for the next 500 years.

Table A16: Regression Results: Carbon-Temperature Conversion Parameter (λ_T)

Dependent: Global Average Temperature $(\mathbb{T}(\mathcal{C}_t))$	(1) 1850 - 2022	(2) 2010 - 2022	(3) 2014 - 2022
$\log_2(\mathcal{C}_t)$	1.556***	1.786***	1.836***
	(0.037)	(0.043)	(0.043)
Observations	173	12	9
R-squared	0.919	0.994	0.996

Notes: Data on global temperature $\mathbb{T}(\mathcal{C}_t)$ in celsius (°C) relative to the year 1850 obtained from Met Office Hadley Centre (2024) and global carbon emissions from Ritchie and Roser (2020). Global carbon stock ($\mathcal{C}_t$) computed based on equation $\mathbb{T}(\mathcal{C}_t) = \lambda_T \log_2(\mathcal{C}_t/\mathcal{S})$. Columns 1, 2, and 3 present regression results for the periods 1850–2022, 2010–2022, and 2014–2022, respectively. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A17: Non-energy Consumption Share (α_n)

	US	China	India	EU	ROW
α_n	0.9632	0.9872	0.9737	0.9680	0.9659

Notes: α_n represents the Non-energy consumption share in economy n.
Source: WIOD 2016 Release. US: United States, EU: European Union, ROW: Rest of the World.

Table A18: Non-energy Share by Sector (α_n^j)

Sector	US	China	India	EU	ROW
AG	0.0051	0.0462	0.1157	0.0130	0.0337
MN	0.1534	0.2741	0.2796	0.1842	0.1978
SR	0.8416	0.6797	0.6047	0.8027	0.7685

Notes: α_n^j represents the Non-energy share of sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A19: Value Added Share ($\gamma_n^{j,V}$)

Sector	US	China	India	EU	ROW
AG	0.4978	0.6156	0.7844	0.4758	0.6458
MN	0.3907	0.2849	0.2827	0.3461	0.3814
SR	0.6405	0.4960	0.7043	0.5861	0.6264
CL	0.8279	0.7805	0.7434	0.7212	0.8414
B	0.8155	0.7001	0.8502	0.5879	0.7399

Notes: $\gamma_n^{j,V}$ represents the value added share for sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A20: Material Input Share ($\gamma_n^{j,k}$)

From (k) $\rightarrow$ To (j)	US	China	India	EU	ROW
AG $\rightarrow$ AG	0.4146	0.3585	0.4965	0.2744	0.3416
MN $\rightarrow$ AG	0.2652	0.5019	0.1755	0.3834	0.3462
SR $\rightarrow$ AG	0.3202	0.1396	0.3280	0.3422	0.3122
AG $\rightarrow$ MN	0.0854	0.0875	0.1062	0.0650	0.0886
MN $\rightarrow$ MN	0.5878	0.7389	0.5618	0.5785	0.6269
SR $\rightarrow$ MN	0.3268	0.1736	0.3320	0.3565	0.2844
AG $\rightarrow$ SR	0.0023	0.0205	0.0499	0.0052	0.0129
MN $\rightarrow$ SR	0.1634	0.4865	0.3603	0.1699	0.2716
SR $\rightarrow$ SR	0.8342	0.4930	0.5898	0.8249	0.7154
AG $\rightarrow$ CL	0.0001	0.0005	0.0003	0.0086	0.0008
MN $\rightarrow$ CL	0.0814	0.4928	0.3150	0.2351	0.3098
SR $\rightarrow$ CL	0.9184	0.5067	0.6847	0.7563	0.6894
AG $\rightarrow$ B	0.0006	0.0014	0.0005	0.0021	0.0015
MN $\rightarrow$ B	0.2391	0.4784	0.3199	0.2701	0.2951
SR $\rightarrow$ B	0.7300	0.3685	0.6043	0.6651	0.5010
CL $\rightarrow$ B	0.0303	0.1517	0.0753	0.0627	0.2023

Notes: $\gamma_n^{j,k}$ represents the material input share from sector k to sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A21: Labor Input Share ($\psi_n^{j,L}$)

Sector	US	China	India	EU	ROW
AG	0.3327	0.9044	0.5378	0.5741	0.6599
MN	0.4656	0.3075	0.2641	0.5428	0.4162
SR	0.5732	0.5199	0.4919	0.5901	0.5443
CL	0.2152	0.0826	0.2546	0.1355	0.1120
B	0.0993	0.1862	0.0749	0.0834	0.1153

Notes: $\psi_n^{j,L}$ represents the labor input share for sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A22: Capital Input Share ($\psi_n^{j,K}$)

Sector	US	China	India	EU	ROW
AG	0.5526	0.0477	0.4456	0.3306	0.2792
MN	0.4590	0.3782	0.4508	0.3387	0.3473
SR	0.3885	0.4059	0.4532	0.3746	0.4012
CL	0.5967	0.1781	0.2698	0.3356	0.2422
B	0.4160	0.2758	0.2434	0.1707	0.4852

Notes: $\psi_n^{j,K}$ represents the capital input share for sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A23: Energy Input Share ($\psi_n^{j,E}$)

Sector	US	China	India	EU	ROW
AG	0.1147	0.0478	0.0166	0.0952	0.0608
MN	0.0754	0.3143	0.2851	0.1185	0.2364
SR	0.0383	0.0743	0.0549	0.0352	0.0546
CL	0.1882	0.7393	0.4755	0.5289	0.6457

Notes: $\psi_n^{j,E}$ represents the energy input share for sector j in economy n.
Source: WIOD 2016 Release. AG: Agriculture, MN: Manufacturing, SR: Services, CL: Clean Energy, B: Brown. US: United States, EU: European Union, ROW: Rest of the World.

Table A24: Fossil Fuel Input Share ($\psi_n^{B,F}$)

	US	China	India	EU	ROW
$\psi_n^{B,F}$	0.4847	0.5380	0.6816	0.7459	0.3995

Notes: $\psi_n^{B,F}$ represents the fossil fuel input share in economy n.
Source: WIOD 2016 Release. US: United States, EU: European Union, ROW: Rest of the World.

Table A25: Exogenous Trade Costs ($d^j_{nm,0}$) - Agriculture, Manufacturing, Services

From \ To	US	China	India	EU	ROW
Agriculture (AG)					
US	1.0000	3.5076	2.8975	2.6888	1.4766
China	2.4547	1.0000	3.6859	3.6420	1.6869
India	3.1896	3.8452	1.0000	3.8841	1.8247
EU	2.3837	2.8882	2.7970	1.0000	1.4708
ROW	2.2668	2.8426	3.0167	2.3440	1.0000
Manufacturing (MN)					
US	1.0000	1.5562	2.6075	1.6877	1.2952
China	2.6936	1.0000	3.8920	2.2264	1.5460
India	2.1878	1.7732	1.0000	1.7996	1.4172
EU	2.0853	1.7659	2.8148	1.0000	1.4741
ROW	1.7958	1.5828	2.6152	1.5765	1.0000
Services (SR)					
US	1.0000	4.4265	5.4879	3.0378	2.3623
China	3.7658	1.0000	4.7236	2.9896	2.4606
India	2.7947	3.8830	1.0000	2.6310	2.2056
EU	2.4649	3.4372	4.7347	1.0000	2.0299
ROW	2.3324	2.6026	3.2605	2.1337	1.0000

Notes: Exogenous trade costs ($d^j_{nm,0}$) are estimated using a fixed effects regression specification (24), which is conditional on the trade elasticity σ, the bilateral trade shares ($\pi^j_{nm,t}$). Bilateral trade shares for the years 2000 to 2014 are computed using the WIOD 2016 Release.

Table A26: Exogenous Trade Costs ($d^j_{nm,0}$) - Clean Energy, Brown Energy

From \ To	US	China	India	EU	ROW
Clean Energy (CL)					
US	1.0000	3.8274	62.3328	3.0329	2.3645
China	8.3960	1.0000	118.3053	4.3355	3.3871
India	9.4558	3.3788	1.0000	3.2306	2.2408
EU	3.8112	3.8161	43.0767	1.0000	1.8855
ROW	5.2842	4.3795	41.5787	2.9739	1.0000
Brown Energy (B)					
US	1.0000	3.4274	4.0656	2.4224	1.3113
China	3.0898	1.0000	2.5712	3.5095	1.3407
India	2.5363	2.4576	1.0000	2.5766	1.0333
EU	1.9383	2.7517	2.8199	1.0000	1.0466
ROW	2.0465	2.6036	2.7203	2.1445	1.0000

Notes: Exogenous trade costs ($d^j_{nm,0}$) are estimated using a fixed effects regression specification (24), which is conditional on the trade elasticity σ, the bilateral trade shares ($\pi^j_{nm,t}$). Bilateral trade shares for the years 2000 to 2014 are computed using the WIOD 2016 Release.

Table A27: Inter-Sectoral Labor Mobility Costs $\left(\frac{\ln(1-\kappa_n^{j,k})}{\rho_L}\right)$ for US, China, and India

From \ To	NE	AG	MN	SR	CL	B
United States (US)						
NE	0	-2.46	-2.12	-1.46	-2.84	-4.32
AG	-2.79	0	-4.46	-2.13	-5.50	-6.04
MN	-3.22	-5.57	0	-1.82	-3.49	-5.55
SR	-3.81	-6.28	-4.89	0	-5.29	-6.64
CL	-3.73	-6.45	-3.44	-2.26	0	-5.97
B	-3.51	-5.97	-4.00	-2.14	-4.07	0
China						
NE	0	-1.02	-1.95	-1.13	-3.84	-2.74
AG	-1.02	0	-2.68	-2.32	-4.19	-3.57
MN	-1.95	-2.68	0	-2.09	-3.21	-3.35
SR	-1.13	-2.32	-2.09	0	-3.07	-3.30
CL	-3.84	-4.19	-3.21	-3.07	0	-5.96
B	-2.74	-3.57	-3.35	-3.30	-5.96	0
India						
NE	0	-0.49	-3.19	-2.41	-4.76	-5.25
AG	-5.35	0	-5.02	-4.28	-8.90	-8.90
MN	-3.19	-5.02	0	-3.58	-4.61	-5.56
SR	-2.41	-4.28	-3.58	0	-3.45	-4.97
CL	-4.76	-8.90	-4.61	-5.55	0	-5.55
B	-5.25	-8.90	-5.56	-4.97	-5.55	0

Notes: Inter-sectoral labor mobility costs ($\frac{\ln(1-\kappa_n^{j,k})}{\rho_L}$) are estimated using the Poisson Pseudo Maximum Likelihood (PPML) method.
Sources: Current Population Survey, China Family Panel Studies, Periodic Labour Force Survey, Continuous Labour Force Survey, Permanent Household Survey, The Household, Income and Labour Dynamics in Australia, Korean Labor & Income Panel Study, Labour Force Survey.

Table A28: Inter-Sectoral Labor Mobility Costs $\left(\frac{\ln(1-\kappa_n^{j,k})}{\rho_L}\right)$ for EU and ROW

From \ To	NE	AG	MN	SR	CL	B
European Union (EU)						
NE	0	-4.87	-5.60	-2.80	-6.61	-9.28
AG	-4.24	0	-7.32	-5.20	-9.28	-9.28
MN	-3.55	-7.32	0	-3.68	-7.26	-8.90
SR	-2.80	-8.67	-8.28	0	-9.28	-8.90
CL	-5.34	-9.28	-8.51	-5.94	0	-9.28
B	-4.94	-9.28	-8.90	-4.45	-9.28	0
Rest of the World (ROW)						
NE	0	-3.38	-2.07	-1.75	-3.98	-4.81
AG	-4.65	0	-4.52	-3.08	-6.94	-7.10
MN	-4.25	-5.65	0	-2.38	-4.59	-5.84
SR	-1.75	-6.09	-4.38	0	-6.04	-7.05
CL	-4.52	-6.55	-3.37	-2.75	0	-5.65
B	-3.92	-4.88	-2.88	-2.00	-4.13	0

Notes: Inter-sectoral labor mobility costs ($\frac{\ln(1-\kappa_n^{j,k})}{\rho_L}$) are estimated using the Poisson Pseudo Maximum Likelihood (PPML) method.
Sources: Current Population Survey, China Family Panel Studies, Periodic Labour Force Survey, Continuous Labour Force Survey, Permanent Household Survey, The Household, Income and Labour Dynamics in Australia, Korean Labor & Income Panel Study, Labour Force Survey.

Table A29: Non-Employment Income Parameter (w_n^{NE})

Economy	w_n^{NE}
United States (US)	0.16
China	0.55
India	0.24
European Union (EU)	0.57
Rest of the World (ROW)	0.45

Notes: Non-employment income (w_n^{NE}) as a share of average labor income is estimated using panel labor force surveys. The estimation involves Mincerian regression of (log) wages on sector-year fixed effects, along with education, age, and gender fixed effects, weighted by individual weights to obtain residualized sector-year-level wages.
Sources: Current Population Survey, China Family Panel Studies, Periodic Labour Force Survey, Continuous Labour Force Survey, Permanent Household Survey, The Household, Income and Labour Dynamics in Australia, Korean Labor & Income Panel Study, Labour Force Survey.

Table A30: Sector-Specific Total Factor Productivity ($z^j_{n,0}$)

Economy	AG	MN	SR	CL	B
United States (US)	1.4632	4.4722	1.0258	0.0348	23.2856
China	0.1474	5.0419	0.7433	0.1696	32.3684
India	0.1378	4.2557	0.4652	0.2352	94.1721
European Union (EU)	0.8037	4.2467	0.8640	0.1282	58.2622
Rest of the World (ROW)	0.2872	5.5169	0.9434	0.1257	12.1399

Notes: Sector-specific total factor productivity at baseline ($z^j_{n,0}$) is estimated by targeting bilateral trade shares and wages across sectors and countries. Sectors: AG (Agriculture), MN (Manufacturing), SR (Services), CL (Clean Energy), B (Brown Energy).
Source: WIOD 2016 Release.

Table A31: Non-Pecuniary Benefits (μ^j_n)

Economy	NE	AG	MN	SR	CL	B
United States (US)	1.0000	1.2866	0.4835	1.7133	0.4425	0.2649
China	1.0000	4.2805	2.2656	3.0521	0.7677	0.9952
India	1.0000	0.2358	0.5618	0.2627	0.1333	0.1778
European Union (EU)	1.0000	6.0513	1.5934	2.0170	1.8822	1.3523
Rest of the World (ROW)	1.0000	1.2706	0.4708	0.8838	0.2981	0.2578

Notes: Non-pecuniary benefits (μ^j_n) are matched to inter-sectoral labor mobility shares. Sectors: NE (Non-Employment), AG (Agriculture), MN (Manufacturing), SR (Services), CL (Clean Energy), B (Brown Energy).
Source: Panel labor force surveys.

Table A32: Weights on Fossil Fuels $\left((\omega^h_n)^{\frac{1}{\eta_2}}\right)$ with $\eta_2 = 2.67$

Economy	Crude Oil	Natural Gas	Coal
United States (US)	0.9999	0.0020	0.0460
China	0.9993	0.0029	0.0958
India	0.9991	0.0019	0.1064
European Union (EU)	0.9994	0.0040	0.0872
Rest of the World (ROW)	0.9999	0.0020	0.0339

Notes: Weights on fossil fuels (ω^h_n) target economy-level fossil fuel production by fuel type. Fossil fuels: Crude Oil, Natural Gas, Coal.
Source: Ritchie and Rosado (2017).

Table A33: Weights on Clean and Brown Energy $\left((\alpha_n^{CL})^{\frac{1}{\eta_1}}, (\alpha_n^{B})^{\frac{1}{\eta_1}}\right)$ with $\eta_1 = 2.28$

Economy	$(\alpha_n^{CL})^{\frac{1}{\eta_1}}$	$(\alpha_n^{B})^{\frac{1}{\eta_1}}$
United States (US)	0.9863	0.2177
China	0.9989	0.0726
India	0.8214	0.6400
European Union (EU)	0.9968	0.1149
Rest of the World (ROW)	0.9979	0.0961

Notes: Weights on clean (α_n^{CL}) and brown (α_n^{B}) energy are matched to the final expenditure on the brown and clean energy sectors, respectively.
Source: WIOD 2016 Release.

Table A34: Environmental Damage Parameter (ξ_n)

Economy	ξ_n
United States (US)	0.0166
China	0.0093
India	0.0262
European Union (EU)	0.0228
Rest of the World (ROW)	0.0313

Notes: Environmental damage parameter (ξ_n) is estimated to match the impulse response of real GDP per capita to a 1°C increase in global temperature.
Source: Bilal and Känzig (2024).

A.9 Simultaneous Policy Design

This section complements Section 4.2 by reporting results for the simultaneous policy design, in which all five economies reallocate their model-implied subsidy expenditure toward the same alternative clean energy instrument at the same time. The purpose is to assess whether the ranking of instruments in Section 4.2 changes when reform is coordinated across economies rather than implemented unilaterally. The scenario definitions, fiscal equivalence conditions, and notation are otherwise identical to those in Section 4.2. In particular, $\widehat{\mathcal{E}}_{n,y}$ denotes cumulative emission changes relative to BAU, $\widehat{\mathcal{E}}_y = \sum_{n \in \mathcal{N}} \widehat{\mathcal{E}}_{n,y}$ denotes the global counterpart, and $\widehat{\mathcal{W}}_{n,y}$ denotes the percentage change in cumulative real consumption relative to BAU.

Table A35: Simultaneous Design: Cumulative Emission and Welfare Changes

	Scenario 1 Model-Implied		Scenario 2 Consumption		Scenario 3 Capital		Scenario 4 Labor	
	2030	2100	2030	2100	2030	2100	2030	2100
Panel A: Cumulative Emissions ($\widehat{\mathcal{E}}_{n,y}$, $GtCO_2$ change from BAU)								
United States	5.7	37.7	-6.1	-40.8	-4.7	-30.5	-5.4	-37.8
China	16.1	120.2	0.1	-6.2	-11.5	-72.1	-2.2	-12.0
India	11.2	60.9	0.5	4.3	0.4	5.0	0.1	0.9
European Union	-2.6	-22.8	2.8	15.9	2.6	14.7	1.5	8.1
Rest of World	0.5	-14.0	-0.9	-7.6	-1.2	-8.9	-2.5	-17.1
Global ($\widehat{\mathcal{E}}_y$)	30.9	182.1	-3.6	-34.4	-14.5	-91.8	-8.6	-57.8
Panel B: Cumulative Welfare ($\widehat{\mathcal{W}}_{n,y}$, % change from BAU)								
United States	-1.16	-0.82	-4.40	-4.86	-5.21	-5.44	-3.27	-4.00
China	-1.93	-1.82	-4.85	-5.50	-9.35	-10.38	0.38	0.19
India	1.32	-2.38	-0.86	-0.54	-2.14	-1.37	0.46	0.36
European Union	0.32	0.23	-0.54	-0.49	-0.68	-0.53	-0.22	-0.22
Rest of World	0.37	-1.39	-1.60	-2.00	-1.56	-1.77	-1.37	-1.74

Notes: Panel A reports cumulative emission changes from 2015 to the specified year relative to BAU. Panel B reports cumulative changes in real consumption relative to BAU, measured as the sum of real consumption across all agents from 2015 to the specified year. Scenario 1 implements the model-implied energy-related policy instruments. Scenarios 2–4 are fiscally equivalent, reallocating the total value of model-implied subsidies toward clean energy subsidies only. Scenario 2 implements clean energy consumption subsidies ($\tau_{n,t}^{Y,CL}$); Scenario 3 implements clean energy capital subsidies ($\tau_{n,t}^{K,CL}$); Scenario 4 implements clean energy labor subsidies ($\tau_{n,t}^{L,CL}$).

Table A35 shows that the welfare and emission tradeoff in Section 4.2 remains largely intact under coordinated reform, but the global effects become more favorable. Under the model-implied instruments, global cumulative emissions rise by 182.1 $GtCO_2$ by 2100 relative to BAU. All three alternative instruments reduce global cumulative emissions, with capital subsidies delivering the largest reduction and labor subsidies the smallest. Thus, the global ranking of instruments is similar to the unilateral case, but coordination makes the global consequences of reform more favorable.

The within-economy patterns also remain close to those in Section 4.2. In the United States, the consumption subsidy again delivers the largest domestic emission reduction, while the labor subsidy minimizes welfare losses. In China, the capital subsidy again

delivers the largest domestic emission reduction, but it also imposes the largest welfare cost, whereas the labor subsidy continues to improve welfare. These results mirror the unilateral design: the capital subsidy is strongest on the emission margin in China but performs poorly on the welfare margin, while the labor subsidy delivers the best welfare outcome.

The remaining economies display a more nuanced pattern. In India, all three alternative instruments still leave cumulative emissions above BAU, but labor subsidies generate the smallest increase and the best welfare outcome. The European Union differs from the unilateral case on the emission margin but not on the welfare margin. Under simultaneous reform, the recovered instruments dominate all three alternatives on both domestic emissions and welfare, suggesting that once all economies reallocate subsidies at the same time, the resulting scale effects are strong enough that the alternatives no longer outperform the EU's relatively stringent baseline policy mix. For the Rest of World, all three alternative instruments reduce emissions relative to BAU, and labor subsidies again perform best on the welfare margin.

Relative to the unilateral design, the primary difference is that simultaneous reallocation pools fiscal effort across all economies, offsetting the adverse cross-border spillovers that arise when individual economies act alone and producing larger global emission reductions. The qualitative tradeoff identified in Section 4.2, however, persists: the instrument that delivers the largest global emission reduction is not the instrument that minimizes welfare costs, and the ranking of instruments by either criterion continues to reflect each economy's production structure and the degree of factor adjustment frictions.

A.10 Derivations

A.10.1 Sectoral Good and Bilateral Trade Share

The representative buyer in economy n solves

$$\min_{y^j_{nm,t}} \quad \sum_{m\in\mathcal{N}} p^j_{nm,t} y^j_{nm,t} \qquad \text{subject to} \qquad Y^j_{n,t} = \left(\sum_{m\in\mathcal{N}} \left(y^j_{nm,t} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}.$$

The Lagrangian is

$$\mathcal{L} = \sum_{m \in \mathcal{N}} p^j_{nm,t} y^j_{nm,t} + \lambda \left(Y^j_{n,t} - \left(\sum_{m \in \mathcal{N}} \left(y^j_{nm,t} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \right).$$

The first-order condition with respect to $y^j_{nm,t}$ is

$$p^j_{nm,t} = \lambda \left(\sum_{m \in \mathcal{N}} \left(y^j_{nm,t} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} \left(y^j_{nm,t} \right)^{-\frac{1}{\sigma}}.$$

Raising both sides to the power $-\sigma$ and substituting the constraint gives

$$y^j_{nm,t} = \left(\frac{p^j_{nm,t}}{\lambda} \right)^{-\sigma} Y^j_{n,t}.$$

Summing over all m and imposing the constraint pins down the Lagrange multiplier as the CES price index $\lambda = P^j_{n,t} \equiv \left(\sum_{m \in \mathcal{N}} (p^j_{nm,t})^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$. Substituting back yields the demand function and bilateral trade share:

$$y^j_{nm,t} = \left(\frac{p^j_{nm,t}}{P^j_{n,t}} \right)^{-\sigma} Y^j_{n,t}, \tag{24}$$

$$\pi^j_{nm,t} \equiv \frac{p^j_{nm,t} y^j_{nm,t}}{P^j_{n,t} Y^j_{n,t}} = \left(\frac{p^j_{nm,t}}{P^j_{n,t}} \right)^{1-\sigma}. \tag{25}$$

A.10.2 Energy Composite Good

The buyer minimizes $(1 + \tau^{CL}_{n,t}) P^{CL}_{n,t} C^{CL}_{n,t} + (1 + \tau^{B}_{n,t}) P^{B}_{n,t} C^{B}_{n,t}$ subject to

$$Y^E_{n,t} = \left((\alpha^{CL}_n)^{\frac{1}{\eta_1}} (Y^{CL}_{n,t})^{\frac{\eta_1 - 1}{\eta_1}} + (\alpha^{B}_n)^{\frac{1}{\eta_1}} (Y^{B}_{n,t})^{\frac{\eta_1 - 1}{\eta_1}} \right)^{\frac{\eta_1}{\eta_1 - 1}}.$$

The first-order conditions are

$$(1 + \tau^{CL}_{n,t}) P^{CL}_{n,t} = \lambda \cdot (Y^E_{n,t})^{\frac{1}{\eta_1}} \cdot (\alpha^{CL}_n)^{\frac{1}{\eta_1}} \cdot (Y^{CL}_{n,t})^{-\frac{1}{\eta_1}},$$
$$(1 + \tau^{B}_{n,t}) P^{B}_{n,t} = \lambda \cdot (Y^E_{n,t})^{\frac{1}{\eta_1}} \cdot (\alpha^{B}_n)^{\frac{1}{\eta_1}} \cdot (Y^{B}_{n,t})^{-\frac{1}{\eta_1}}.$$

Dividing these two equations and rearranging gives the optimal input ratio

$$Y_{n,t}^{CL} = \left(\frac{\alpha_n^{CL}}{\alpha_n^B}\right)\left(\frac{(1+\tau_{n,t}^{CL})P_{n,t}^{CL}}{(1+\tau_{n,t}^B)P_{n,t}^B}\right)^{-\eta_1} Y_{n,t}^B.$$

Substituting this ratio into the constraint and simplifying identifies the energy price index

$$P_{n,t}^E = \left(\alpha_n^{CL}\left((1+\tau_{n,t}^{CL})P_{n,t}^{CL}\right)^{1-\eta_1} + \alpha_n^B\left((1+\tau_{n,t}^B)P_{n,t}^B\right)^{1-\eta_1}\right)^{\frac{1}{1-\eta_1}}. \tag{26}$$

The demand for brown energy is then

$$Y_{n,t}^B = \alpha_n^B\left(\frac{(1+\tau_{n,t}^B)P_{n,t}^B}{P_{n,t}^E}\right)^{-\eta_1} Y_{n,t}^E. \tag{27}$$

A.10.3 Material

The firm minimizes $\sum_{k\in\mathcal{J}}(1+\tau_{n,t}^k)P_{n,t}^k M_{n,t}^{j,k}$ subject to $M_{n,t}^j = \prod_{k\in\mathcal{J}}(M_{n,t}^{j,k})^{\gamma_n^{j,k}}$. The first-order condition for input k is

$$(1+\tau_{n,t}^k)P_{n,t}^k = \lambda \cdot \gamma_n^{j,k} \cdot M_{n,t}^j \cdot (M_{n,t}^{j,k})^{-1}.$$

Solving for $M_{n,t}^{j,k}$ and substituting into the constraint gives

$$\lambda = \prod_{k\in\mathcal{J}}\left((1+\tau_{n,t}^k)P_{n,t}^k/\gamma_n^{j,k}\right)^{\gamma_n^{j,k}} \equiv P_{n,t}^{M,j}.$$

The input demand function is

$$M_{n,t}^{j,k} = \frac{\gamma_n^{j,k} \cdot P_{n,t}^{M,j}}{(1+\tau_{n,t}^k)P_{n,t}^k} M_{n,t}^j. \tag{28}$$

A.10.4 Final Good

The representative capitalist minimizes $\sum_{j\in\mathcal{J}_N}(1+\tau^j_{n,t})P^j_{n,t}C^j_{n,t}+P^E_{n,t}C^E_{n,t}$ subject to

$$Y_{n,t}=\mathcal{D}_n(\mathcal{C}_t)\Big(\prod_{j\in\mathcal{J}_N}(C^j_{n,t})^{\alpha^j_n}\Big)^{\alpha_n}(C^E_{n,t})^{1-\alpha_n}.$$

The first-order conditions imply

$$C^j_{n,t}=\frac{\alpha_n\alpha^j_n\lambda Y_{n,t}}{(1+\tau^j_{n,t})P^j_{n,t}},\qquad C^E_{n,t}=\frac{(1-\alpha_n)\lambda Y_{n,t}}{P^E_{n,t}}.$$

Substituting these into the constraint and solving for λ identifies the final good price index

$$P_{n,t}=\Big(\mathcal{D}_n(\mathcal{C}_t)\Big)^{-1}\left(\prod_{j\in\mathcal{J}_N}\Big(\frac{(1+\tau^j_{n,t})P^j_{n,t}}{\alpha_n\alpha^j_n}\Big)^{\alpha^j_n}\right)^{\alpha_n}\Big(\frac{P^E_{n,t}}{1-\alpha_n}\Big)^{1-\alpha_n}. \tag{29}$$

The demand functions are

$$C^j_{n,t}=\frac{\alpha_n\alpha^j_nP_{n,t}}{(1+\tau^j_{n,t})P^j_{n,t}}Y_{n,t},\qquad C^E_{n,t}=\frac{(1-\alpha_n)P_{n,t}}{P^E_{n,t}}Y_{n,t}. \tag{30}$$

A.11 Brown Energy Variety

The producer of variety m of brown energy solves

$$\begin{aligned}\min\quad & w^B_{n,t}L^B_{mn,t}+r^B_{n,t}K^B_{mn,t}+s_{n,t}F^B_{mn,t}+P^{M,B}_{n,t}M^B_{mn,t}\\ \text{subject to}\quad & y^B_{mn,t}=\Big(\frac{z^B_{n,t}}{d^B_{mn,t}}\Big)\Big((F^B_{mn,t})^{\psi^{B,F}_n}(L^B_{mn,t})^{\psi^{B,L}_n}(K^B_{mn,t})^{\psi^{B,K}_n}\Big)^{\gamma^{B,V}_n}(M^B_{mn,t})^{1-\gamma^{B,V}_n}.\end{aligned}$$

The first-order conditions give input demands proportional to output:

$$\begin{aligned}L^B_{mn,t}&=\lambda_{n,t}y^B_{mn,t}\frac{\gamma^{B,V}_n\psi^{B,L}_n}{w^B_{n,t}},\quad K^B_{mn,t}=\lambda_{n,t}y^B_{mn,t}\frac{\gamma^{B,V}_n\psi^{B,K}_n}{r^B_{n,t}},\\ F^B_{mn,t}&=\lambda_{n,t}y^B_{mn,t}\frac{\gamma^{B,V}_n\psi^{B,F}_n}{s_{n,t}},\quad M^B_{mn,t}=\lambda_{n,t}y^B_{mn,t}\frac{1-\gamma^{B,V}_n}{P^{M,B}_{n,t}}.\end{aligned}$$

Since the Lagrange multiplier equals marginal cost, i.e. $\lambda_{n,t} = (w^B_{n,t} L^B_{mn,t} + r^B_{n,t} K^B_{mn,t} + s_{n,t} F^B_{mn,t} + P^{M,B}_{n,t} M^B_{mn,t})/y^B_{mn,t}$, substituting the input demands into the production constraint and solving for $\lambda_{n,t}$ yields

$$mc^B_{mn,t} = \left(\frac{d^B_{mn,t}}{z^B_{n,t}}\right)\left[\left(\frac{s_{n,t}}{\gamma^{B,V}_n \psi^{B,F}_n}\right)^{\psi^{B,F}_n}\left(\frac{w^B_{n,t}}{\gamma^{B,V}_n \psi^{B,L}_n}\right)^{\psi^{B,L}_n}\left(\frac{r^B_{n,t}}{\gamma^{B,V}_n \psi^{B,K}_n}\right)^{\psi^{B,K}_n}\right]^{\gamma^{B,V}_n}\left(\frac{P^{M,B}_{n,t}}{1-\gamma^{B,V}_n}\right)^{1-\gamma^{B,V}_n}. \tag{31}$$

A.12 Clean Energy and Non-energy Varieties

The producer of variety m in sector $j \in \{\text{non-energy, clean energy}\}$ solves an analogous problem with energy input $Q^j_{mn,t}$ in place of fossil fuel:

$$\begin{aligned} \min \quad & w^j_{n,t} L^j_{mn,t} + r^j_{n,t} K^j_{mn,t} + P^E_{n,t} Q^j_{mn,t} + P^{M,j}_{n,t} M^j_{mn,t} \\ \text{subject to} \quad & y^j_{mn,t} = \left(\frac{z^j_{n,t}}{d^j_{mn,t}}\right)\left[(Q^j_{mn,t})^{\psi^{j,E}_n}(L^j_{mn,t})^{\psi^{j,L}_n}(K^j_{mn,t})^{\psi^{j,K}_n}\right]^{\gamma^{j,V}_n}(M^j_{mn,t})^{1-\gamma^{j,V}_n}. \end{aligned}$$

Proceeding identically to the brown energy case, the marginal cost is

$$mc^j_{mn,t} = \left(\frac{d^j_{mn,t}}{z^j_{n,t}}\right)\left[\left(\frac{P^E_{n,t}}{\gamma^{j,V}_n \psi^{j,E}_n}\right)^{\psi^{j,E}_n}\left(\frac{w^j_{n,t}}{\gamma^{j,V}_n \psi^{j,L}_n}\right)^{\psi^{j,L}_n}\left(\frac{r^j_{n,t}}{\gamma^{j,V}_n \psi^{j,K}_n}\right)^{\psi^{j,K}_n}\right]^{\gamma^{j,V}_n}\left(\frac{P^{M,j}_{n,t}}{1-\gamma^{j,V}_n}\right)^{1-\gamma^{j,V}_n}. \tag{32}$$

A.12.1 Labor Mobility Share

The lifetime utility of a worker currently in sector j satisfies

$$\mathcal{U}^j_{n,t} = \ln(c^j_{n,t}) + \ln(\mu^j_n) + \max_{k\in\mathcal{J}}\left\{\beta V^k_{n,t+1} + \ln(1-\kappa^{j,k}_n) + \rho_L \varepsilon^k_{n,t}\right\}.$$

Define $A^{j,k}_{n,t} \equiv \beta V^k_{n,t+1} + \ln(1-\kappa^{j,k}_n)$. The idiosyncratic preference shocks $\varepsilon^k_{n,t}$ are i.i.d. Gumbel with CDF $F(x) = \exp(-\exp(-(x+\gamma)))$ and PDF $f(x) = \exp(-x-\gamma)\exp(-e^{-x-\gamma})$, where γ is the Euler–Mascheroni constant. The probability of transitioning to sector k

from sector j is

$$m_{n,t}^{j,k} = \int_{-\infty}^{\infty} f(\varepsilon^k) \prod_{h \neq k} F\Big(\frac{A_{n,t}^{j,k} - A_{n,t}^{j,h}}{\rho_L} + \varepsilon^k\Big) d\varepsilon^k$$
$$= \int_{-\infty}^{\infty} \exp(-\varepsilon^k - \gamma) \exp\Bigg(- e^{-\varepsilon^k - \gamma} \sum_{h \in \mathcal{J}} \exp\Big(\frac{A_{n,t}^{j,h} - A_{n,t}^{j,k}}{\rho_L}\Big)\Bigg) d\varepsilon^k.$$

Applying the change of variables $x = \varepsilon^k + \gamma$ and $y = \log \sum_h \exp\big((A_{n,t}^{j,h} - A_{n,t}^{j,k})/\rho_L\big)$, the integral becomes $\int_{-\infty}^{\infty} e^{-x-e^{y-x}} dx$. A second substitution $z = x - y$ then gives

$$m_{n,t}^{j,k} = e^{-y} \int_{-\infty}^{\infty} e^{-z-e^{-z}} dz = e^{-y},$$

where the integral equals unity since it is the normalizing constant of the Gumbel distribution. Substituting back the definition of y yields the multinomial logit formula

$$m_{n,t}^{j,k} = \frac{\exp\Big(\beta V_{n,t+1}^k + \ln(1 - \kappa_n^{j,k})\Big)^{\frac{1}{\rho_L}}}{\sum_{h \in \mathcal{J}} \exp\Big(\beta V_{n,t+1}^h + \ln(1 - \kappa_n^{j,h})\Big)^{\frac{1}{\rho_L}}}. \tag{33}$$

A.12.2 Expected Lifetime Utility of a Worker

The expected future value of a worker in sector j who moves to sector k is

$$\tilde{V}_{n,t+1}^{j,k} = \int_{-\infty}^{\infty} \Big(A_{n,t}^{j,k} + \rho_L \varepsilon^k\Big) \exp(-\varepsilon^k - \gamma) \exp\Bigg(- e^{-\varepsilon^k - \gamma} \sum_h \exp\Big(\frac{A_{n,t}^{j,h} - A_{n,t}^{j,k}}{\rho_L}\Big)\Bigg) d\varepsilon^k.$$

Applying the same two changes of variable as above ($x = \varepsilon^k + \gamma$, then $z = x - y$) gives

$$\tilde{V}_{n,t+1}^{j,k} = \int_{-\infty}^{\infty} \Big(A_{n,t}^{j,k} + \rho_L(z + y - \gamma)\Big) e^{-(z+y)-e^{-z}} dz$$
$$= A_{n,t}^{j,k} e^{-y} \underbrace{\int_{-\infty}^{\infty} e^{-z-e^{-z}} dz}_{=1} + \rho_L e^{-y} \underbrace{\int_{-\infty}^{\infty} z\, e^{-z-e^{-z}} dz}_{=\gamma} + \rho_L e^{-y}(y - \gamma) \underbrace{\int_{-\infty}^{\infty} e^{-z-e^{-z}} dz}_{=1}$$
$$= e^{-y}\Big(A_{n,t}^{j,k} + \rho_L y\Big).$$

Since $e^{-y} = m^{j,k}_{n,t}$ and $\rho_L y = -A^{j,k}_{n,t} + \rho_L \log \sum_h \exp(A^{j,h}_{n,t}/\rho_L)$, this simplifies to $\tilde{V}^{j,k}_{n,t+1} = m^{j,k}_{n,t}\rho_L \log \sum_h \exp(A^{j,h}_{n,t}/\rho_L)$. Summing over all destination sectors k and using $\sum_k m^{j,k}_{n,t} = 1$, the expected future value of a worker in sector j is

$$\tilde{V}^{j}_{n,t+1} = \rho_L \ln\left(\sum_{k\in\mathcal{J}} \exp\Big(\frac{\beta}{\rho_L} V^k_{n,t+1} + \frac{1}{\rho_L}\ln(1-\kappa^{j,k}_n)\Big)\right). \tag{34}$$

A.12.3 Fossil Fuel Extraction Problem

The fossil fuel producer solves

$$\min_{f^h_{n,t}} \quad \sum_h (1+\tau^h_{n,t}) s^h_{n,t} f^h_{n,t} \qquad \text{subject to} \qquad F_{n,t} = \left(\sum_h (\omega^h_n)^{\frac{1}{\eta_2}} (f^h_{n,t})^{\frac{\eta_2-1}{\eta_2}}\right)^{\frac{\eta_2}{\eta_2-1}}.$$

The first-order condition for type h is

$$(1+\tau^h_{n,t}) s^h_{n,t} = \lambda_{n,t} (\omega^h_n)^{\frac{1}{\eta_2}} (f^h_{n,t})^{-\frac{1}{\eta_2}} \left(\sum_h (\omega^h_n)^{\frac{1}{\eta_2}} (f^h_{n,t})^{\frac{\eta_2-1}{\eta_2}}\right)^{\frac{1}{\eta_2-1}}.$$

Taking the ratio of conditions for fuels h and k and rearranging gives $f^h_{n,t} = (\omega^h_n/\omega^k_n)\big((1+\tau^k_{n,t})s^k_{n,t}/((1+\tau^h_{n,t})s^h_{n,t})\big)^{\eta_2} f^k_{n,t}$. Substituting into the constraint and solving identifies the aggregate fossil fuel price index

$$s_{n,t} = \Big(\sum_h \omega^h_n \big((1+\tau^h_{n,t})s^h_{n,t}\big)^{1-\eta_2}\Big)^{\frac{1}{1-\eta_2}}, \tag{35}$$

and the demand policy function

$$f^k_{n,t} = \omega^k_n \Big(\frac{(1+\tau^k_{n,t})s^k_{n,t}}{s_{n,t}}\Big)^{-\eta_2} F_{n,t}. \tag{36}$$

A.12.4 Capitalist Optimization Problem

The capitalist maximizes $\sum_{t=0}^{\infty} \beta^t \mathbb{E}_0[u(C^K_{n,t})]$ subject to the budget constraint

$$C^K_{n,t} + W_{n,t+1} = R_{n,t} W_{n,t} + \frac{\Omega_{n,t}}{P_{n,t}},$$

where $R_{n,t} = v_{n,t}/P_{n,t} + (1 - \delta^K_n)$. The first-order conditions are

$$\frac{\beta^t}{C^K_{n,t}} = \lambda_{n,t} P_{n,t}, \qquad \lambda_{n,t+1}\Big(v_{n,t+1} + (1-\delta^K_n) P_{n,t+1}\Big) = \lambda_{n,t} P_{n,t},$$

which together imply the Euler equation $C^K_{n,t+1} = \beta R_{n,t+1} C^K_{n,t}$. The present discounted value of future non-capital income satisfies the recursion

$$h_{n,t} = (R_{n,t+1})^{-1} \frac{\Pi_{n,t+1} + \Omega_{n,t+1}}{P_{n,t+1}} + (R_{n,t+1})^{-1} h_{n,t+1}.$$

Conjecturing that consumption is linear in total wealth, the guess $C^K_{n,t} = (1-\beta)(R_{n,t} W_{n,t} + \Omega_{n,t}/P_{n,t} + h_{n,t})$ can be verified by substitution into the Euler equation. Using the recursion for $h_{n,t}$, verification requires $R_{n,t+1} h_{n,t} = (\Pi_{n,t+1} + \Omega_{n,t+1})/P_{n,t+1} + h_{n,t+1}$, which holds by construction. The optimal policies are therefore

$$C^K_{n,t} = (1-\beta)\Big(R_{n,t} W_{n,t} + \frac{\Omega_{n,t}}{P_{n,t}} + h_{n,t}\Big), \tag{37}$$

$$W_{n,t+1} = \beta\Big(R_{n,t} W_{n,t} + \frac{\Omega_{n,t}}{P_{n,t}} + h_{n,t}\Big) - h_{n,t}. \tag{38}$$

A.12.5 Capital Allocation Problem

Each capitalist draws an idiosyncratic capital-use efficiency $\zeta^j_{n,t}$ from a Fréchet distribution with scale $a^j_{n,t}$ and shape ρ_K, i.e. $F^j(\zeta) = \exp(-((\zeta)/a^j_{n,t})^{-\rho_K})$, and invests in the sector offering the highest return $\zeta^j_{n,t} P_{n,t} r^j_{n,t}$. The probability of investing in sector j is

$$\Phi^j_{n,t} = \int_0^\infty \prod_{k \neq j} F^k\Big(\frac{r^j_{n,t}\zeta}{r^k_{n,t}}\Big) f^j(\zeta)\, d\zeta = \int_0^\infty \rho_K (a^j_{n,t})^{\rho_K} \zeta^{-\rho_K - 1} e^{-\alpha \zeta^{-\rho_K}}\, d\zeta,$$

where $\alpha = (r^j_{n,t})^{-\rho_K} \sum_k (r^k_{n,t} a^k_{n,t})^{\rho_K}$. The substitution $u = \zeta^{-\rho_K}$ reduces the integral to $\int_0^\infty e^{-\alpha u}\, du = \alpha^{-1}$, giving

$$\Phi^j_{n,t} = \frac{(r^j_{n,t} a^j_{n,t})^{\rho_K}}{\sum_{k \in \mathcal{J}} (r^k_{n,t} a^k_{n,t})^{\rho_K}}. \tag{39}$$

Letting $v^j_{n,t} = \zeta^j_{n,t} P_{n,t} r^j_{n,t}$, the returns across all sectors follow a Fréchet distribution with scale $\Psi^{1/\rho_K}_{n,t}$ where $\Psi_{n,t} = \sum_j (a^j_{n,t} P_{n,t} r^j_{n,t})^{\rho_K}$. The expected aggregate return is obtained by integrating against this distribution. Using the substitution $y = \Psi_{n,t} v^{-\rho_K}$ and recognizing the resulting integral as the Gamma function gives

$$v_{n,t} = \Gamma\Big(1 - \tfrac{1}{\rho_K}\Big) \Big(\sum_{j \in \mathcal{J}} (a^j_{n,t} P_{n,t} r^j_{n,t})^{\rho_K} \Big)^{\frac{1}{\rho_K}}. \tag{40}$$

To derive sector-specific capital supply, one computes $K^j_{n,t} = W^j_{n,t} \mathbb{E}[\zeta^j_{n,t} \mid \text{invest in } j]$. The conditional distribution of returns given investment in j is Fréchet with scale $\Psi^{1/\rho_K}_{n,t}$, so the same Gamma-function argument applies. After two changes of variable and substituting $W^j_{n,t} = \Phi^j_{n,t} W_{n,t}$, one obtains

$$K^j_{n,t} = \left(\Gamma\Big(\frac{\rho_K - 1}{\rho_K}\Big) \right)^{\rho_K} (a^j_{n,t})^{\rho_K} \Big(\frac{P_{n,t} r^j_{n,t}}{v_{n,t}} \Big)^{\rho_K - 1} W_{n,t}. \tag{41}$$

Taking logs and substituting the law of motion $\ln(a^j_{n,t}) = \ln(a^j_n) + \rho_\Phi \ln(K^j_{n,t-1}/W_{n,t-1})$ yields

$$\begin{aligned} \ln(K^j_{n,t}) = {} & (\rho_K - 1) \ln\Big(\frac{P_{n,t} r^j_{n,t}}{v_{n,t}} \Big) + \ln(W_{n,t}) \\ & + \rho_K \rho_\Phi \ln\Big(\frac{K^j_{n,t-1}}{W_{n,t-1}} \Big) + \rho_K \ln(a^j_n) + \rho_K \ln \Gamma\Big(\frac{\rho_K - 1}{\rho_K} \Big). \end{aligned} \tag{42}$$

A.12.6 Consumption-based Carbon Emissions

Emission intensity per unit of fossil fuel input and per unit of brown energy output are defined as

$$\phi^F_{n,t} = \frac{\sum_{h\in\{o,g,c\}} \phi^h f^h_{n,t}}{F_{n,t}}, \qquad \phi^B_{n,t} = \frac{\phi^F_{n,t} F_{n,t}}{y^B_{n,t}},$$

where $y^B_{n,t} \equiv \sum_{m\in\mathcal{N}} y^B_{mn,t}$. Country-level consumption-based and production-based emissions are then

$$E^C_{n,t} = \sum_{m\in\mathcal{N}} \phi^B_{m,t} y^B_{nm,t}, \qquad E_{n,t} = \sum_{h\in\{o,g,c\}} \phi^h f^h_{n,t}. \tag{43}$$

Global emissions satisfy $E_t = \sum_{n\in\mathcal{N}} E_{n,t} = \sum_{n\in\mathcal{N}} E^C_{n,t}$, confirming consistency between production-based and consumption-based accounting.

A.12.7 Estimation Equation for Labor Mobility Costs

The labor mobility share expression implies

$$\ln\Big(\frac{L^{j,k}_{n,t}}{L^j_{n,t}}\Big) = \frac{\beta}{\rho_L} V^k_{n,t+1} + \frac{1}{\rho_L}\ln(1-\kappa^{j,k}_n) - \ln\left(\sum_{h\in\mathcal{J}} \exp\Big(\beta V^h_{n,t+1} + \ln(1-\kappa^{j,h}_n)\Big)^{\frac{1}{\rho_L}}\right),$$

which motivates the regression specification

$$L^{j,k}_{n,t} = \exp\Big(C^{j,k}_n + \lambda^k_{n,t} + \lambda^j_{n,t}\Big) + \xi^{j,k}_{n,t}, \tag{44}$$

where $C^{j,k}_n \equiv \ln(1-\kappa^{j,k}_n)/\rho_L$, $\lambda^k_{n,t} \equiv (\beta/\rho_L)V^k_{n,t+1}$, $\lambda^j_{n,t}$ is a sector-time fixed effect, and $\xi^{j,k}_{n,t}$ is the residual. To separately identify mobility costs, note that for any pair (j,k) the lifetime utility of a worker in sector j can be written as

$$V^j_{n,t} = \ln\Big(\frac{w^j_{n,t}}{P_{n,t}}\Big) + \ln(\mu^j_n) + \beta V^k_{n,t+1} + \ln(1-\kappa^{j,k}_n) - \rho_L \ln(m^{j,k}_{n,t}).$$

Setting $k = j$ and then taking differences across pairs (j, k) and (k, j) eliminates the future value terms, giving

$$\rho_L \ln\Big(\frac{L_{n,t}^{k,j} L_{n,t}^{j,k}}{L_{n,t}^{j,j} L_{n,t}^{k,k}}\Big) = \ln(1 - \kappa_n^{j,k}) + \ln(1 - \kappa_n^{k,j}),$$

which motivates the symmetric regression $L_{n,t}^{j,k} L_{n,t}^{k,j} / (L_{n,t}^{j,j} L_{n,t}^{k,k}) = \exp(\tilde{C}_n^{j,k}) + \tilde{\xi}_{n,t}^{j,k}$, with $\tilde{C}_n^{j,k} \equiv \rho_L^{-1}[\ln(1 - \kappa_n^{j,k}) + \ln(1 - \kappa_n^{k,j})]$.

A.13 Derivations for Estimation Equations

A.13.1 Estimation Equation for Dispersion of Non-pecuniary Preferences

Using the lifetime utility expression derived in Section A.12.7, taking the difference of utility equations for sectors j and k gives

$$\rho_L \ln\Big(\frac{m_{n,t}^{j,k}}{m_{n,t}^{j,j}}\Big) = \beta\Big(V_{n,t+1}^k - V_{n,t+1}^j\Big) + \ln(1 - \kappa_n^{j,k}).$$

Substituting the forward-looking expressions for $V_{n,t+1}^k$ and $V_{n,t+1}^j$ and eliminating $V_{n,t+2}^k$ yields

$$V_{n,t+1}^k - V_{n,t+1}^j = \ln\Big(\frac{w_{n,t+1}^k}{w_{n,t+1}^j}\Big) + \ln\Big(\frac{\mu_n^k}{\mu_n^j}\Big) + \rho_L \ln\Big(\frac{m_{n,t+1}^{j,k}}{m_{n,t+1}^{k,k}}\Big) - \ln(1 - \kappa_n^{j,k}).$$

Substituting this expression back gives

$$\ln\Big(\frac{L_{n,t}^{j,k}}{L_{n,t}^{j,j}}\Big) = \Big(\frac{1-\beta}{\rho_L}\Big) \ln(1 - \kappa_n^{j,k}) + \Big(\frac{\beta}{\rho_L}\Big) \ln\Big(\frac{w_{n,t+1}^k}{w_{n,t+1}^j}\Big) + \beta \ln\Big(\frac{\mu_n^k}{\mu_n^j}\Big) + \beta \ln\Big(\frac{L_{n,t+1}^{j,k} L_{n,t+1}^k}{L_{n,t+1}^j L_{n,t+1}^{k,k}}\Big).$$

This motivates the regression specification

$$y_{n,t}^{j,k} = \omega_n^k + \omega_n^j + \Big(\frac{\beta}{\rho_L}\Big) \ln\Big(\frac{w_{n,t+1}^k}{w_{n,t+1}^j}\Big) + \eta_{n,t+1}^{j,k}, \tag{45}$$

where $y^{j,k}_{n,t} \equiv \ln(L^{j,k}_{n,t}/L^{j,j}_{n,t}) - (1-\beta)C^{j,k}_n - \beta\ln(L^k_{n,t+1}L^{j,k}_{n,t+1}/(L^j_{n,t+1}L^{k,k}_{n,t+1}))$, $\omega^k_n \equiv \beta\ln(\mu^k_n)$, $\omega^j_n \equiv -\beta\ln(\mu^j_n)$, and $\eta^{j,k}_{n,t+1}$ is the forecast error.

A.13.2 Estimation Equation for Capital Supply Elasticities

From the capital allocation derived in Section A.12.5, taking log differences between sectors j and k gives

$$\ln(K^j_{n,t}/K^k_{n,t}) = \ln(a^j_{n,t}/a^k_{n,t}) + \left(\frac{\rho_K - 1}{\rho_K}\right)\ln(\Phi^j_{n,t}/\Phi^k_{n,t}).$$

Since $\ln(\Phi^j_{n,t}/\Phi^k_{n,t}) = \rho_K[\ln(a^j_{n,t}/a^k_{n,t}) + \ln(r^j_{n,t}/r^k_{n,t})]$, substituting yields

$$\ln(K^j_{n,t}/K^k_{n,t}) = \rho_K\ln(a^j_{n,t}/a^k_{n,t}) + (\rho_K - 1)\ln(r^j_{n,t}/r^k_{n,t}).$$

Substituting the law of motion $\ln(a^j_{n,t}) = \ln(a^j_n) + \rho_\Phi\ln(K^j_{n,t-1}/W_{n,t-1})$ and normalizing by a reference sector o gives the regression specification

$$\ln\left(\frac{K^j_{n,t}}{K^o_{n,t}}\right) = \tilde{\omega}^j_n + (\rho_K - 1)\ln\left(\frac{r^j_{n,t}}{r^o_{n,t}}\right) + \rho_K\rho_\Phi\ln\left(\frac{K^j_{n,t-1}}{K^o_{n,t-1}}\right) + \tilde{\eta}^j_{n,t}, \tag{46}$$

where $\tilde{\omega}^j_n \equiv \rho_K\ln(a^j_n/a^o_n)$ and $\tilde{\eta}^j_{n,t}$ is the residual.

A.13.3 Estimation Equation for Energy Demand Elasticity

From the energy cost-minimization problem, relative demand satisfies

$$\ln\left(\frac{Y^{CL}_{n,t}}{Y^B_{n,t}}\right) = \ln\left(\frac{\alpha^{CL}_n}{\alpha^B_n}\right) - \eta_1\ln\left(\frac{1+\tau^{CL}_{n,t}}{1+\tau^B_{n,t}}\right) - \eta_1\ln\left(\frac{P^{CL}_{n,t}}{P^B_{n,t}}\right).$$

Taking first differences $\Delta x_{n,t} \equiv \ln(x_{n,t}) - \ln(x_{n,0})$ and treating the tax term as an error component yields

$$\Delta Y^{CL}_{n,t} - \Delta Y^B_{n,t} = -\eta_1(\Delta P^{CL}_{n,t} - \Delta P^B_{n,t}) + \iota_{n,t}, \tag{47}$$

where $\iota_{n,t} \equiv -\eta_1\Delta(1+\tau^{CL}_{n,t}) + \eta_1\Delta(1+\tau^B_{n,t})$.

A.13.4 Estimation Equation for Fossil Fuel Elasticity

Relative fossil fuel demand satisfies

$$\ln\Big(\frac{f^k_{n,t}}{f^h_{n,t}}\Big) = \ln\Big(\frac{\omega^k_n}{\omega^h_n}\Big) - \eta_2 \ln\Big(\frac{s^k_{n,t}}{s^h_{n,t}}\Big) - \eta_2 \ln\Big(\frac{1+\tau^k_{n,t}}{1+\tau^h_{n,t}}\Big).$$

This motivates the regression specification

$$\ln\Big(\frac{f^k_{n,t}}{f^h_{n,t}}\Big) = \delta^k_n - \eta_2 \ln\Big(\frac{s^k_{n,t}}{s^h_{n,t}}\Big) - \eta_2 \ln\Big(\frac{1+\tau^k_{n,t}}{1+\tau^h_{n,t}}\Big) + \varepsilon^k_{n,t}, \tag{48}$$

where $\delta^k_n \equiv \ln(\omega^k_n/\omega^h_n)$ are country–fuel fixed effects and $\varepsilon^k_{n,t}$ is an error term.

A.14 Solution Algorithm

This section describes the algorithm used to solve the model. The algorithm iterates over three nested blocks until convergence.

Initialization. Load parameters and set convergence objects $dev_\star$, $tol_\star$, $smth_\star$, $iter_\star$, $itermax_\star$. Initialize guesses for wages w^j_{nt}, rental rates r^j_{nt}, labor transition probabilities m^{jk}_{nt}, and fossil fuel extraction costs $(\varphi^*)^h_{nt}$.

Outer Loop: repeat until $dev_1 < tol_1$ or $iter_1 = itermax_1$

1. Price Block: repeat until $dev_{NN} < tol_{NN}$ or $iter_{NN} = itermax_{NN}$
   (a) Compute fossil fuel price s^h_{nt} and after-tax fossil fuel price $\tilde{s}^h_{nt}$.
   (b) Compute aggregate fossil fuel price s_{nt}.
   (c) Compute marginal costs and variety prices p^j_{nnt} for non-energy, clean energy, and brown energy sectors.
   (d) Compute trade costs d^j_{nmt} and destination-origin variety prices p^j_{nmt}.
   (e) Compute sector good prices P^j_{nt} and after-tax prices $\tilde{P}^j_{nt}$.
   (f) Compute energy price P^E_{nt}, final good price P_{nt}, and material price $P^{M,j}_{nt}$.
   (g) Compute bilateral trade share π^j_{nmt} and capital allocation share Φ^j_{nt}.
   (h) Compute rental rate of aggregate capital v_{nt} and gross return R_{nt}.

(i) Normalize prices using the US final good as numeraire; update dev_{NN} and increment $iter_{NN}$.

2. Quantity and Value Block

   (a) Compute the present-discounted value of non-capital income h_{nt}.

   (b) Compute capitalist consumption C^K_{nt} and capitalist investment W_{nt} (with capital irreversibility constraint).

   (c) Compute sector-specific capital supply K^j_{nt} under imperfect capital mobility.

   (d) Compute worker consumption c^j_{nt} and demand for final goods Y_{nt}.

   (e) Compute input-output linkages via the Leontief inverse; obtain industry expenditure X^j_{nt} and output Y^j_{nt}.

   (f) Compute demand for varieties y^j_{nmt}, gross revenues GR^j_{nt}, and factor demands $L^{D,j}_{nt}$, $K^{D,j}_{nt}$, Q^j_{nt}, M^j_{nt}.

   (g) Compute aggregate fossil fuel demand F_{nt}, fuel-type demands f^h_{nt}, and update reserves D^h_{nt}.

   (h) Compute government transfers Ω_{nt} from tax revenues on goods, fossil fuels, labor, and capital net of non-employment benefits.

3. Labor Mobility Block: repeat until $dev_m < tol_m$ or $iter_m = itermax_m$

   (a) Compute worker lifetime utility V^j_{nt} by backward induction from the terminal period.

   (b) Compute updated labor transition probabilities $(m')^{jk}_{nt}$ using the logit formula.

   (c) Compute deviation dev_m; update m^{jk}_{nt} via smoothing and increment $iter_m$.

4. Aggregate Labor and Capital

   (a) Apply labor transition probabilities to compute inter-sectoral flows L^{jk}_{nt} and update sectoral labor stocks L^j_{nt} via the law of motion.

   (b) Update sector-specific capital-use efficiency a^j_{nt} using the law of motion with adjustment toward data capital shares.

5. Environment Block

   (a) Compute emission intensity per fossil fuel input ϕ^F_{nt} and per brown energy output ϕ^B_{nt}.

   (b) Compute production-based emissions E_{nt} and consumption-based emissions E^C_{nt}.

   (c) Update the global carbon stock $\mathcal{C}_t$, temperature $\mathbb{T}(\mathcal{C}_t)$, and environmental damage factor $\mathcal{D}_n(\mathcal{C}_t)$.

6. Market Clearing and Updating

   (a) Update wages w^j_{nt} proportionally to the ratio of labor demand to labor supply in each sector.

   (b) Scale wages by the income-expenditure ratio to satisfy Walras' law.

   (c) Update rental rates r^j_{nt} proportionally to the ratio of capital demand to capital supply in each sector.

   (d) Apply stability bounds to limit the step size of wage and rental rate updates.

   (e) Set non-employment income as a fixed fraction ϖ of the average wage.

   (f) Smooth wages and rental rates: $w^j_{nt} \leftarrow smth_w \cdot w^j_{nt} + (1 - smth_w) \cdot (w')^j_{nt}$, and analogously for r^j_{nt}.

   (g) Compute $dev_1 = \max(dev_w, dev_r)$; increment $iter_1$.

Convergence. The algorithm terminates when $dev_1 < tol_1$ and $dev_a < tol_a$.